How Much Can You Make? Misprediction and Biased Memory in Gig Jobs*

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Abstract

Flexibility is an increasingly prominent feature of many jobs. In the gig economy, workers can choose their work hours and face wages that vary across hours and weeks. This increased complexity adds challenges to predicting and understanding job outcomes. Incomplete information or behavioral biases can then lead to inaccurate beliefs about pay and labor supply. We test this hypothesis by collecting novel survey data on 454 delivery and ride share gig workers in the United States. Comparing gig workers' beliefs with data on their actual job performance, we find they overestimate their predictions (43%) and their recalls (31%) of weekly pay, despite it being reported prominently in their earnings statements. Furthermore, gig workers underestimate expenses and overestimate hours worked. The results are consistent with selective recall: when forming and updating their beliefs in noisy environments, workers overweight past high-paying periods. We then examine how biased beliefs affect labor market decisions. We derive predictions from a behavioral labor supply model and test them. We conclude that job choices and labor supply decisions are likely influenced by mistaken beliefs in flexible gig jobs.

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Many jobs have some level of flexibility, with work hours and pay that vary from month to month. Examples include shift choices and tipping in service sector jobs and short-term contracts with piece rate pay in developing countries. A more prominent case are gig economy jobs, which have surged in popularity in the past decade (Collins et al., 2019). Over 9 percent of adults in the United States work in a gig job for an online platform such as Uber, DoorDash, Instacart or TaskRabbit in a given year (Pew Research Center, 2021). According to the gig economy literature (Chen et al., 2019; Koustas, 2018), flexible jobs generate surpluses of thousands of dollars per year, as workers are able to adjust work hours to accommodate changes in reservation wages and in demand.

Standard economic models assume that agents have accurate expectations of their labor market outcomes. Yet, the extra complexity involved in understanding and predicting pay and work hours in flexible jobs may lead workers to hold incorrect beliefs. For example, workers may not fully comprehend how predictable supply and demand shocks influence variation in earnings and hours. In this case, labor market decisions will not be optimal and prior estimates of worker surplus created by flexibility may no longer hold. For instance, a worker who misperceives how much they make may choose a sub-optimal number of work hours and allocate them inefficiently across the month. Furthermore, they might misunderstand expected pay differentials across jobs and make incorrect job choices. Inefficient selection can also occur: workers who overvalue flexible jobs may stay at them longer.

In this paper, we examine whether workers in flexible jobs misperceive their labor market outcomes and analyze how this impacts their decision making. We collect novel data from 454 ride share and delivery gig workers in the United States through online surveys. We begin by eliciting recalls and forecasts of key outcomes such as pay, hours, and expenses. We then collect actual job performance data by asking workers to submit screenshots from the gig platform app. We measure errors in remembering and predicting job outcomes by comparing workers' stated beliefs with their actual outcomes.

We find that gig workers overestimate their job performance by economically meaningful

amounts. We first consider gross weekly pay, which is an important outcome for gig workers and is featured prominently in earnings statements. On average, gig workers overestimate a forecast (incentivized for accuracy with a bonus of up to $5) of next week's weekly pay by 43.7%. Likewise, a recall of weekly pay of either the last week or the last month is overestimated by, on average, 31.3%. These errors amount to around $85 to $110 per week. More than 70% of our sample overestimates each measure. We show that forecast and recall mistakes are strongly correlated, suggesting a link between prediction errors and biased memory.

Gig workers' perceptions of their *post-expenses* pay is what should guide consumption choices and other decisions. Thus, to have a better understanding of mistaken beliefs, we estimate drivers' expected costs when doing ride share and delivery gig work. Our measure is a function of a car's category, such as small sedan or medium SUV, and age. We consider only variable costs, including maintenance and repair, and make conservative assumptions regarding taxes. Using our estimate of expenses, we find that recalls of net weekly pay are exaggerated by, on average, 46.3%.

Mistaken beliefs about net weekly pay can be broken down into three components. Indeed, the net weekly pay is the product of the average gross hourly pay, an expenses discount, and hours worked per week. We find that overestimation of labor supply and underestimation of expenses are the most influential factors in explaining aggregate errors. Misperceptions of gross hourly pay play only a minor role. In particular, gig workers over-recall work hours by, on average, 33.1% or 5.8 hours, while underestimating expenses by 22%. In addition, a majority of gig workers report ignoring several categories of costs, including depreciation and expected repairs, when calculating their take home pay. We argue that our findings are robust to potential concerns about sample selection and social desirability bias.

The next step is to investigate why gig workers misperceive their job outcomes. The explanation should provide a reason for pay being consistently *over*-estimated, rather than equally misunderstood in both directions. Our finding that forecast and recall errors are

correlated suggests that memory biases are important in the development of overoptimistic beliefs. We propose selective positive memory as a leading explanation. In this framework, past high-paying periods stand out in workers' memories much more than low-paying ones. Consistent with this mechanism, we show that a pay increase of $100 in the highest-earning week out of the last four is associated with a rise in recall of weekly pay of $57, compared to $26 in the lowest-earning week. Such recall biases may arise either from motivated beliefs like self-deception (Bénabou and Tirole, 2002, 2016) or from more fundamental cognitive limitations (Ba et al., 2024; Bordalo et al., 2023). Selective recall of high-earning weeks may also lead workers to overestimate hours if earnings and hours are remembered together as features of the same work episode, consistent with associative accounts of episodic memory (Kahana, 2012).

We show gig workers update their pay beliefs asymmetrically, reacting more after realizations of weekly pay that are greater than their previous belief. This finding provides a rationale for the persistence of overoptimistic beliefs over time. Consistent with this conclusion, we document a positive correlation between gig work experience and overestimation of pay. That is, gig workers who are experienced overestimate their pay by *more*. We show this striking result reflects both incomplete learning and a selection of biased workers in gig jobs over time.

When labor market outcomes are noisier, selective memory has greater scope to distort beliefs by overweighting atypical events. Supporting this prediction, we show that overestimation is increasing in the variance of previous realizations of weekly pay. Increasing the coefficient of variation (standard deviation over the mean) of the four previous weeks by 0.1 is associated with a statistically significant rise in weekly pay over-recalling of around $9. This result supports the view that the flexibility of gig jobs is the key aspect behind misperception of job outcomes. It can also explain why full-time salaried workers do not overestimate their earnings (Moore et al., 2000; Rothbaum, 2015): there is no way in these settings to select memories of being paid above the usual salary.

Misperceptions about job outcomes will likely not significantly reduce gig workers' welfare unless their labor market decisions are affected. To this end, we discuss the labor market consequences of mistaken beliefs in gig jobs. We derive predictions from a behavioral labor supply model. We incorporate biased memory related to pay and disutility of work, a key determinant of labor supply, and systematic overestimation of net hourly earnings (which workers overestimate by 20.7%, on average) into agents' belief updating process. The model predicts that biased gig workers will: (i) sometimes fail to switch to a higher-paying alternative job; (ii) inefficiently allocate work hours across their household budget cycle, working relatively little early in the cycle due to overly optimistic predictions about future earnings; (iii) work too many hours overall; and (iv) save less than anticipated due to systematically lower-than-expected earnings.

We then test these predictions. We begin by investigating whether gig workers with biased beliefs make sub-optimal job choices. We find that 45% of the workers in our sample move from being above to being below their stated reservation wage when we use their actual (rather than perceived) net hourly pay. This number is probably an overestimate and can be partially explained by other factors. Nevertheless, even in our most conservative specification, we find that 17% of gig workers would be put below their outside option if they considered their actual take home pay. These results imply that a significant share of gig jobs are sustained by mistaken beliefs about pay.

We next examine the model's prediction of excess labor supply among biased workers by analyzing how hours worked relate to misperceptions of earnings and expenses. Supporting this prediction, we find that overestimating hourly pay is associated with working more hours: a \$5 increase in hourly pay overestimation is associated with working an additional 1 to 2 hours per week. Although not conclusive, these findings are indicative of gig workers working more hours than desired due to believing pay is higher than it actually is.

Finally, we test for under-smoothing of work hours within household budget cycles. Using information on workers' primary bill payment dates, we find that biased gig workers increase

their labor supply near when bills are due. Specifically, workers who overestimate their pay work 10 to 15 percent more hours in the week before bills are due compared to 3 or 4 weeks away. This pattern holds across multiple measures of overestimation and suggests that biased beliefs about earnings distort workers' intertemporal labor supply decisions.

Our paper builds on the behavioral economics literature on the existence and persistence of biased beliefs such as overconfidence.[1] In particular, we closely relate to the theoretical (Bénabou and Tirole, 2002, 2016; Köszegi, 2006) and empirical (Eil and Rao, 2011; Gödker et al., 2022; Gottlieb, 2010; Möbius et al., 2022; Saucet and Villeval, 2019; Sial et al., 2024; Zimmermann, 2020) literature linking mistaken beliefs to the functioning of memory.[2] We contribute to this literature by being one of the first to apply these ideas to the labor market (Hoffman and Burks, 2020; Huffman et al., 2022), by emphasizing the importance of flexibility in generating mistaken beliefs, and by documenting more thoroughly the implications of mistaken beliefs on labor market decisions.

This paper also relates to a growing literature in labor economics documenting how workers lack information about several variables necessary to make optimal labor supply decisions. For instance, they can have biased beliefs about their outside options (Jäger et al., 2024) and job market prospects (Bandiera et al., 2021; Banerjee and Sequeira, 2020; Conlon et al., 2018; Cortés et al., 2022; Mueller et al., 2021), as well as how their performance (Huffman et al., 2022) and compensation compare with that of their peers (Card et al., 2012; Cullen and Perez-Truglia, 2022). These beliefs have been shown to affect their search behavior, employment decisions (Bergman and Jenter, 2007; Larkin and Leider, 2012; Oyer and Schaefer, 2005) and other labor market choices. We contribute to this literature by investigating a parallel under-explored question: whether workers have accurate beliefs about their inside option – i.e., their current job.

Our work also contributes to the literature on flexible jobs (Camerer and Lovallo, 1999;

[1] See, for instance, Camerer and Lovallo (1999); Grubb and Osborne (2015); Malmendier and Tate (2015); Moore and Healy (2008); Puri and Robinson (2007); Sharot (2011).

[2] In addition, see Adler and Pansky (2020); Carlson et al. (2020); Chammat et al. (2017); Enke et al. (2020); Mischel et al. (1976); Schacter et al. (2008); Di Tella et al. (2015).

Crawford and Meng, 2011; Farber, 2015; Fehr and Goette, 2007; Thakral and Tô, 2021) and the gig economy (Angrist et al., 2021; Bernhardt et al., 2022; Chen et al., 2019; Collins et al., 2019; Cook et al., 2021; Hall et al., 2021; Katz and Krueger, 2019; Koustas, 2018; Mas and Pallais, 2017; Reich and Parrott, 2020). These jobs allows workers to choose their work schedule and are characterized by pay that fluctuates over time. A key question in this literature is how labor supply decisions are made. Our paper provides additional insight into this topic by finding that workers misunderstand key job outcomes in this setting, and that these biases can influence labor supply choices.

The paper is structured as follows. Section 2 presents the survey design and describes the data. Section 3 presents the results on mistakes in recalls and forecasts of job outcomes. Section 4 argues these mistakes can be explained by selective recall. Section 5 derives and tests the empirical predictions of a behavioral labor supply model in which workers have mistaken beliefs. Section 6 concludes and shows directions for future work.

2 Survey Design and Data

We study ride share and delivery gig workers for online platforms in the United States. These jobs are characterized by the use of platform apps to accept gigs, pay that varies with demand and skill, full flexibility in hours and a responsibility for workers of paying for most expenses. We only consider companies for which the earnings page in the gig platform app includes not only information on pay but also on hours worked. This choice, made to increase the range of outcomes we observe at these jobs, implies we do not study gig work done for Postmates, Amazon Flex, Shipt, and others.

We consider in our data work done for five gig economy companies: Uber, Lyft, Uber Eats, DoorDash and Instacart. During our study's time frame, Uber and Lyft combined to be almost the entirety of the ride share market, while DoorDash and Uber Eats represented over 80% of the food delivery market in the United States. Finally, Instacart had a market

share of around 45% of the American grocery delivery market (Bloomberg Second Measure, 2022).

We collected online survey data from gig workers from April to November of 2022. We recruited participants to our study using social media posts, social media ads, and gig economy newsletters. We invited participants with the following prompt: "*If you're a driver in the gig economy, answer our online survey to get $10 and a personalized report*".[3] Only people who worked in the past 3 months for at least one of the five gig companies we consider could take part in our study.

If a participant finishes our baseline survey, we invite them through email to answer two follow-up surveys: the midline and the endline surveys. The midline happens 1 to 2 weeks after the baseline, depending on when the first full week (starting on a Monday) following the baseline survey ends. The endline, in turn, happens 2 to 5 months after the baseline survey. We collected all endline surveys in October and November of 2022. Figure 1 summarizes the design of our surveys. We paid participants $10 for participating in our baseline and midline surveys and $20 for answering our endline survey.[4]

2.1 Belief Elicitation

Unless otherwise noted, all beliefs of job outcomes elicited over our three surveys refer exclusively to the gig company that the worker worked the most for in the previous 3 months. We start the baseline survey by eliciting recalls from workers about their job outcomes. We ask two thirds of respondents for a recall of the last month they worked for the gig company and the other one third for a recall of the last week. In our setting, weeks are defined to

[3]We also used other similar formulations: "*Drivers in the gig economy: Join our online academic study and receive a $10 gift card.*", "*Do you do food delivery or rideshare gig work? Participate in our online survey to receive at least $10.*" and "*Complete a 15 min survey from a UC Berkeley graduate student researcher for US drivers of DoorDash, Uber Eats, Instacart, Uber or Lyft. Receive a $10 Amazon gift card and see how you compare to other drivers.*"

[4]Participants that answered our follow-up surveys were also given a personalized report summarizing (i) their own pay and (ii) the average pay in our sample at their gig company. As described in Appendix E, half of participants actually received information (i) at the end of the baseline survey in the form of a randomized information treatment.

always start on a Monday and end on a Sunday.

We focus on recalls of gross weekly pay, weekly expenses, weekly hours, gross hourly pay and net hourly pay. We construct the recall of net weekly pay by subtracting the recall of weekly costs from the recall of weekly gross pay. We calculate the recall of expenses share out of total pay by dividing the recall of weekly costs by the recall of weekly pay. Figure A1 shows an example of our recall belief elicitation questions.

Next, we ask workers to forecast their job outcomes in the next week (starting on the following Monday) after the baseline survey. We only ask for forecasts of workers who say they are very or somewhat likely to work during this week, which is around 85% of our sample. The forecast for gross weekly pay is incentivized. In particular, we use a Quadratic Scoring Rule to define a payoff based on accuracy, with a bonus that goes up to $5 and is $0 if the absolute value of the prediction error is equal to around $160 or higher. Thus, workers have an incentive for truth-telling.[5] Forecasts for weekly hours and gross hourly pay are also elicited but are not incentivized. Figure A2 shows an example of our forecast belief elicitation questions.[6]

Both the midline and the endline surveys are simplified versions of the baseline survey. In the midline survey, we measure the accuracy of the forecasts elicited in the baseline. We again elicit recalls for key job outcomes. The recall period is the same week as the forecasts refer to. We then show participants how their forecast of weekly pay matched reality, and inform them of their accuracy bonus.

In the endline survey, we elicit recalls referring to the average of the four previous weeks in which participants worked for the gig company. We only elicit beliefs about the baseline gig company during the follow-up surveys if participants report doing work for that company during the relevant time period. Throughout all surveys, we gather a rich set of covariates

[5]We used the Quadratic Scoring Rule (QSR) due to its simplicity. In particular, the payoff function is defined as $\max\left\{5 - \frac{(X-\tilde{X})^2}{5000}, 0\right\}$, where X is the actual weekly pay for the relevant week and $\tilde{X}$ is its forecast. Participants have to click a button to see an example and the formula of the payoff function. The QSR has been shown to not be robust to risk-aversion, but we believe more complex alternatives would be more problematic, as discussed in Charness et al. (2021); Danz et al. (2020).

[6]The final part of the baseline survey is a randomized information treatment, detailed in Appendix E.

referring to demographics, job market history, work habits, and other secondary beliefs.

2.2 Labor Market Outcomes

We obtain information on labor market outcomes by asking workers to submit screenshots of the weekly earnings page from the gig company app. As with beliefs, this gig company is the one they worked the most for in the past three months. In all three surveys, we ask for these screenshots only *after* gig workers state their beliefs, and have no possibility of modifying them. Figure 2 shows examples of the screenshots we request. They have information on, among other variables, weekly hours and gross weekly pay for a particular week. Each screenshot prominently displays the gross weekly pay in large font at the top.

As part of the baseline survey, we require workers to submit a screenshot of their weekly earnings page for their most recent week of work with the gig company. Participants receive a bonus of $2 if they agree to submit an additional three screenshots, referring to their three previous weeks working for the same gig company. In the midline survey, we ask for a screenshot of the weekly earnings page for the forecast week. In the endline survey, we ask workers to upload screenshots of the weekly earnings page for up to twelve weeks in the period between the midline and endline surveys. In this survey, we pay participants a bonus of $1 per screenshot submitted beyond the first one.[7]

Our measure of weekly hours is the total amount of hours spent online in the app (that is, time spent actively on gigs plus time spent waiting for gigs) in a given week. This is the same definition of labor supply as some of the previous literature on the gig economy (Angrist et al., 2021; Chen et al., 2019; Cook et al., 2021). It is also a measure available for all the gig companies we consider. In addition, this measure captures the nature of a standard job, in which only part of the time is spent actively working. We calculate the average gross hourly pay by dividing the gross weekly pay by weekly hours.

[7]In some cases, workers are asked for recalls referring to the past month but only agree to submit screenshot information referring to one week. When studying mistaken beliefs about job outcomes, we will use this one week as a proxy for the last four.

2.3 Accounting for Expenses

We do not observe work expenses for gig workers in our sample. As a result, we use an estimate of expected costs of driving to do gig work to calculate the expected actual net pay, a measure relevant for consumption and work decisions. We take a conservative approach and only consider the main variable costs, which are: fuel, maintenance and repair, variable depreciation and taxes. Our cost measures for maintenance, repair and variable depreciation are based on the AAA Your Driving Costs 2022 guide. Fuel costs are an average of gas prices, also from AAA, in the three months before the baseline survey.

Using survey information on which car is used to do gigs, we estimate expected expenses across 27 groups, combining 9 car categories and 3 car age groups (0-5, 5-10 and 10+). Estimates from the AAA Your Driving Costs guide apply only to the first 5 years of owning a car. We adjust for variation of maintenance, repair and depreciation costs over a car's lifespan by using information from CarEdge. In particular, we find the ratio of how (i) depreciation and (ii) maintenance and repair costs compare for a car that is either 5 to 10 or over 10 years old and a car that is between 0 and 5 years old. We then apply this ratio to the estimates from the AAA guide.

We use the IRS mileage rate deduction in 2022 ($.585/mile) to calculate self-employment taxes. We estimate federal income taxes by combining reported yearly household income, an estimate of gig income over the year and by applying the standard deduction. We ignore state income taxes for simplicity. Table B1 shows an example of our calculation of expected expenses. We do not estimate expenses for workers who rent a car to do gig work or that use a bike or a scooter. We discuss additional details of our expenses estimation in Appendix B.

2.4 Sample Selection and Descriptive Statistics

We screen out non-active gig workers from our sample by only including participants who submit valid screenshots from the gig company app. For a screenshot to be considered valid, it has to satisfy three conditions: (i) weekly pay is visible and legible, (ii) the screenshot

is not findable on reverse image search, and (iii) the week at the top of the screenshot is at most 3 months before to when the baseline survey was answered. A total of 13 survey responses with identical emails or identical screenshots as other responses were also excluded from our sample.

When we make these restrictions, we end up with 454 baseline survey responses. 51% of workers in our sample have DoorDash as their main gig company for the past three months. This number is 20% for Instacart, 13% for Uber Eats, 10% for Uber and 5% for Lyft. Furthermore, we have 210 midline survey responses and 202 endline survey responses. These numbers imply a response rate of around 50% for the follow-up surveys. We trim the top and bottom 1% of all forecasts, recalls and actual job outcomes. Our results are robust to changing this trimming cut-off to 2% or 3%. We analyze selective attrition across our three surveys in Table A1 and Table A2. We cannot reject that the mean observable characteristics are overall the same across surveys.

Summary statistics for our baseline survey sample are presented in Table 1. In Panels (A) through (C), we present information for our main covariates. Most gig workers in our sample are between 18 and 54 years old. They are over 70% white and majority female. Around 38% of them have at least a complete college degree, and a little over half of our sample has a household income of $40,000 per year or less. We find that 42% of respondents are struggling financially and 83% of them consider the pay from their gig work to be essential.

Over half of gig workers in our sample has another job – either a gig or a non-gig one. Around 10% were unemployed before starting doing gigs, while 36% had a full-time job. In addition, 23% of workers have at least a year of gig ride share experience, while that number is 57% for gig delivery experience. The five most common cars that gig workers in our sample use are: Hyundai Elantra, Toyota Corolla, Honda Civic, Honda Accord and Hyundai Sonata. About 80% of participants drives either a small sedan, a medium sedan, a compact SUV or a medium SUV. In addition, the median car age is 8 years old.

In Panels (D) through (F) of Table 1, we present summary statistics for our main outcome

variables. Gross pay is, on average, $18 per hour and $284 per week. Average hours worked is equal to 17 per week. Note this includes only weeks with a positive amount of work hours. Our estimated expected expenses share is, on average, 32% of total gross earnings.

We observe that, generally, recalls and forecasts have higher averages than actual job outcomes.[8] For instance, the average forecast of gross weekly pay is around $376/week, while the average recall of weekly hours is around 22 hours per week. In addition, recalls and forecasts of job outcomes are similar. Outcome variables vary widely across workers, reflecting the nature of flexible gig work.

Table A3(A) compares the share of workers in our sample who have done any gigs for different gig companies to how many people each of those companies contract in the United States. We find evidence that DoorDash workers, and delivery workers more broadly, are over-represented in our sample relative to Uber and Lyft drivers. Table A3(B) shows how key summary statistics in our sample of gig workers compares to other surveys on the gig economy (DoorDash, 2021; Pew Research Center, 2021; Reich and Parrott, 2020). We find suggestive evidence that our sample is younger, more white and more educated. In addition, participants in our study appear to work more hours than the representative gig economy worker.[9]

3 Mistaken Beliefs about Job Outcomes

3.1 Forecasting Errors

We now explore whether job outcomes are misperceived by gig workers. We begin by looking at forecasts of gross weekly pay for the next week. This is a consequential belief, essential for labor supply and consumption choices. Weekly pay is also the most salient job variable when checking the gig platform app, as the screenshots in Figure 2 show. Further-

[8] As in the rest of the paper, we pool together recalls for the past week and the past month.

[9] We discuss the implications for our estimation in Sections 3.5 and 3.6, where we also examine heterogeneity in our main results across different types of gig workers.

more, our sample of gig workers had a special incentive to get this prediction right, as we provided them with an accuracy bonus of up to $5.

Figure 3(A) shows a scatter plot comparing the forecast and the actual gross weekly pay for each gig worker. The forecast refers to the first week following the baseline survey. We only include participants who answered the midline survey, as otherwise we cannot measure the accuracy of their forecast. We draw a 45 degree line to represent the case where forecasts exactly match actual realizations. Points above this line indicate overestimation, while points below it imply underestimation of gross weekly pay.

Due to noise and variation in gig pay over time, points are unlikely to be positioned neatly on the 45 degree line even if workers' beliefs are correctly calibrated. However, in the absence of systematic biases, points should be positioned *symmetrically* around this line. Looking at the results, we find that points are concentrated above the 45 degree line, indicating that the majority of the sample over-predicted their weekly pay. Furthermore, this asymmetry holds across most of the distribution and is not dependent on outliers.

We define overestimation of a job outcome (either for a recall or a forecast) as the belief minus the actual realization. We say a worker underestimates a job outcome if this measure is negative. We present statistics using this variable in Table 2 Panel (A).[10] We find that overestimation of forecasts of gross weekly pay is significant: it is 43% of the actual weekly pay, or around $110, on average. 72% of our sample is identified as overestimating their forecast. These measures are statistically significant at 1% against the null of no overestimation. As seen in Figure 3(A), the slope of a linear regression of forecasts on the actual weekly pay is around 0.85. This implies that forecasts are significantly related to actual outcomes and that overestimation is overall decreasing on actual weekly pay.

[10]The number of observations varies across variables because the same trimming rule can result in slightly different sample sizes due to ties, and because some submitted screenshots contain incomplete information. In particular, some screenshots contain earnings information but not hours worked. Moreover, we do not estimate expenses for certain categories of drivers, such as those who use bikes or scooters.

3.2 Recall Errors

We have shown that workers make large mistakes when they predict their weekly pay. Nevertheless, forecasting job outcomes is by nature a complex task in flexible jobs. As such predictions require taking into account a complex set of shocks and understanding their exact relationship with hours and pay, it might not be surprising that workers can fail to do them correctly.

We now tackle a much simpler problem: *recalling* gross weekly pay. A gig worker has easy access to the actual value of this variable for all periods in which they worked for the company. That is, they can know their exact gross total pay in every week. Consequently, errors in recall will indicate additional difficulties in understanding gig pay.

In our analysis, we pool recalls of the previous week and month before the baseline survey. On Figure 3(B), we see a similar pattern as with forecasts: points are concentrated above the 45 degree line, and most participants seem to over-recall their weekly pay. The statistics in Table 2 confirm this suspicion. Overestimation of recalls is both statistically and economically significant. We find that gig workers exaggerate their actual weekly pay by over 30%, or around \$90/week. Over 70% of gig workers in our sample overestimate their recall of weekly pay. These measures are statistically significant at 1% against the null of no overestimation.

Thus, biases in forecasting and recalling weekly pay are similar in magnitude. This is true despite the fact that weekly pay predictions included incentives for accuracy while recalls did not. This is consistent with workers stating their true beliefs even without being incentivized. The similarity between recall and forecast errors may also indicate that mistaken beliefs about the future are tied to incorrectly remembering the past.

This suspicion is tested in Figure 4(B) by graphing, at the individual level, forecast overestimation against recall overestimation for weekly pay. The slope coefficient of the regression between these two types of mistakes is 0.5, which is statistically significant at 1%. Figure 4(A) re-does this analysis for forecast versus recall *beliefs*. Here, the relationship is

even stronger (coefficient of 0.84), suggesting that gig workers see these two beliefs as nearly equivalent.[11] From now on, we will sometimes focus on recalls as a short-hand for both types of errors due to the similarity between them and to allow us to take advantage of our larger sample of recall errors.

3.3 Gross versus Net Pay

The previous sections have documented that gig workers overestimate their future and past weekly pay. However, we have not taken into account the expenses and taxes associated with gig work. Costs are a significant margin at these jobs, since workers are generally responsible for paying them. This adds a new layer of complexity to understanding pay. At the same time, the net weekly pay, which represents the amount of income from gig work left over for consumption, is likely to be of special importance to workers.

Taking expenses into account is also relevant when thinking about the consequences of errors in understanding job outcomes in gig work. For instance, a gig worker that overestimates both gross weekly pay and expenses can end up holding correct beliefs about their expected net weekly pay. In this scenario, their mistaken beliefs might not result in sub-optimal labor supply and consumption decisions.

As previously mentioned, we calculate a measure of expected costs by car category and car age group, taking into account variable operating expenses and taxes (see Appendix B for more details). Our estimates will likely not reflect true costs at the individual level in particular weeks, but they are designed to approximate reality on average. We calculate the actual expected net weekly pay by discounting the gross weekly pay by our measure of the expected expenses share.[12]

We find more evidence of mistaken beliefs after adjusting for expenses. In Table 2, we see that workers significantly overestimate their recall of net weekly pay by about 46%, or

[11] We elicited recalls and forecasts at different points in the survey and on pages with very different layouts, as one can see by comparing Figure A1 and Figure A2. Therefore, confusion or inertia is not likely to be the cause for this result.

[12] Our estimated expected expenses share is, on average, 32% of total gross earnings.

around $90 per week. As shown in Figure A3, a scatter plot relating beliefs and the actual expected net weekly pay reveals a sizable grouping of individuals above the 45 degree line, where recalls are greater than the expected net pay.[13]

3.4 Decomposition of Mistakes

Net weekly pay is the product of three factors: average gross hourly pay, an expenses discount and weekly hours worked. This allows us to decompose total errors in understanding this outcome into individual errors along these three dimensions. Prior to discussing this decomposition, we examine the average overestimation for each outcome separately. Table 2 Panel (B) summarizes recall results once again. Overestimation of weekly hours is around 6 hours a week, or 33% of actual hours worked. Gig workers underestimate expenses by, on average, about 22% or 7 percentage points. Gross hourly pay is overestimated only by a small non-statistically significant magnitude. However, this analysis is based on recalling the average hourly pay over entire weeks. Our finding does not imply that workers are able to recall or predict variation in gross hourly pay over particular hours, for instance. Finally, we find that net hourly pay is over-recalled by, on average, 20.7%.[14]

Table 2 Panel (A) shows forecast errors for these outcomes. Weekly hours are overestimated by about 8 hours, or 63% of actual hours driven. Once again, gross hourly pay is overestimated by only a small, non-significant amount. For both recalls and forecasts, about 70 percent of gig workers overestimate (or underestimate in the case of expenses) each individual outcome, with the exception of average gross hourly pay. These shares are statistically different from 50% at the 1% significance level. Thus, we find strong evidence of significant overestimation in recalls and forecasts across most job outcomes in gig jobs.[15]

[13] Our inability to capture variation in expenses across time partially explains the dispersion seen in this plot.

[14] Appendix C shows an analysis of beliefs about the average worker at the same gig company.

[15] Figure A3 and Figure A4 present additional scatter plots relating beliefs to actual job outcomes. Table A4 shows that overestimation is generally positively correlated across different job outcomes, with the exception being hourly pay and hours. Figure A5 and Figure A6 show a significant relationship between forecast and recall errors and forecast and recall beliefs for both gross hourly pay and weekly hours. This

We now decompose the total error in recalling net weekly pay into three categories: average gross hourly pay, the expenses discount and weekly hours worked. Multiplying the recall of each of these components gives us the *implied* recall of net weekly pay. The average gig worker overestimates this implied measure by 53.3% of the expected actual net weekly pay. This is greater by about \$50/week to what we found using another definition of net weekly pay recall (recall of gross weekly pay minus recall of weekly expenses).

In the next step, we replace, in turn, each element of the implied net weekly pay recall with its correct equivalent. Figure 5 presents the results of this exercise. In this figure, we show the average pay for four possibilities related to this decomposition of net weekly pay: (i) implied recall, (ii) correct gross hourly, (iii) correct weekly hours; and (iv) correct expenses. By replacing recalls by their actual equivalent for each individual outcome, the implied net weekly pay drops by 25% for hours, 15% for expenses and 7% for gross hourly pay. Hence, underestimation of expenses and overestimation of hours explain most of why workers overestimate their take home weekly pay.

3.5 Heterogeneity Analysis

In Table A5, we examine whether overestimation is heterogeneous by worker characteristics. Within each table, we regress our measure of recall overestimation for different job outcomes on variables summarizing key worker attributes. In Panel (A), we divide the sample into (i) workers who are entirely or somewhat certain of their recall beliefs and (ii) workers who are neither certain nor uncertain, somewhat uncertain, or not certain at all of their recall beliefs. We find some evidence that overestimation is larger for workers more certain of their recalls. This is problematic, as this group may be less likely to seek out information about their actual gig pay and work hours.

Panel (B) examines whether errors vary by whether workers are full-time gig workers or not. Full-time gig work is defined as working at least 30 hours per week, on average. In

means that the strong ties between the two types of beliefs extend beyond weekly pay.

our sample, 23% of workers satisfy this definition. The results indicate that full-time gig workers overestimate their hourly wages by more, but underestimate their weekly hours. The combination of these effects implies the same amount of overestimation of weekly pay recalls.

Panel (C) of Table A5 focuses on financial need. Results show that workers who rely more on gig pay tend to overestimate their hours and weekly pay by more. This is significant, since these workers are likely to face more consequences from not knowing their pay. In Panel (D), we examine demographic characteristics (age, gender, race, education, and household income). Overall, we find no statistically significant relationship between these characteristics and mistakes in recalling job outcomes. In Panel (E), looking at labor market information, we show that gig workers previously employed full-time overestimate their weekly pay and hours by larger margins.

Finally, in Table 3, we examine how misestimation varies with gig work experience. We find that gig workers with more than one year of experience in both rideshare and delivery overestimate their weekly pay by around \$70 *more* than inexperienced ones.[16] Thus, not only is experience not associated with gig workers holding more accurate beliefs, it is correlated with larger errors. As we discuss below, this result likely reflects a mix of incomplete learning and selection of biased workers over time at these jobs.[17]

3.6 Robustness

In this section, we discuss alternative explanations for some of our findings and conduct robustness checks. We categorize possible issues into three categories: selection and attrition, measurement of beliefs, and measurement of outcomes. We address each set of potential problems in turn.

We first argue against the concern that our overestimation results come from dispropor-

[16]We provide an alternative analysis in Table A6, by separately considering delivery and ride share experience. We also include other job outcomes. We find similar results.

[17]We discuss additional results in Appendix D.

tionately selecting biased workers into our sample. First, our sample is likely to be more sophisticated and more informed than the average gig worker, as we partially recruited participants from gig worker groups and gig economy newsletters. Next, our heterogeneity analysis in Table A5 has shown that mistakes are widespread across a wide range of demographic and other characteristics. For this reason, selection along many observable dimensions would not be enough to explain our findings.

Third, a pilot study conducted without monetary incentives for participation ($N = 49$, January-April 2021) found overestimation of job outcomes of a similar magnitude as our main study.[18] While these different incentive structures likely attracted different types of gig workers, the similarity in results across both samples suggests our findings are not driven by sample selection bias (Dutz et al., 2021, 2023).

We further examine this issue in Table A7(C) by comparing overestimation patterns between survey respondents who did and did not accept an additional $2 payment for providing three additional weekly earnings screenshots from their gig platform app. This choice to provide screenshots may indicate differences in financial motivations for survey participation, which could correlate with unobserved characteristics affecting our overestimation measures. Yet, overestimation patterns are again similar across both groups.

Another potential concern is selective attrition in the follow-up surveys. As noted above, Table A1 and Table A2 provide no clear evidence of selective attrition along observable characteristics. Follow-up respondents and nonrespondents are broadly similar across a set of demographic, financial, and labor-market characteristics, as well as job outcomes and recalls. In addition, attrition across surveys does not affect many central results identified using only the baseline sample, including the evidence on recall errors.

Table A8 presents a sensitivity analysis of forecast errors in job outcomes, our main result potentially affected by selective attrition. We assess how extreme nonrespondents' forecast errors would need to be to substantially reduce or eliminate the estimated biases. Reducing

[18] In this pilot, workers overestimated their weekly earnings by $90.22 (s.e. = 30.90, $p < 0.01$), their weekly hours by 6.82 hours (s.e. = 1.77, $p < 0.01$) and their hourly wages by $1.61 (s.e. = 1.09, $p = 0.147$).

the implied full-sample bias by half would still leave economically significant overprediction of approximately $55 and 4 hours per week, while requiring mean errors among nonrespondents near the 25th percentile of the observed forecast- and recall-error distributions. Reducing the bias to one-quarter would require nonrespondent means near the 10th to 15th percentiles, while eliminating it would require mean *under*prediction of $109 and 8 hours per week, near the 5th percentile. Selective attrition can therefore overturn these findings only if nonrespondents have belief errors drawn from the very bottom of the observed distributions.

We believe our findings are not the result of measuring recall and forecast beliefs incorrectly. First, we explicitly ask gig workers to consider all elements of pay, including tips, bonuses and platform fees. Another possibility is that workers round up their beliefs when answering our survey. Table A7(B) shows that considering only workers with beliefs that are not round numbers implies qualitatively similar overestimation results.

We elicit beliefs about job outcomes only for one gig company. However, a significant number of gig workers in our sample (42%) worked for more than one platform in the previous three months. If workers do not correctly understand our survey questions, they might report total pay and hours across multiple companies. Table A7(A) shows overestimation of job outcomes is still economically and statistically significant if we only consider the group who worked for only one gig company.

As a result of social desirability bias, workers may intentionally overstate their beliefs. We do not believe this is driving our results for a number of reasons. First, remember that predictions of weekly pay were incentivized with a monetary bonus for accuracy. This raises the costs of inflating beliefs and should make beliefs more accurate in the presence of a desire to impress researchers. Nevertheless, we found substantial overestimation in weekly pay forecasts. Second, work hours are less likely to be overstated in this way. Indeed, working additional hours implies a lower hourly pay and does not clearly carry a self-image benefit (required for social desirability bias). We have shown, however, that workers significantly

overestimate their hours worked.[19]

Finally, during the follow-up surveys, workers are aware that we might ask them to later submit screenshots containing their job performance. As such, inflating beliefs for the sake of impressing researchers makes less sense, since participants can infer that their answers can be directly compared to reality. Table A13 shows that, in the endline survey, gig workers still overestimate job outcomes by large margins.

We discuss robustness checks for our measurement of job outcomes, including a discussion on the distinction between online and active hours, in Appendix D. To test the robustness of our measure of expected gig work costs, we now propose an alternative formulation of errors in understanding this outcome. We first ask workers which categories of costs they consider when calculating their net pay. Results are shown in Figure A7. Many gig workers report ignoring types of costs such as maintenance (45%), taxes (65%) and depreciation (83%).

Based on this information, we calculate an implied measure of belief about expenses. Specifically, we assume drivers use only the cost categories they self-report to take into account. We then input *our own* expenses estimates for each of these categories. In this alternate estimate, only errors resulting from ignoring some types of expenses are considered. In reality, errors also arise from wrong beliefs about the expected cost of repairs, for instance. Using this measure, we still find that gig drivers underestimate expenses by about 5 percentage points, or 15%.

4 Explaining Mistaken Beliefs

In the previous section, we documented that gig workers overestimate both their prior and future earnings by significant margins. This section attempts to understand why these errors occur. The proposed mechanism should be able to explain why beliefs are biased toward overestimation, rather than being inaccurate in both directions. In addition, we

[19] Appendix C documents that gig workers in our sample do not believe they earn more than the average gig worker. This piece of evidence is also inconsistent with the standard formulation of the social desirability bias.

found before that forecasting and recall errors are correlated. In light of this, we must also take into account the connection between inaccurate memories and wrong beliefs about the future.

We propose that misperceptions are developed and maintained by selective positive recall: favorable memories are more easily accessed than unfavorable ones. By relying on a biased memory, forward-looking beliefs used in making decisions are then distorted. For example, a gig driver with selective recall may remember high-earning weeks – e.g., during which major business conferences and sporting events happen – more readily than typical weeks. By doing so, they reinforce the perception of being well-compensated. This will then reduce the accuracy of earnings forecasts.

By a similar logic, new information does not necessarily lead to more accurate beliefs: updating is not Bayesian, but it reacts disproportionately to positive news over time, which have a more lasting effect on beliefs. Therefore, incorrect views can persist and additional experience may not make mistakes disappear. There are limits to belief bias in this framework. Nevertheless, these constraints can be relaxed when signals are less informative and the environment is noisier, since this allows for additional distortions through the overweighting of unrepresentative memories.

Recent empirical literature finds extensive evidence for overly optimistic beliefs tied to selective memory. Many domains have been studied, including investment decisions (Gödker et al., 2022), exercising (Sial et al., 2024), school grades (Roy-Chowdhury, 2024), intelligence (Möbius et al., 2022; Zimmermann, 2020), beauty (Eil and Rao, 2011), and generosity and altruism (Carlson et al., 2020; Saucet and Villeval, 2019; Di Tella et al., 2015).

While we remain agnostic about the ultimate mechanism behind selective memory, motivated beliefs (Bénabou and Tirole, 2002, 2016) offer one framework for understanding it. Under this theory, individuals maintain incorrect beliefs either for direct hedonic utility or as a motivational device. For example, people may derive pleasure from believing they are highly compensated, intelligent, or productive, with biased recall facilitating these self-

enhancing beliefs.[20] Alternatively, the patterns we observe could arise from direct cognitive imperfections: high-paying weeks might simply be more memorable due to their salience, due to how frequently they are revisited or rehearsed, or because they are more commonly cued by current events (Adler and Pansky, 2020; Ba et al., 2024; Bordalo et al., 2023; Conlon, 2024; Graeber et al., 2024; Mullainathan, 2002).

4.1 Evidence of Biased Memory

We now test for evidence of biased memory in our setting. Our primary outcome of interest is weekly pay. Unless otherwise noted, we use data collected in the baseline survey.

Biased recall of high-paying periods. First, we examine whether high-paying periods influence recalls more than low-paying ones. This may happen if gig workers remember high-paying weeks more easily. We only consider workers asked to recall the average of the last four weeks. Our first specification regresses recalls of gross weekly pay on the maximum and the minimum of the last four actual realizations. In the absence of biases, a one-unit increase to either the maximum or the minimum should affect recalls equally. Indeed, the recall *should* be a simple average of the weekly pay in the last four weeks. We find that this is not the case. Results are shown in the first column of Table 4(A).[21] Increasing pay in the workers' highest paying week by \$100 is associated with a rise in recall of \$57, compared to \$26 for a similar increase in the lowest paying week out of the four.

In column (3), we regress the recall of weekly pay on both the actual mean of the past 4 weeks and the maximum of these same weeks. If workers rationally form their beliefs, the coefficient on the maximum week should equal 0, since the mean is a sufficient statistic for the recall. Yet, we find that not to be the case. Indeed, the maximum week has a statistically significant coefficient that is about forty percent as large as the coefficient on the mean. Thus, the best-paying periods are overweighted by gig workers in belief formation,

[20] Figures 3(A) and 3(B) show evidence that workers with higher actual weekly earnings have more accurate beliefs. While this pattern is suggestive of self-deception – since workers earning more have less motivation to deceive themselves – it is also consistent with pure selective recall of high-paying weeks.

[21] Table A12 shows the same set of results, but for forecasts.

as predicted by selective recall of favorable events.

We now evaluate whether workers update their beliefs asymmetrically when presented with new information about their pay. We estimate how the weekly pay recall at the midline survey relates to the relative comparison of (1) the actual weekly pay in the week that the recall refers to, and (2) the previous recall belief, elicited in the baseline survey. We expect workers to update more strongly to positive than to negative realizations.

We apply this idea in a simplified way by dividing weekly pay realizations into two: those above or equal to the previous recall, and those below. We then run regressions of the midline weekly pay recall on the previous recall and two binary variables, equal to one when the actual weekly pay realization is (i) above or equal to or (ii) below the original recall belief.

Results are shown in Table 4(B). We find that gig workers update their beliefs more strongly after a positive realization. The recall of weekly pay is estimated to increase by \$208 when the actual weekly pay is above or equal to the previous belief, and to fall by only \$48 when the opposite is true. This asymmetry in reacting to new information can justify the persistence of biased beliefs. These findings are consistent with the previously discussed Table 3, which examines how mistaken beliefs vary with gig work experience. We found that not only is gig experience not associated with gig workers holding more accurate beliefs, it is correlated with larger errors.

However, our correlational estimates of experience effects are influenced by both selection and learning. To disentangle learning from selection effects, we examine overestimation levels in our endline survey, conducted two to five months after our initial survey, where we ask for recalls of the past 4 weeks. Comparing these belief errors with those from the baseline survey allows us to estimate learning effects. As shown in Table A13, the average level of overestimation across job outcomes is 30% to 40% lower at the endline survey (compared to one-month recalls shown in Table D1). This indicates that some learning, likely hindered by biased recall, occurs over this time frame. At the same time, this appears insufficient to

counter the likely selection over time of overconfident workers at gig jobs.

Flexibility in pay and biased recall. We now evaluate whether greater variation in job outcomes may amplify overestimation of weekly pay by increasing the scope for memory distortions. Indeed, when outcomes are more uncertain, selective recall of unrepresentative events can have larger effects on beliefs. The same mechanism might also explain why workers in non-flexible jobs do not appear to overestimate their earnings.[22] In fact, studies comparing survey data with tax data reveal a slight *under*-reporting of labor income in surveys (Moore et al., 2000; Rothbaum, 2015).

We measure wage variability using the coefficient of variation (standard deviation divided by mean) of weekly earnings over four-week periods. Results are presented in Table 5. We find that a higher coefficient of variation leads to more overestimation: increasing this variable by 0.1 (mean of 0.48) is associated with an increase in weekly pay recall overestimation of around $9, which is statistically significant at 1%. This result supports the view that the flexibility of gig jobs is the key aspect behind overestimation of job outcomes.

Other outcomes. We re-do our previous analyses for weekly hours and gross hourly pay in Table A9 and Table A11. Generally, the results are similar for hours. Overestimation of hours may appear to challenge the selective positive memory mechanism, since longer work hours may not themselves be viewed positively. However, if earnings and hours are encoded as associated features of the same work episode[23], recalling a high-earning week may also bring its associated hours to mind, consistent with associative and context-based accounts of episodic memory (Kahana, 2012). Selective recall of such episodes can therefore generate upward bias in beliefs about both weekly pay and hours, without requiring workers to value longer hours intrinsically.

Table A10 provides an extension of the selective recall analysis of weekly hours reported in Table A9. We distinguish between overweighting of high-hours weeks from overweighting of high-earning weeks. We find that the weight on maximum hours more than doubles

[22] See, for instance, Appendix Figure C.4 in Cullen and Perez-Truglia (2022).

[23] Indeed, in the data, high-earning weeks are typically also high-hours weeks.

when using the highest-earning week rather than the highest-hours week, and this variable also becomes statistically significant. This pattern supports the interpretation that workers recall hours as part of high-earning episodes rather than overweighting high-hours weeks independently.

Our results for hourly pay are broadly consistent with a lack of significant overestimation in this variable, as we don't find much evidence for the mechanisms discussed above. Alternatively, this can be explained by each realization of hourly pay in our data being only its weekly average, not allowing us to not incorporate intra-week variation.

We have not discussed the mechanisms underlying underestimation of gig job expenses. This is because we do not observe variation in actual expenses over time at the individual level. We believe it is reasonable to assume similar mechanisms than the ones just discussed are at play. For instance, when forming their beliefs, gig workers may overlook periods when their car broke down or when they had to pay taxes.

5 Consequences of Mistaken Beliefs

In this section, we explore the consequences of workers' systematic overestimation of key job outcomes, such as gross and net pay. This is important, as these mistakes matter more if they lead to welfare-reducing labor market decisions. We develop a simple behavioral labor supply model to derive testable predictions, which we then evaluate empirically.

5.1 A Model of Behavioral Labor Supply

Consider a model in which a worker makes consumption (c) and work (h) decisions over a series of two-period household budget cycles in either a gig job or a non-flexible job.[24] At the end of each cycle, the worker leaves savings (s) and the next cycle begins.

Period t decisions. Consider a gig worker at the start of a cycle, in period t. Given

[24]The cycles' structure in our model – and its size of two periods – is meant as a simplified representation of joint work and consumption decisions made in households across periods.

our setup, we study the worker's optimization problem within a single cycle only, starting at t and ending at $t+1$. A gig job G has net hourly pay equal to $w_{G,t} = \mu_G + \varepsilon_t$, where $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$. The gig worker maximizes his (perceived) value function by choosing c_t and h_t, while making plans for c_{t+1}, h_{t+1} and s_{t+1}:

$$\max_{c_t, c_{t+1}, h_t, h_{t+1}, s_{t+1}} u(c_t) + u(c_{t+1}) + g(s_{t+1}) - \eta_t c(h_t) - \tilde{E}[\eta_{t+1}] c(h_{t+1}) \tag{1}$$

such that

$$c_t + c_{t+1} + s_{t+1} = w_{G,t}(1+\theta) \cdot h_t + \tilde{E}[w_{G,t+1}] \cdot h_{t+1} + (1+r) \cdot s_{t-1} \tag{2}$$

Where $u(c)$ is a concave consumption utility function and $c(h)$ is a convex cost of effort function. Furthermore, s_{t+1} are non-negative savings, to be used after period $t+1$, and $g(s_{t+1})$ is the continuation value of s_{t+1}, with $\frac{dg(s)}{ds} > 0$. In addition, s_{t-1} are savings from the previous household cycle, and r is the real interest rate. For simplicity, we assume no time discounting.

Furthermore, η_t is the realized period t value for a shifter of the cost of effort function. Intuitively, it influences the disutility cost of working in a given period. $\tilde{E}[\eta_{t+1}]$ is the (potentially biased) expected value for η at $t+1$. The random variable η_t satisfies $\eta_t = 1 + \nu_t$, where $\nu_t \sim N(0, \sigma_\nu^2)$. In addition, $w_{G,t}$ is the realized net hourly gig pay at period t and $\tilde{E}[w_{G,t+1}]$ is the (potentially biased) expected net hourly gig pay at period $t+1$, detailed below. The parameter θ, with $\theta \geq 0$, denotes overestimation of net gig pay.

Represent period t choices by $\tilde{c}_t^*, \tilde{h}_t^*$ and period t plans for $t+1$ choices by $\tilde{c}_{t+1}^p, \tilde{h}_{t+1}^p$ and s_{t+1}^p. The first-order conditions of equation (1) imply that

$$u'(\tilde{c}_t^*) = u'(\tilde{c}_{t+1}^p) = g'(s_{t+1}^p) = \frac{\eta_t c'(\tilde{h}_t^*)}{w_{G,t}(1+\theta)} = \frac{\tilde{E}[\eta_{t+1}] c'(\tilde{h}_{t+1}^p)}{\tilde{E}[w_{G,t+1}]} \tag{3}$$

Equation (3) highlights three different biases that can be matched to our results from

previous sections: (1) $\theta > 0$: Overestimation of the marginal benefit of working by believing current pay is higher than it actually is; (2) $\tilde{E}[w_{G,t+1}] > E[w_{G,t+1}]$: Overestimation of the average future marginal benefit of working by believing average future pay is higher than it actually will be; (3) $\tilde{E}[\eta_{t+1}] < E[\eta_{t+1}]$: Underestimation of the average future disutility of work, leading to over-predictions of work hours at a given wage.

Period $t+1$ decisions. At the start of period $t+1$, η_{t+1} and $w_{G,t+1}$ are realized and the worker has an opportunity to revise their plans for c_{t+1} and h_{t+1}. They again maximize their perceived value function by choosing c_{t+1}, h_{t+1} and making plans for s_{t+1}. Represent period $t+1$ choices by $\tilde{c}^*_{t+1}$ and $\tilde{h}^*_{t+1}$ and period $t+1$ plans for savings as $\tilde{s}^p_{t+1}$.

Finally, at the end of period $t+1$, we assume the worker learns the *actual* (potentially negative) amount of money saved after period $t+1$, $\tilde{s}_{t+1}$:[25]

$$\tilde{s}_{t+1} = w_{G,t} \cdot \tilde{h}^*_t + w_{G,t+1} \cdot \tilde{h}^*_{t+1} - \tilde{c}^*_t - \tilde{c}^*_{t+1} + (1+r) \cdot s_{t-1} \tag{4}$$

At the start of the next household budget cycle, and before learning η_{t+2} and $w_{G,t+2}$, gig workers decide whether they want to move to the non-flexible job O. We assume they make this decision by comparing their perceived expected utility from one two-period cycle in the gig job ($\tilde{E}[V_G]$) and in the non-flexible job ($E[V_O]$).[26]

Biased memory and learning. Gig workers learn (in a potentially biased way) about their expected net hourly pay w_G over time. A gig job in period t has net hourly pay equal to $w_{G,t} = \mu_G + \varepsilon_t$, where $\varepsilon_t \sim N(0, \sigma^2_\varepsilon)$. Assume σ^2_ε is known. Assume a gig worker's prior $\tilde{w}_{G,0}$ satisfies $\tilde{w}_{G,0} \sim N(\mu_0(1+\theta), \sigma^2_0)$. The gig worker receives a signal each period for μ_G, which is $\tilde{w}_{G,t} = w_{G,t}(1+\theta)$. If $\theta > 0$, this signal will be biased and generate, in itself, persistent optimism in beliefs.

We assume an additional bias in the processing of signals, which depends on their rank

[25] This might happen, for instance, by the worker observing the balance on their bank account when having to pay a bill. We assume no additional learning occurs from this reveal. Adding a learning component associated to this step would add a countervailing force to holding biased beliefs about net hourly pay and weekly hours worked in gig jobs.

[26] We discuss the job choice decision in more detail in Appendix F.

within the distribution of observed signals. In particular, let $\phi_{t,j} = 1 + \alpha F_t(\tilde{w}_{G,j})$, where $\alpha \geq 0$ and $F_t(\tilde{w}_{G,j})$ is the empirical cumulative distribution function of signals up to period t, referring to $\tilde{w}_{G,j}$. This is intended to model a bias in recall that overweights past high-paying periods. Adapting the standard formula for normal Bayesian learning, we find the (perceived) posterior mean of $w_{G,t+1}$, which we call $\tilde{E}[w_{G,t+1}]$, to now be:

$$\tilde{E}[w_{G,t+1}] = \frac{\sigma_\varepsilon^2 \mu_0 (1+\theta) + \sigma_0^2 \sum_{j=1}^{t} \phi_{t,j} \tilde{w}_{G,j}}{\sigma_\varepsilon^2 + \sigma_0^2 \sum_{j=1}^{t} \phi_{t,j}} = (1+\theta) \cdot \frac{\sigma_\varepsilon^2 \mu_0 + \sigma_0^2 \sum_{j=1}^{t} \phi_{t,j} w_{G,j}}{\sigma_\varepsilon^2 + \sigma_0^2 \sum_{j=1}^{t} \phi_{t,j}} \tag{5}$$

Note that, if $\alpha = \theta = 0$, we return to the standard Bayesian learning model, such that the posterior mean eventually converges to μ_G. The memory bias ($\alpha > 0$) leads to overestimation of future net pay by assigning greater weights to higher values of past signals, biasing beliefs upward. This mechanism is stronger the more variation there is in past signals, which in our setting is akin to greater pay flexibility.

The learning process for the work disutility shifter η, which influences predictions of future labor supply, follows an analogous mechanism. The memory bias similarly overweights certain periods – in this case, those when work disutility was relatively low (i.e., when working more hours felt easier). We detail this process in Appendix F. Thus, the model can generate overestimation of hours without requiring workers to value longer work hours intrinsically. In this logic, periods of lower fatigue, fewer competing obligations, or more convenient schedules are both more favorable and associated with greater labor supply.

Figure A9 simulates how workers' beliefs evolve in our learning model. We examine the posterior mean of net hourly pay and the work disutility shifter η under varying levels of memory bias (α) and misperception of net hourly pay (θ). When both parameters equal zero, beliefs converge to the true values. Adding $\theta > 0$ is sufficient to cause overestimation of net gig pay. With a memory bias, beliefs progressively deviate from the truth over time before reaching a biased steady state.

5.2 Model-Based Implications of Mistaken Beliefs

Our model allows us to analyze how misperceptions of hours, pay, and expenses in gig work affect both labor supply and consumption decisions. We identify four key inefficiencies, which we now list and discuss:[27]

1. *Work too many hours.* Gig workers decide their labor supply in period t through the use of the first-order conditions in equation (3). If the worker overestimates net pay ($\theta > 0$), they believe the benefits from working one additional hour are higher than they actually are. Due to the assumed convexity of the cost of effort function, they will work too many hours relative to the rational benchmark.

2. *Under-smoothing of labor supply.* If the gig worker overweights past favorable memories ($\alpha > 0$), they will, in period t, over-forecast their expected net pay and labor supply in $t+1$. As a consequence, they will decrease period t labor supply $\tilde{h}_t^*$ and increase period t consumption. These channels push in the same direction of reducing available income in period $t+1$, when compared to a scenario with no biases. This, in turn, increases the marginal benefits of working in period $t+1$, raising $\tilde{h}_{t+1}^*$. Thus, the gig worker inefficiently works too much late in the household budget cycle and not enough at its beginning.

3. *Incorrect choice of employer.* A worker is mistakenly at a gig job when the perceived gig job value $\tilde{E}[V_G]$ is larger than the outside job value $E[V_O]$, which in turn is larger than the true gig job value $E[V_G]$.[28] A biased gig worker will overestimate the perceived value of the gig job $\tilde{E}[V_G]$. In addition, the same biased beliefs decrease $E[V_G]$ by leading them to make sub-optimal labor supply and consumption decisions. Both of these mechanisms increase the gap between the perceived and the actual value of a gig

[27] We show model simulation results in Appendix F.

[28] For some individuals, flexible work arrangements may impair time management and exacerbate procrastination, diminishing their ability to align work hours with their long-run welfare preferences. This possibility would further reduce welfare in gig jobs. In our framework, this is equivalent to assuming a systematic divergence in values of η between the actual and the perceived gig job value functions.

job. Thus, the more biased a gig worker is, the larger the range of values of the outside job net pay for which $\tilde{E}[V_G] > E[V_O] > E[V_G]$.[29]

4. *Savings are too low.* A gig worker that overestimates net hourly pay ($\theta > 0$) overestimates their labor income and consumes more than they would otherwise. In our model, they only realize this at the end of each household budget cycle. As such, biased gig workers overestimate their savings, and its realization is sub-optimal and unexpectedly low.

5.3 Empirical Evidence of Consequences of Mistaken Beliefs

Job choice. We begin by testing our prediction that biased gig workers will sometimes not move to a superior alternative job. As a proxy for the expected payoff from their outside option, we use workers' reservation wages, equal to their self-reported lowest acceptable net hourly pay to keep working in their current gig job. Figure 6 shows scatter plots relating the reservation wage to the net hourly pay, for either its recall (left panel) or its actual expected value (right panel). We find that, as we move from the recall to the actual expected net hourly pay, many gig workers are moved under the 45 degree line. In other words, their net pay falls below their reservation wage.

This impression is confirmed in Table 6. First, we find that 23 percent of workers have a higher reservation wage than recall of net hourly pay. This relatively high percentage may reflect a mix of noise in beliefs and actual job outcomes, confusion over the outside option, and perhaps real plans to stop working for the gig company in the future. When comparing the reservation wage with the *actual* expected net hourly pay, however, this percentage increases substantially, to 68%.

In other words, 45 percent of gig workers move from being above to being below their

[29]Figure A10 simulates job transitions over time for gig workers in our model. For a range of outside job wages, unbiased workers exit gig work while biased ones largely remain. Under some conditions, workers with a memory bias become increasingly less likely to transition out over time, as selective recall of high-paying periods leads them to progressively overvalue gig work.

reservation wage when their actual net hourly pay is considered, versus their belief of it. Gig workers for whom this is true can be characterized as making a job choice mistake. We find similar qualitative results by looking at the averages (Table A14) or the share of workers $5 below their stated reservation wage.

The other rows of Table 6 provide robustness checks to our results. First, we may exaggerate the magnitude of gig workers not choosing the best available job if they also overestimate their outside options. To address this, we first consider workers less likely to have another gig job – where mistaken beliefs are more likely to occur – as their outside option. Next, we assume that the reservation wage is reported with an error equal to half of the error related to understanding the net hourly pay. Our results remain similar when using these alternate measures.

We then re-run our analysis using a different proxy for the reservation wage. We use worker's reported gross hourly wages in either a previous job or in another current job. Using this measure, we find an increase in the group positioned under this proxy of the outside option pay of approximately 20 percentage points when using the actual expected net hourly pay. Next, fluctuations in gig pay may temporarily place workers below their reservation wages, even if this is not the case over longer periods of time. We provide a robustness check to this by assuming the actual pay is the highest average net hourly pay over the four previous weeks.

We should keep in mind that the outside option for the gig work we consider might be to work for another gig company. Consequently, workers who quit their current gig jobs in the event of being de-biased would not necessarily leave the gig economy. Our current analysis ignores errors in valuing gig work stemming from the overestimation of labor supply. Due to this channel, gig income will likely be lower than expected, lowering the value of gig work. With these caveats in mind, our findings indicate that a significant share of gig jobs are sustained by mistaken beliefs about pay.

Total hours worked. Our model predicts that biased gig workers supply excess hours

of work relative to the optimal. To test this, we examine the relationship between hours worked and measures of overestimation in Table 7. Panel (A) documents a strong positive relationship between hours worked and misestimation across job outcomes, with coefficients of approximately 0.4 for both gross and net hourly pay overestimation (significant at the 1 percent level) and 0.05 for expenses underestimation in percentage points (marginally significant). This implies that raising net hourly pay overestimation by $5 is associated with working an additional 2 hours per week.

However, any systematic link between actual hours and actual hourly pay could generate a spurious correlation between hours and overestimation.[30] To address this concern, we estimate specifications using non-contemporaneous measures of hours worked and recall misestimation in Panels (B) and (C). These specifications reduce mechanical linkages between the two variables at the cost of a reduced sample size. We still find a positive relationship between gross and net hourly pay overestimation and hours worked, albeit with smaller magnitudes of around 0.2. While these patterns are correlational, the results broadly support our model's prediction of excess labor supply among biased gig workers.[31]

Allocation of hours worked. Our model predicts that biased gig workers work relatively more hours in the latter part of the household budget cycle. This occurs as these workers overestimate future earnings, leading them to work fewer hours early in the cycle. When their pay turns out to be lower than anticipated, they then increase their labor supply to compensate. This distortion can be heightened if workers are budget-constrained and face end-of-cycle payment obligations.

We test this prediction using information on workers' primary monthly bill payment date (Figure A8). To conduct this analysis, we construct a worker-week panel and measure the distance between each week's end (Sunday) and the worker's next primary bill due date. We

[30] This concern does not apply to the underestimation of expenses, as the construction of this variable is independent of actual hours worked.

[31] As a further test of our model's predictions, we ran a randomized information treatment, in which we informed gig workers in the treatment group what their *actual* expected net hourly pay is. The effects of this intervention should be similar to moving our model's overestimation parameter θ towards zero. In general, we are under-powered to identify small to moderate effects. See Appendix E for more details.

then divide the household budgeting cycle into four weeks based on this metric. Figure 7 shows how labor supply varies within this cycle. In Panel (A), we split workers by their over-forecast of weekly pay. Workers who over-forecast increase their weekly hours by around 10 percent across the cycle, from 11.7 hours when bills are furthest away to 13 hours in the week before due dates. In contrast, workers who either correctly forecast or under-forecast their weekly pay reduce their hours worked over the cycle. This latter pattern is consistent with, among other things, workers who underestimate their future pay and work more early in the cycle to compensate.

In Panel (B) of Figure 7, the same heterogeneity is explored, but now using over-recall of net weekly pay. Workers who overestimate this measure increase their hours across the cycle, while those who do not overestimate it show no clear pattern. Panel (C) uses over-recall of net hourly pay. We again find similar results. Although the estimates are noisy, the consistency of our findings across all three measures provides support for the model's prediction.

Note that we cannot rule out that unobservable characteristics related to which days gig workers pay their bills partially explain our findings. Another caveat is that our definition of the household budget cycle, based on the main monthly bill due date, abstracts from other important dimensions of households' financial planning and labor supply choices.

6 Conclusion

In this study, we examine whether gig workers correctly understand their job outcomes. First, we collected data on beliefs and on actual job performance. When comparing the two, we find that gig workers consistently overestimate recalls and forecasts of their gross and net weekly pay. We then document that recall errors and forecast errors are strongly correlated. In addition, we show that overestimation of labor supply and underestimation of expenses are key drivers of aggregate mistakes.

Our findings are consistent with selective recall of high-paying weeks. According to this mechanism, errors stem from biased formation and updating of beliefs, do not necessarily improve with experience and increase with the noisiness of job outcomes. We find evidence for these implications in our setting. We develop a behavioral labor supply model and derive predictions for the consequences of overestimating pay in gig jobs, which we then test. The magnitude of the errors we identify is enough to move a significant share of gig workers below their stated reservation wage, indicating potential mistakes in job choices. We also document that gig labor supply choices and its allocation across the budget cycle are affected by overestimation of gig job outcomes, leading to sub-optimal decisions.

Our findings imply that the economic modeling of flexible jobs needs to incorporate behavioral biases in understanding pay and hours. Otherwise, welfare calculations and predictions will be inaccurate. Moreover, our work motivates policies that provide summarized job performance information to workers in flexible jobs. By doing so, mistakes in understanding job outcomes can be reduced. As technological advances allow for greater flexibility across a wider range of jobs, these issues become more relevant.

Our study can be interpreted as saying that flexible gig jobs are less valuable than previously thought. In spite of this, we believe these jobs may still provide significant surpluses, especially to individuals with strong preferences for flexibility. A study of how mistaken beliefs affect firms' decisions and their market power is left for future research.

References

Adler, Orly and Ainat Pansky, "A "rosy view" of the past: Positive memory biases," in "Cognitive biases in health and psychiatric disorders," Elsevier, 2020, pp. 139–171.

Angrist, Joshua D, Sydnee Caldwell, and Jonathan V Hall, "Uber versus taxi: A driver's eye view," *American Economic Journal: Applied Economics*, 2021, *13* (3), 272–308.

Ba, Cuimin, J Aislinn Bohren, and Alex Imas, "Over-and underreaction to information," *Available at SSRN 4274617*, 2024.

Bandiera, Oriana, Vittorio Bassi, Robin Burgess, Imran Rasul, Munshi Sulaiman, and Anna Vitali, "The search for good jobs: evidence from a six-year field experiment in Uganda," *Available at SSRN 3910330*, 2021.

Banerjee, Abhijit V and Sandra Sequeira, "Spatial mismatches and imperfect information in the job search," 2020.

Bénabou, Roland and Jean Tirole, "Self-confidence and personal motivation," *The quarterly journal of economics*, 2002, *117* (3), 871–915.

and , "Mindful economics: The production, consumption, and value of beliefs," *Journal of Economic Perspectives*, 2016, *30* (3), 141–64.

Bergman, Nittai K and Dirk Jenter, "Employee sentiment and stock option compensation," *Journal of financial Economics*, 2007, *84* (3), 667–712.

Bernhardt, Annette, Christopher Campos, Allen Prohofsky, Aparna Ramesh, and Jesse Rothstein, "Independent Contracting, Self-Employment, and Gig Work: Evidence from California Tax Data," Technical Report, National Bureau of Economic Research 2022.

Bloomberg Second Measure, Technical Report 2022.

Bordalo, Pedro, John J Conlon, Nicola Gennaioli, Spencer Y Kwon, and Andrei Shleifer, "Memory and probability," *The Quarterly Journal of Economics*, 2023, *138* (1), 265–311.

Camerer, Colin and Dan Lovallo, "Overconfidence and excess entry: An experimental approach," *American economic review*, 1999, *89* (1), 306–318.

Card, David, Alexandre Mas, Enrico Moretti, and Emmanuel Saez, "Inequality at work: The effect of peer salaries on job satisfaction," *American Economic Review*, 2012, *102* (6), 2981–3003.

Carlson, Ryan W, Michel André Maréchal, Bastiaan Oud, Ernst Fehr, and Molly J Crockett, "Motivated misremembering of selfish decisions," *Nature communications*, 2020, *11* (1), 1–11.

Chammat, Mariam, Imen El Karoui, Sébastien Allali, Joshua Hagège, Katia Lehongre, Dominique Hasboun, Michel Baulac, Stéphane Epelbaum, Agnès Michon, Bruno Dubois et al., "Cognitive dissonance resolution depends on episodic memory," *Scientific reports*, 2017, *7* (1), 1–10.

Charness, Gary, Uri Gneezy, and Vlastimil Rasocha, "Experimental methods: Eliciting beliefs," *Journal of Economic Behavior & Organization*, 2021, *189*, 234–256.

Chen, M Keith, Peter E Rossi, Judith A Chevalier, and Emily Oehlsen, "The value of flexible work: Evidence from Uber drivers," *Journal of political economy*, 2019, *127* (6), 2735–2794.

Collins, Brett, Andrew Garin, Emilie Jackson, Dmitri Koustas, and Mark Payne, "Is gig work replacing traditional employment? Evidence from two decades of tax returns," *Unpublished paper, IRS SOI Joint Statistical Research Program*, 2019.

Conlon, John J, "Memory Rehearsal and Belief Biases," Technical Report, Working Paper 2024.

, Laura Pilossoph, Matthew Wiswall, and Basit Zafar, "Labor market search with imperfect information and learning," Technical Report, National Bureau of Economic Research 2018.

Cook, Cody, Rebecca Diamond, Jonathan V Hall, John A List, and Paul Oyer, "The gender earnings gap in the gig economy: Evidence from over a million rideshare drivers," *The Review of Economic Studies*, 2021, *88* (5), 2210–2238.

Cortés, Patricia, Jessica Pan, Ernesto Reuben, Laura Pilossoph, and Basit Zafar, "Gender Differences in Job Search and the Earnings Gap: Evidence from the Field and Lab," 2022.

Crawford, Vincent P and Juanjuan Meng, "New York City cab drivers' labor supply revisited: Reference-dependent preferences with rational-expectations targets for hours and income," *American Economic Review*, 2011, *101* (5), 1912–32.

Cullen, Zoë and Ricardo Perez-Truglia, "How much does your boss make? the effects of salary comparisons," *Journal of Political Economy*, 2022, *130* (3), 766–822.

Danz, David, Lise Vesterlund, and Alistair J Wilson, "Belief elicitation: Limiting truth telling with information on incentives," Technical Report, National Bureau of Economic Research 2020.

DoorDash, "DoorDash ESG Report 2021," Technical Report 2021.

Dutz, Deniz, Ingrid Huitfeldt, Santiago Lacouture, Magne Mogstad, Alexander Torgovitsky, and Winnie Van Dijk, "Selection in surveys: Using randomized incentives to detect and account for nonresponse bias," Technical Report, National Bureau of Economic Research 2021.

, Michael Greenstone, Ali Hortaçsu, Santiago Lacouture, Magne Mogstad, Danae Roumis, Azeem M Shaikh, Alexander Torgovitsky, and Winnie van Dijk, "Selection bias in voluntary random testing: Evidence from a COVID-19 antibody study," in "AEA Papers and Proceedings," Vol. 113 2023, pp. 562–566.

Eil, David and Justin M Rao, "The good news-bad news effect: asymmetric processing of objective information about yourself," *American Economic Journal: Microeconomics*, 2011, *3* (2), 114–38.

Enke, Benjamin, Frederik Schwerter, and Florian Zimmermann, "Associative memory and belief formation," Technical Report, National Bureau of Economic Research 2020.

Farber, Henry S, "Why you can't find a taxi in the rain and other labor supply lessons from cab drivers," *The Quarterly Journal of Economics*, 2015, *130* (4), 1975–2026.

Fehr, Ernst and Lorenz Goette, "Do workers work more if wages are high? Evidence from a randomized field experiment," *American Economic Review*, 2007, *97* (1), 298–317.

Gödker, Katrin, Peiran Jiao, and Paul Smeets, "Investor Memory," *SSRN Electronic Journal*, 2022.

Gottlieb, Daniel, "Will you never learn? self deception and biases in information processing," *Unpublished Manuscript, Princeton University*, 2010.

Graeber, Thomas, Christopher Roth, and Florian Zimmermann, "Stories, statistics, and memory," *The Quarterly Journal of Economics*, 2024, p. qjae020.

Grubb, Michael D and Matthew Osborne, "Cellular service demand: Biased beliefs, learning, and bill shock," *American Economic Review*, 2015, *105* (1), 234–71.

Hall, Jonathan V, John J Horton, and Daniel T Knoepfle, "Pricing in designed markets: The case of ride-sharing," Technical Report, Tech. rep., Working paper, Massachusetts Institute of Technology 2021.

Hoffman, Mitchell and Stephen V Burks, "Worker overconfidence: Field evidence and implications for employee turnover and firm profits," *Quantitative Economics*, 2020, *11* (1), 315–348.

Huffman, David, Collin Raymond, and Julia Shvets, "Persistent overconfidence and biased memory: Evidence from managers," *American Economic Review*, 2022, *112* (10), 3141–75.

Jäger, Simon, Christopher Roth, Nina Roussille, and Benjamin Schoefer, "Worker beliefs about outside options," *The Quarterly Journal of Economics*, 2024, p. qjae001.

Kahana, Michael J., *Foundations of Human Memory*, New York: Oxford University Press, 2012.

Katz, Lawrence F and Alan B Krueger, "The rise and nature of alternative work arrangements in the United States, 1995–2015," *ILR review*, 2019, *72* (2), 382–416.

Köszegi, Botond, "Ego utility, overconfidence, and task choice," *Journal of the European Economic Association*, 2006, *4* (4), 673–707.

Koustas, Dmitri, "Consumption insurance and multiple jobs: Evidence from rideshare drivers," *Unpublished working paper*, 2018.

Larkin, Ian and Stephen Leider, "Incentive schemes, sorting, and behavioral biases of employees: Experimental evidence," *American Economic Journal: Microeconomics*, 2012, *4* (2), 184–214.

Malmendier, Ulrike and Geoffrey Tate, "Behavioral CEOs: The role of managerial overconfidence," *Journal of Economic Perspectives*, 2015, *29* (4), 37–60.

Mas, Alexandre and Amanda Pallais, "Valuing alternative work arrangements," *American Economic Review*, 2017, *107* (12), 3722–59.

Mischel, Walter, Ebbe B Ebbesen, and Antonette M Zeiss, "Determinants of selective memory about the self.," *Journal of consulting and clinical Psychology*, 1976, *44* (1), 92.

Möbius, Markus M, Muriel Niederle, Paul Niehaus, and Tanya S Rosenblat, "Managing self-confidence: Theory and experimental evidence," *Management Science*, 2022.

Moore, Don A and Paul J Healy, "The trouble with overconfidence.," *Psychological review*, 2008, *115* (2), 502.

Moore, Jeffrey C, Linda L Stinson, and Edward J Welniak, "Income measurement error in surveys: A review," *Journal of Official Statistics-Stockholm-*, 2000, *16* (4), 331–362.

Mueller, Andreas I, Johannes Spinnewijn, and Giorgio Topa, "Job seekers' perceptions and employment prospects: Heterogeneity, duration dependence, and bias," *American Economic Review*, 2021, *111* (1), 324–63.

Mullainathan, Sendhil, "A memory-based model of bounded rationality," *The Quarterly Journal of Economics*, 2002, *117* (3), 735–774.

Oyer, Paul and Scott Schaefer, "Why do some firms give stock options to all employees?: An empirical examination of alternative theories," *Journal of financial Economics*, 2005, *76* (1), 99–133.

Pew Research Center, "The state of gig work in 2021," Technical Report 2021.

Puri, Manju and David T Robinson, "Optimism and economic choice," *Journal of financial economics*, 2007, *86* (1), 71–99.

Reich, Michael and James A Parrott, "A Minimum Compensation Standard for Seattle TNC Drivers," 2020.

Rothbaum, Jonathan L, "Comparing Income Aggregates: How do the CPS and ACS Match the National Income and Product Accounts, 2007-2012," *US Census Bureau, SEHSD Working Paper*, 2015, *1*.

Roy-Chowdhury, Vivek, "Self-confidence and motivated memory loss: Evidence from schools," *Available at SSRN 4986405*, 2024.

Saucet, Charlotte and Marie Claire Villeval, "Motivated memory in dictator games," *Games and Economic Behavior*, 2019, *117*, 250–275.

Schacter, Daniel L, Donna Rose Addis, and Randy L Buckner, "Episodic simulation of future events: Concepts, data, and applications," *Annals of the New York Academy of Sciences*, 2008, *1124* (1), 39–60.

Sharot, Tali, "The optimism bias," *Current biology*, 2011, *21* (23), R941–R945.

Sial, Afras, Justin Sydnor, and Dmitry Taubinsky, "Biased Memory and Perceptions of Self-Control," *Working paper*, 2024.

Tella, Rafael Di, Ricardo Perez-Truglia, Andres Babino, and Mariano Sigman, "Conveniently upset: Avoiding altruism by distorting beliefs about others' altruism," *American Economic Review*, 2015, *105* (11), 3416–42.

Thakral, Neil and Linh T. Tô, "Daily Labor Supply and Adaptive Reference Points," *American Economic Review*, August 2021, *111* (8), 2417–2443.

Zimmermann, Florian, "The dynamics of motivated beliefs," *American Economic Review*, 2020, *110* (2), 337–61.

Tables

Table 1: Summary statistics

Baseline Survey	**Full Sample**	
	Mean	Std. Dev.
Panel (A): Demographics (%)		
Age 18-34	42.07	(49.42)
Age 35-54	47.14	(49.97)
White	72.69	(44.61)
Male	41.19	(49.27)
College Degree	38.55	(48.72)
Panel (B): Financial Situation (%)		
Household Income $0-$40k	54.19	(49.88)
No Household Budget	21.59	(41.19)
Struggling Financially	42.73	(49.52)
Gig Pay is Essential	83.48	(37.18)
Panel (C): Labor Market (%)		
Has Other Gig Job	35.68	(47.96)
Has Non-Gig Job	17.18	(37.76)
Employed Full-Time Prior to Gig	36.34	(48.15)
Employed Part-Time Prior to Gig	20.26	(40.24)
Unemployed Prior to Gig	10.13	(30.21)
Experience Delivery (12+ mo.)	57.49	(49.49)
Experience Rideshare (12+ mo.)	23.57	(42.49)
Panel (D): Actual Outcomes		
Weekly Pay	284.2	(250.4)
Weekly Hours	17.41	(14.81)
Hourly Pay	17.94	(7.515)
Expected Expenses Share	32.19	(5.188)
Panel (E): Recall Outcomes		
Weekly Pay	372.8	(265.8)
Weekly Hours	22.69	(13.63)
Hourly Pay	18.85	(6.251)
Panel (F): Forecast Outcomes		
Weekly Pay	376.8	(324.9)
Weekly Hours	23.24	(14.05)
Hourly Pay	19.36	(6.798)

Notes: Sample of delivery and ride share gig workers in the United States from our baseline survey ($N = 454$). The mean and standard deviation are shown for each variable. Variables in Panels (A)-(C) are measured in percentage units. *Struggling Financially* is defined as receiving calls from collectors, contemplating bankruptcy, or struggling to pay the bills. Panel (D) shows actual job outcomes, collected from screenshots of the gig economy apps that workers submit. *Expenses Share* is an estimation of expected expenses by car category and car age group. Panel (E) presents pooled recalls of job outcomes for the previous week and month. Panel (F) shows information on forecasts about the first week (starting on a Monday) after the baseline survey.

Table 2: Forecast and recall overestimation of main job outcomes

	N	Mean of Actual	Overestimation (Belief - Actual) Mean	Mean (%)	Share
Panel (A): Forecast					
Weekly Pay	155	$260	$113.7***	43.7%	.72***
Weekly Hours	142	13.5	8.4***	62.3%	.8***
Hourly Pay	125	$20.2	$.5	2.4%	.54
Panel (B): Recall					
Weekly Pay	434	$284.2	$88.9***	31.3%	.73***
Net Weekly Pay	408	$198.8	$92.1***	46.3%	.74***
Expenses Share	396	32.2p.p.	-7.1p.p***	-22%	.29***
Weekly Hours	392	17.4	5.8***	33.1%	.75***
Hourly Pay	386	$17.9	$.7*	3.9%	.53
Net Hourly Pay	338	$12.4	$2.6***	20.7%	.68***

Notes: Panel (A) shows errors in forecasting job outcomes for the first full week (starting on a Monday) after the baseline survey. Panel (B) shows pooled errors in recalling the week and the month before the baseline survey. We include, for each variable, the number of observations used for calculating overestimation. This number varies across variables because the same trimming rule can result in slightly different sample sizes due to ties, and because some submitted screenshots contain incomplete information. In particular, some screenshots contain earnings information but not hours worked. Moreover, we do not estimate expenses for certain categories of drivers, such as those who use bikes or scooters. *Mean of Actual* is the mean of the actual job outcome. Overestimation is defined as the recall or forecast belief minus the actual job outcome. *Mean* is the mean overestimation (including negative values) for each outcome. *Mean (%)* overestimation is the ratio of the mean overestimation and the mean actual job outcome in our sample. *Share* is the share of workers for whom overestimation is positive. We test whether the average overestimation is equal to 0 and the share overestimating is equal to 50%. Stars are used to denote the statistical significance of these tests (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). *Expenses Share* recall is defined as the ratio of recalls for weekly costs and weekly pay. Actual *Expenses Share*, *Net Weekly Pay* and *Net Weekly Pay* use an estimation of expected expenses by car category and car age group.

Table 3: Relationship of gig work experience with overestimation of job outcomes

	Overestimation (Belief - Actual)	
	(1)	(2)
	Weekly Pay	Net Weekly Pay
Inexperienced (Less than 6 Months)	40.2*	64.3***
	(22.0)	(22.2)
Some Experience (Between 6 and 12 Months)	87.2***	93.6***
	(8.57)	(8.90)
Experienced (Over 12 Months)	132.0***	104.7***
	(18.7)	(18.4)
Observations	439	400
p-value(Some Experience = Inexperienced)	0.047	0.22
p-value(Experienced = Inexperienced)	0.0016	0.16

Notes: We regress the overestimation of weekly pay on binary variables for experience in gig work. The regressions do not have a constant term. Overestimation of each outcome is defined as the recall belief minus the actual job outcome for the same time period. We pool recalls of the week and the month before the baseline survey. *Inexperienced* is equal to 1 if a gig worker has less than six months of experience in both delivery and ride share. *Experienced* is equal to 1 if a gig worker has more than one year of experience in both delivery and ride share. *Some Experience* is equal to 1 if both *Inexperienced* and *Experienced* are equal to 0. Robust standard errors in parentheses. Stars are used to denote the statistical significance of each coefficient (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). Under each regression,we present the p-value for a test of whether (i) the coefficients on *Experienced* and *Inexperienced* are equal, and (ii) the coefficients on *Some Experience* and *Inexperienced* are equal.

Table 4: Tests of positive selective recall and asymmetric updating

	Recall Belief Weekly Pay			
	(1)	(2)	(3)	(4)
Last 4 Weeks (Actual)				
Maximum	0.57***	0.55***	0.24**	0.22*
	(0.058)	(0.059)	(0.12)	(0.12)
Minimum	0.26***	0.26***		
	(0.086)	(0.087)		
Mean			0.60***	0.59***
			(0.14)	(0.14)
Observations	320	320	320	320
Demographic Controls		✓		✓
p-value(Max=Min)	0.026	0.039		

(A) Selective Recall

	Belief$_t$ Weekly Pay	
	(1)	(2)
Belief$_{t-1}$	0.69***	0.67***
	(0.049)	(0.053)
$1\{\text{Actual}_t \geq \text{Belief}_{t-1}\}$	208.3***	166.5**
	(33.6)	(69.6)
$1\{\text{Actual}_t < \text{Belief}_{t-1}\}$	-47.5*	-81.8
	(25.9)	(68.6)
Observations	155	155
Demographic Controls		✓
p-value(Above=Below)	0.0016	0.53

(B) Belief Updating

Notes: Robust standard errors in parentheses. Stars are used to denote the statistical significance of each coefficient (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). Our set of demographic controls consists of binary variables for male, white, age between 18-34, age between 35-54, at least a college degree and household income below $40,000/year. Outcome variables in Panel (A) are recalls referring to the month before the baseline survey. Panel (A) shows regressions of recalls of weekly pay on functions of the 4 previous weeks of actual weekly pay. We define the minimum, the maximum and the mean for this set of four weeks. We present the p-value for a test of whether the coefficients for the maximum and the minimum variables are the same. In Panel (B), only participants that replied to our midline survey are included. $Belief_t$ is the recall of weekly pay of the first full week (starting on a Monday) after the baseline survey. $Belief_{t-1}$ is the pooled recall for each job outcome for the week or the month before the baseline survey. $Actual_t$ refers to the actual job outcome (obtained from a screenshot submitted from the gig platform app) of the first full week after the baseline survey. Each column shows the result of the regression of $Belief_t$ on $Belief_{t-1}$ and two binary variables: a variable equal to 1 if $Actual_t$ is above or equal to $Belief_{t-1}$, and a variable equal to 1 if $Actual_t$ is below $Belief_{t-1}$. There is no constant term in the regressions in Panel (B). We present the p-value for a test of whether the coefficients for the two indicator variables are the same.

Table 5: Relationship between variance of pay and overestimation

	Overestimation (Belief - Actual) Weekly Pay	
	(1)	(2)
Last 4 Weeks (Actual)		
Coefficient of Variation (CV)	0.85***	0.94***
	(0.30)	(0.31)
Constant	62.3***	60.3
	(17.4)	(45.7)
Observations	232	232
Demographic Controls		✓
Average CV (SD/Mean)	0.48	0.48

Notes: Robust standard errors in parentheses. Stars are used to denote the statistical significance of each coefficient (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). Overestimation of a job outcome is defined as the recall belief minus the actual job outcomes for the same time period. We show regressions of the overestimation of weekly pay on the coefficient of variation for the 4 previous weeks of weekly pay. The coefficient of variation is the standard deviation over the mean. This variable is normalized so that a 1 unit increase is equal to an increase of 0.01. We present the average coefficient of variation underneath each column. Our set of demographic controls consists of binary variables for male, white, age between 18-34, age between 35-54, at least a college degree and household income below $40,000/year.

Table 6: Relationship between net hourly pay and reservation wage

	Net Hourly Pay		
	Belief	Actual	Difference (SE)
	(1)	(2)	(2) - (1)
Panel (A): Share with Pay below Reservation Wage			
Full Sample	23%	68%	45% (2.6)
Alternative Calculations:			
Outside Option Not Gig Work	22%	57%	35% (4.6)
Overestimated Outside Option	23%	55%	32% (2.5)
Maximum Actual Net Hourly Pay	23%	54%	31% (2.4)
Wage at Other or Previous Jobs	61%	78%	17% (3.3)
Panel (B): Share with Pay $5 or More below Reservation Wage			
Full Sample	6%	29%	22% (2.2)
Alternative Calculations:			
Outside Option Not Gig Work	4%	22%	18% (3.6)
Overestimated Reservation Wage	6%	21%	15% (1.9)
Maximum Actual Net Hourly Pay	6%	22%	16% (1.9)
Wage at Other or Previous Jobs	32%	51%	20% (3.6)

Notes: Belief of net hourly pay is the pooled net hourly pay recall of the week and the month before the baseline survey. The actual net hourly pay refers to the same period. Our reservation wage proxy is the answer to the following question: "What is the lowest acceptable hourly pay after taxes and expenses that would accept to keep working for [gig company]?". In the two panels, we calculate the share of workers for whom the reservation wage proxy is above the net hourly pay by either 0 or 5 dollars. The final column shows the difference between these shares when the belief and actual net hourly pay are used. The first row includes our full sample and is our base measure. For *Outside Option Not Gig Work*, we include only workers that did not do gigs before their current gig job and that do not work for other gig companies. For *Overestimated Reservation Wage*, we assume the reservation wage is measured with half as much error as the net hourly pay. For *Wage at Other or Previous Jobs*, we use the worker's self-reported gross hourly wages in either a previous job or in another current job as the reservation wage proxy. This measure is elicited in 5-dollar bins, for which we take the midpoint. For *Maximum Actual Net Hourly Pay*, we assume the actual net hourly pay is the maximum weekly average of the net hourly pay in the past 4 weeks.

Table 7: Relationship between pay overestimation and hours worked

	Dependent Variable: **Weekly Hours Worked**					
	Hourly Pay		Net Hourly Pay		Expenses Pct.	
	(1)	(2)	(3)	(4)	(5)	(6)
Overestimation (Belief - Actual)	0.35***	0.37***	0.39***	0.39***	0.062*	0.052
	(0.097)	(0.095)	(0.11)	(0.11)	(0.034)	(0.034)
Observations	387	387	341	341	345	345
Demographic Controls		✓		✓		✓

(A) Base Specification

	Dependent Variable: **Weekly Hours Worked**					
	Hourly Pay		Net Hourly Pay		Expenses Pct.	
	(1)	(2)	(3)	(4)	(5)	(6)
Overestimation (Belief - Actual)	0.19	0.23**	0.14	0.18	0.068*	0.060*
	(0.12)	(0.12)	(0.14)	(0.13)	(0.037)	(0.036)
Observations	239	239	204	204	228	228
Demographic Controls		✓		✓		✓

(B) Future Hours Worked

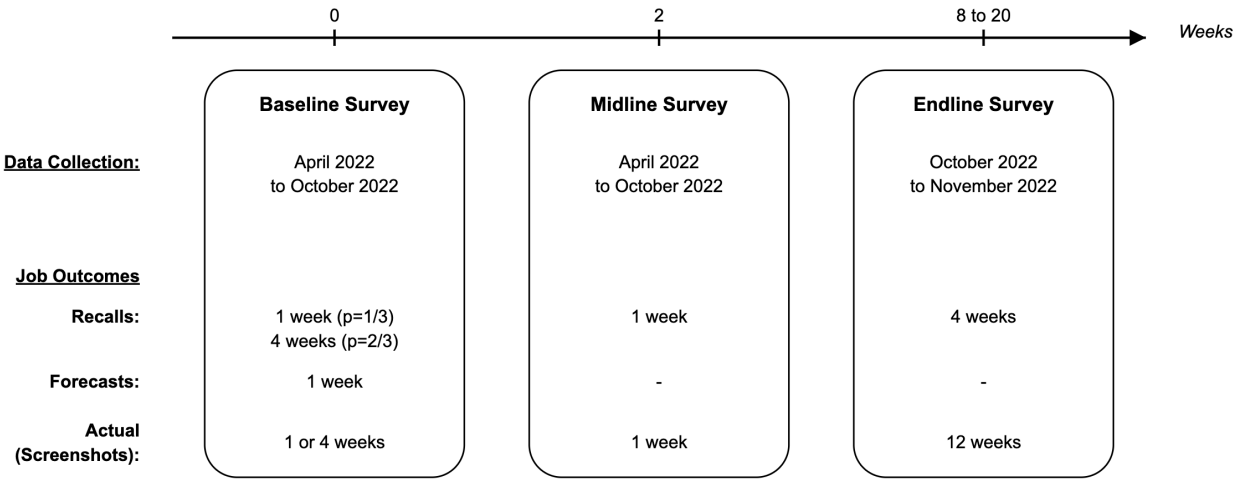
	Dependent Variable: **Weekly Hours Worked**					
	Hourly Pay		Net Hourly Pay		Expenses Pct.	
	(1)	(2)	(3)	(4)	(5)	(6)
Overestimation (Belief - Actual)	0.22	0.26**	0.27*	0.33**	0.049	0.044
	(0.13)	(0.13)	(0.14)	(0.13)	(0.041)	(0.040)
Observations	217	217	190	190	206	206
Demographic Controls		✓		✓		✓

(C) Hours Worked in Different Weeks

Notes: Robust standard errors in parentheses. Stars are used to denote the statistical significance of each coefficient (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). We regress measures of weekly hours worked on overestimation of job outcomes, indicated above the model numbers, and a constant. *Overestimation* is the recall belief minus the actual job outcome. Thus, a positive (negative) value implies overestimation (underestimation) of a job outcome. The exception is the variable *Expenses Pct.*, which measures the underestimation, in percentage points, of expected expenses. Panel (A) uses the average hours worked before the baseline survey, for either one or four weeks, pooling recall overestimation from the baseline survey for past one and four weeks. Panel (B) uses the same recall overestimation measure but uses actual hours worked in the week following the baseline survey (obtained from the midline survey), or when missing, average hours worked from the endline survey data. Panel (C) uses hours worked in the week following the baseline survey (obtained from the midline survey), or when missing, hours worked from weeks two to four before the baseline survey for cases where recall overestimation is measured for the previous working week.

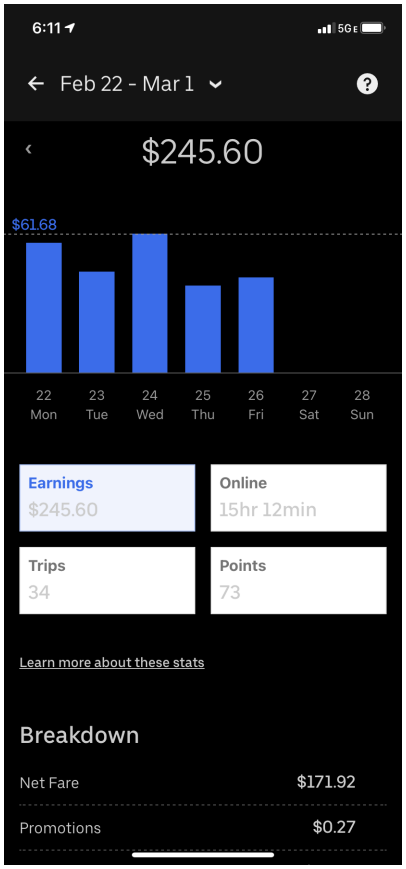
Figures

Figure 1: Survey timeline

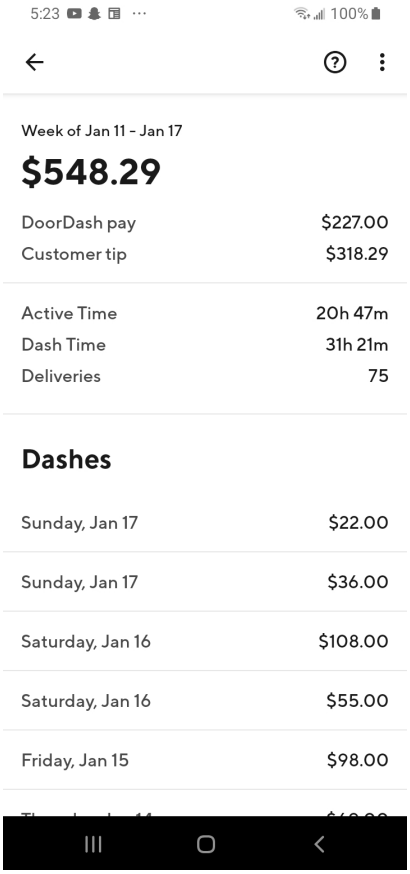


Notes: We survey gig workers in three occasions: the baseline, midline and endline surveys. For two-thirds of the sample, we elicit a one month recall in the baseline survey, while for another one-third, we ask about the previous week. Beliefs refer to full weeks, from Monday to Sunday. We elicit a one week recall in the midline survey and a four weeks recall in the endline survey. We obtain information of actual job outcomes based on screenshots from gig platform apps. In the baseline survey, this information refers to one or four weeks before the survey. In the midline survey, it refers to the week following the baseline survey. In the endline survey, this information is on up to twelve weeks before that survey.

Figure 2: Examples of valid screenshots from gig platform apps



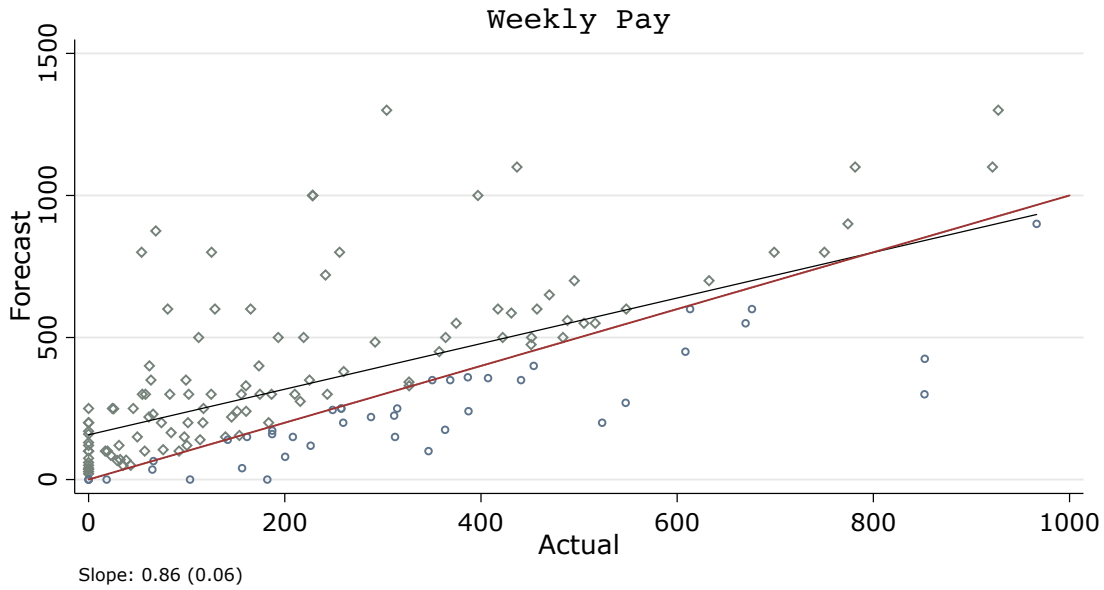
(A) Uber/Uber Eats



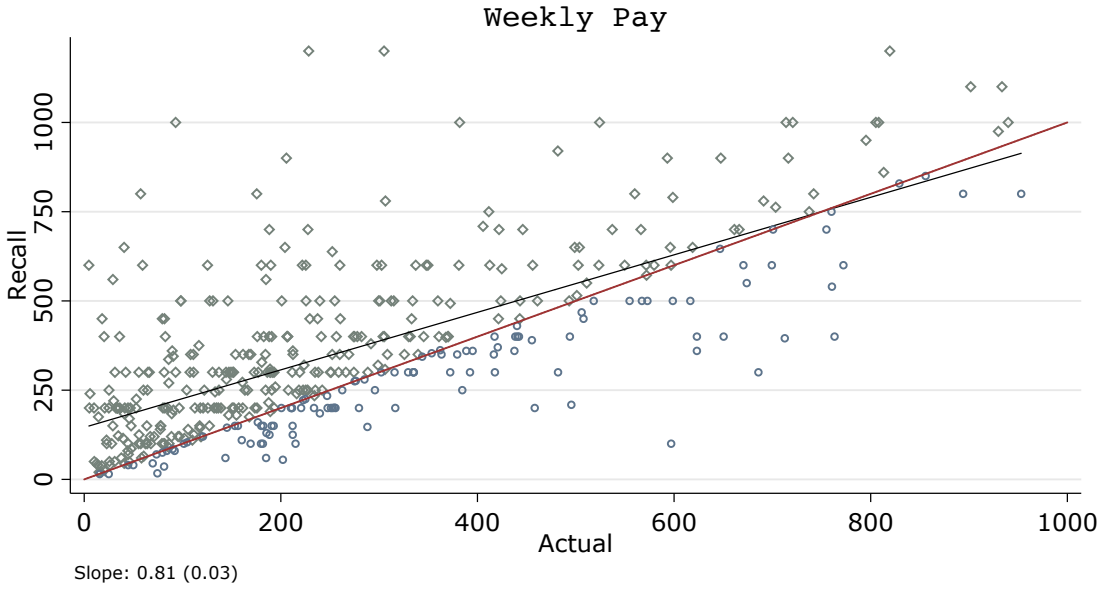
(B) Doordash

Notes: We show examples of valid screenshots from the weekly earnings page in gig platform apps. Panel (A) is an example of an Uber or Uber Eats screenshot, while Panel (B) is an example of a DoorDash screenshot. In each screenshot, among other things, we have information on the gross weekly pay on top (including tips, bonuses and platform fees), one or two measures of work hours and information on the week that the screenshot refers to.

Figure 3: Scatter plots relating beliefs to actual weekly pay



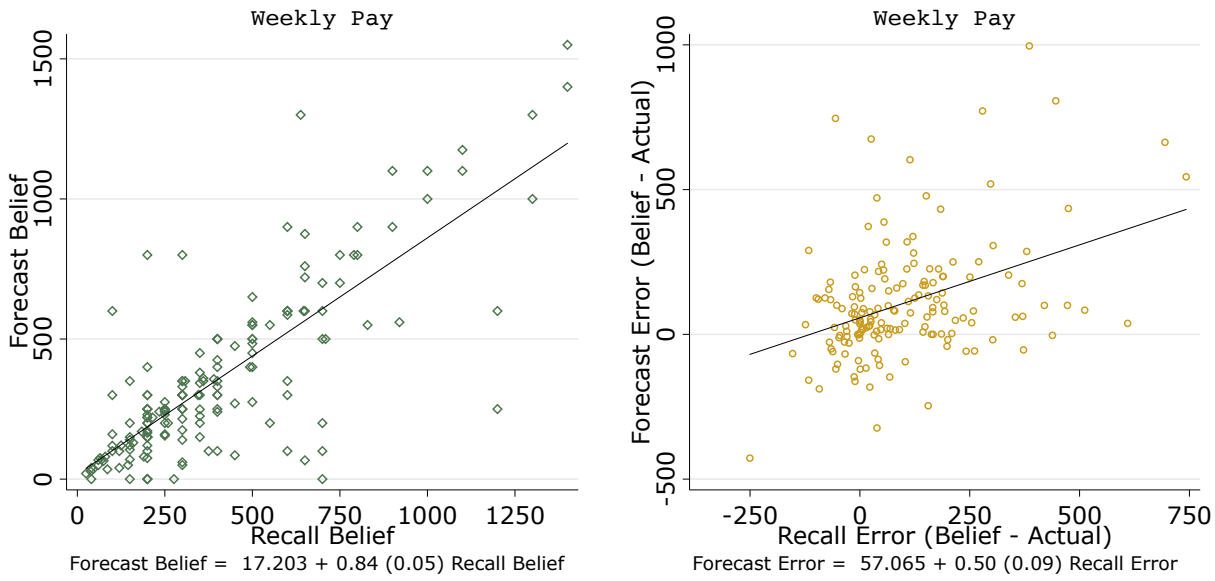
(A) Forecast



(B) Recall

Notes: In each graph, we draw a 45 degree line and the estimated slope of the regression of the belief on the actual weekly pay. Underneath each graph, we present the slope from the regression of the recall or forecast belief on the actual outcome, with the associated standard error in parentheses. We exclude outliers from each graph. Points above the 45 degree line indicate overestimation, whereas points below the 45 degree line indicate underestimation of weekly pay. Panel (A) shows forecasts for the first full week (starting on a Monday) after the baseline survey. In Panel (B), we pool recalls of the week and the month before the baseline survey. The actual weekly pay refers to the same time period as either the forecast or the recall and are obtained using information from screenshots from gig platform apps.

Figure 4: Relationship between forecasts and recalls of weekly pay

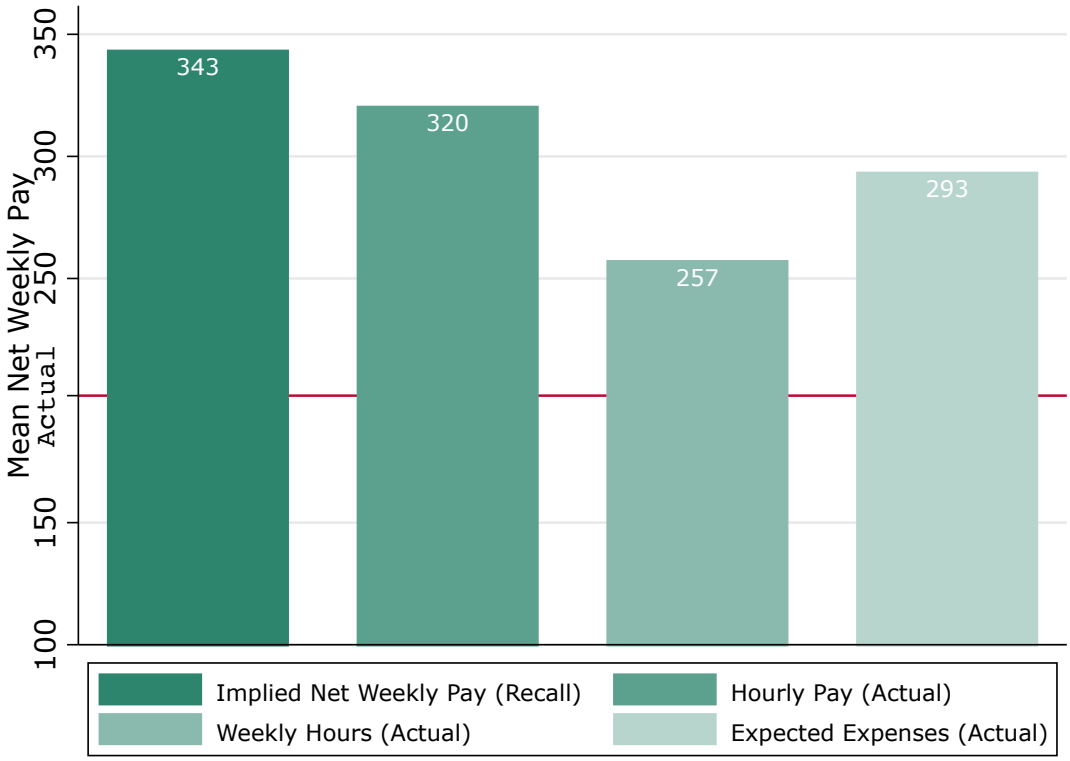


(A) Beliefs

(B) Errors (Belief - Actual)

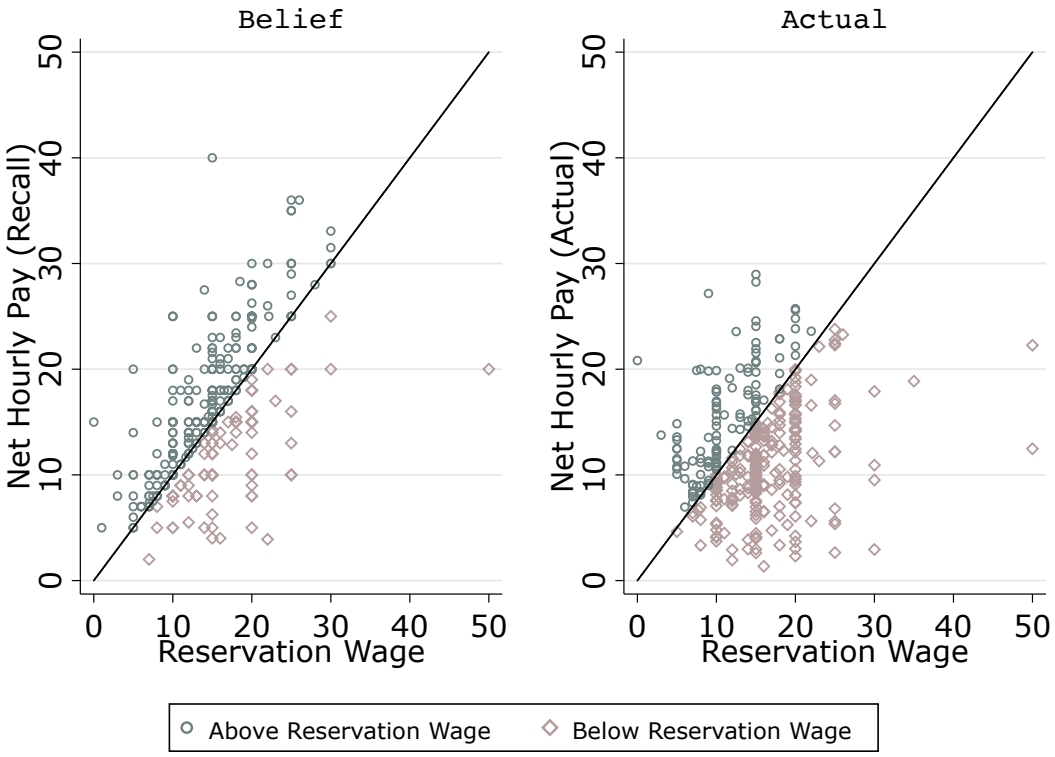
Notes: In Panel (A), we draw the fitted line of the regression relating the forecast belief and the recall belief of weekly pay. In Panel (B), we show the fitted line of the regression relating the forecast overestimation and the recall overestimation of weekly pay. The estimated equation is shown underneath each plot. Overestimation is measured as the recall or forecast minus the actual job outcome. We exclude outliers from each plot. We pool recalls of the week and the month before the baseline survey. The forecast refers to the first full week (starting on a Monday) following the baseline survey.

Figure 5: Decomposing factors that explain errors in recalling net weekly pay



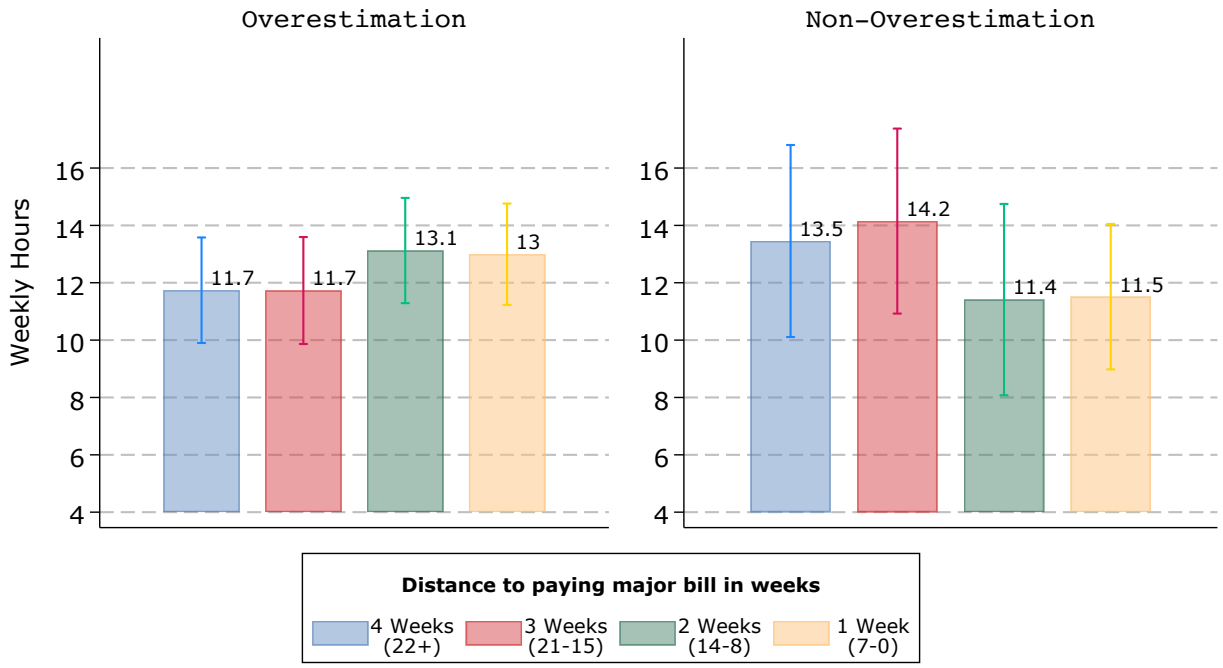
Notes: We plot the average of different measures of net weekly pay. *Implied Net Weekly Pay (Recall)* is the product of the recall of three variables: gross hourly pay, weekly hours and one minus the expenses share. We pool recalls of the week and the month before the baseline survey. The next three variables replace one element only of the implied net weekly pay recall with its correct equivalent. *Hourly Pay (Actual)* is found by replacing the recall for the actual gross hourly pay, while keeping the recalls for weekly hours and expenses. *Weekly Hours (Actual)* and *Expected Expenses (Actual)* are constructed in an analogous manner. Actual job outcomes refer to the same time period as the recalls and are obtained using both information from screenshots from gig platform apps and a calculation of expected expenses by car category and car age group. We plot the average actual expected net weekly pay as a red line on the y-axis.

Figure 6: Relationship between belief and actual net hourly pay with reservation wages

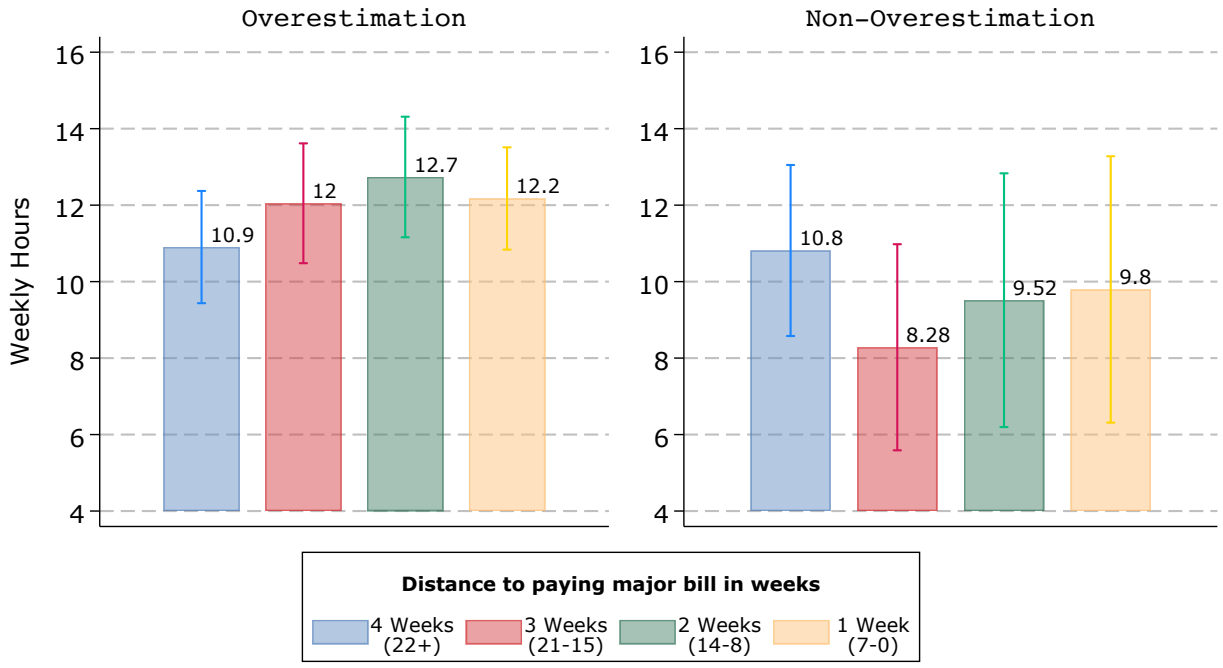


Notes: In each graph, we relate a measure of net hourly pay with workers' stated reservation wage. The reservation wage is the answer to the question: "What is the lowest hourly pay after taxes and expenses that would accept and keep working for [gig company]?" Each chart includes a 45 degree line. Points above this line represent workers for whom the net hourly pay measure is above the reservation wage, with the opposite being true for points below this line. Plotted on the y-axis on the left chart is a recall of net hourly pay, which pools the week and the month before the baseline survey. Plotted on the y-axis on the right chart is the actual expected net hourly pay, referring to the same time period as the recalls and obtained using a calculation of expected expenses and information from screenshots from gig platform apps.

Figure 7: Hours worked depending on distance to bills' due date and overestimation measures

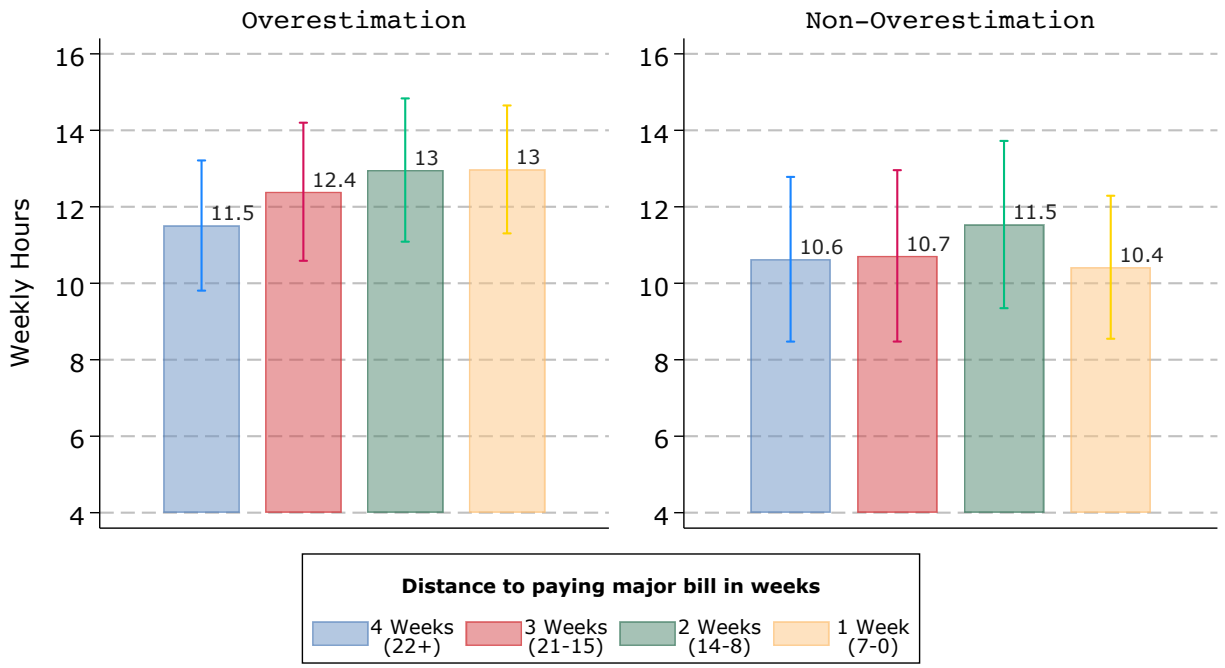


(A) Prediction of Weekly Pay



(B) Recall of Net Weekly Pay

Figure 7: Hours worked depending on distance to bills' due date and overestimation measures (cont.)



(C) Recall of Net Hourly Pay

Notes: We use information on workers' primary monthly bill payment date (Figure A8). We construct a worker-week panel and measure the distance between each week's end (Sunday) and the worker's next primary bill due date. We divide the household budgeting cycle into four periods based on this metric: weeks falling more than 21 days before the due date (4 Weeks), 15 to 21 days before (3 Weeks), 8 to 14 days before (2 Weeks), and within 7 days of the due date (1 Week). Actual work hours data is derived from screenshots from the gig platform app and include information from all surveys. In each panel, the sample is split into workers who overestimate a particular measure (measure is strictly positive) versus those who do not (measure is zero or negative). Panel (A) refers to overestimation of predicted gross weekly pay, Panel (B) refers to overestimation of recalled net weekly pay and Panel (C) refers to overestimation of recalled net hourly pay.

Appendix A: Additional Tables and Figures

Table A1: Selective attrition on midline survey

Answered Midline survey?		**No**			**Yes**		
	N	Mean	SD	N	Mean	SD	**Diff.**
Age 18-34	244	0.46	0.50	210	0.38	0.49	-0.083*
Age 35-54	244	0.43	0.50	210	0.52	0.50	0.098**
White	244	0.75	0.44	210	0.70	0.46	-0.041
Male	244	0.43	0.50	210	0.39	0.49	-0.040
College Degree	244	0.36	0.48	210	0.41	0.49	0.054
HHold Income 0−40k	244	0.59	0.49	210	0.49	0.50	-0.104**
No Household Budget	244	0.24	0.43	210	0.19	0.39	-0.056
Struggling Financially	244	0.47	0.50	210	0.38	0.49	-0.086*
Experience Delivery (12+ mo.)	244	0.52	0.50	210	0.63	0.48	0.109**
Experience Rideshare (12+ mo.)	244	0.21	0.41	210	0.26	0.44	0.049
Gig Pay is Essential	244	0.84	0.37	210	0.83	0.37	-0.003
Employed Full-Time Prior to Gig	244	0.38	0.49	210	0.35	0.48	-0.029
Employed Part-Time Prior to Gig	244	0.18	0.39	210	0.23	0.42	0.048
Unemployed Prior to Gig	244	0.10	0.30	210	0.10	0.30	-0.002
Has Other Gig Job	244	0.36	0.48	210	0.35	0.48	-0.008
Has Non-Gig Job	244	0.17	0.37	210	0.18	0.38	0.008
Hourly Pay	210	17.57	7.40	190	18.33	7.64	0.761
Hourly Net Pay	193	12.19	5.32	167	12.61	5.20	0.424
Weekly Hours	213	17.87	16.19	191	16.90	13.12	-0.975
Weekly Pay	240	271.04	227.91	205	299.60	274.24	28.559
Hourly Pay (Recall)	236	18.61	6.34	200	19.14	6.14	0.523
Net Hourly Pay (Recall)	219	15.48	6.04	193	15.43	6.61	-0.044
Weekly Pay (Recall)	236	362.15	240.15	203	385.23	292.92	23.074
Weekly Hours (Recall)	238	22.58	14.00	202	22.82	13.21	0.239

Notes: We present the number of observations, the mean and standard deviation of observable characteristics for the group of individuals who either did not (*No*) or did (*Yes*) reply to the midline survey. The last column shows the difference in means between the two groups for each variable. Stars are used to denote the statistical significance of this difference (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). The data in this table was collected in our baseline survey. *Struggling Financially* is defined as receiving calls from collectors, contemplating bankruptcy, or struggling to pay the bills. The four final rows present pooled recalls of job outcomes for the previous week and the previous month. The four rows before that show actual job outcomes, collected from screenshots of gig economy apps that workers submit.

Table A2: Selective attrition on endline survey

Answered Endline survey?		**No**			**Yes**		
	N	Mean	SD	N	Mean	SD	**Diff.**
Age 18-34	218	0.42	0.50	190	0.42	0.50	-0.001
Age 35-54	218	0.44	0.50	190	0.51	0.50	0.065
White	218	0.70	0.46	190	0.73	0.45	0.024
Male	218	0.39	0.49	190	0.43	0.50	0.046
College Degree	218	0.36	0.48	190	0.40	0.49	0.038
HHold Income 0−40k	218	0.57	0.50	190	0.48	0.50	-0.085*
No Household Budget	218	0.22	0.41	190	0.22	0.41	0.000
Struggling Financially	218	0.43	0.50	190	0.41	0.49	-0.021
Experience Delivery (12+ mo.)	218	0.56	0.50	190	0.61	0.49	0.055
Experience Rideshare (12+ mo.)	218	0.27	0.44	190	0.22	0.42	-0.045
Gig Pay is Essential	218	0.83	0.37	190	0.82	0.38	-0.014
Employed Full-Time Prior to Gig	218	0.33	0.47	190	0.41	0.49	0.080*
Employed Part-Time Prior to Gig	218	0.18	0.38	190	0.23	0.42	0.047
Unemployed Prior to Gig	218	0.11	0.31	190	0.09	0.29	-0.016
Has Other Gig Job	218	0.35	0.48	190	0.34	0.48	-0.011
Has Non-Gig Job	218	0.13	0.34	190	0.22	0.41	0.087**
Hourly Pay	194	17.48	7.87	168	18.47	7.12	0.987
Hourly Net Pay	175	12.30	5.55	151	12.47	4.84	0.175
Weekly Hours	192	18.27	16.87	173	16.62	12.81	-1.657
Weekly Pay	216	279.26	251.44	184	282.62	238.69	3.364
Hourly Pay (Recall)	207	18.87	6.46	184	19.06	6.20	0.183
Net Hourly Pay (Recall)	194	15.74	6.49	178	15.48	5.98	-0.251
Weekly Pay (Recall)	208	369.80	256.66	185	376.68	273.98	6.881
Weekly Hours (Recall)	208	22.63	14.29	187	22.71	13.13	0.081

Notes: We present the number of observations, the mean and standard deviation of observable characteristics for the group of individuals who either did not (*No*) or did (*Yes*) reply to the endline survey. The last column shows the difference in means between the two groups for each variable. Stars are used to denote the statistical significance of this difference (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). The data in this table was collected in our baseline survey. *Struggling Financially* is defined as receiving calls from collectors, contemplating bankruptcy, or struggling to pay the bills. The four final rows present pooled recalls of job outcomes for the previous week and the previous month. The four rows before that show actual job outcomes, collected from screenshots of gig economy apps that workers submit.

Table A3: Comparison of our sample with other studies and the United States gig market

	Mean of: Worked for Company		
	Sample (percent)	US (Millions)	US (Sample)
DoorDash	64.76	2.00	2.00
Postmates	5.29	0.10	0.15
Grubhub	16.96	0.45	0.53
Uber Eats	35.02	0.80	1.09
Instacart	32.16	0.60	1.03
Uber	16.08	1.50	0.50
Lyft	10.79	1.00	0.34

(A) Company distribution

	Summary Stats Across Studies			
	Sample	Pew (2021)	Parrott-Reich (2020)	Doordash (2021)
Hourly Pay	17.94		23.23	25.00
10+ Hours	0.61	0.37		0.10
20+ Hours	0.34		0.56	
Age 18-34	0.42		0.37	
Age 35-54	0.47		0.53	
White	0.73	0.51	0.45	0.62
Male	0.42	0.44	0.83	0.53
Has Another Job	0.50	0.69	0.59	0.54
College Degree	0.38	0.22		

(B) Sample comparison

Notes: In Panel (A), we show how the distribution of companies in our sample compares to the distribution of gig workers for each gig company in the United States. The data from our sample is the answer to the question: "For which gig companies did you work in the past 3 months?". We obtain information on contracting for gig companies in the United States from multiple sources referring to 2020 or 2021. We use market share information and our own calculations to reach these figures. In the third column of Panel (A), we normalize the values from the first column so that the number of workers working for DoorDash in our sample matches the number of DoorDash workers in the United States (2 million). Comparing these numbers with the second column says whether other companies in our sample are over or underrepresented relative to DoorDash. In Panel (B), we compare mean characteristics in our sample to previous studies of the gig economy. Parrott and Reich (2020) survey both Uber and Lyft drivers in Seattle, Doordash (2021) is a corporate DoorDash survey and Pew (2021) is a study from Pew Research Center of all gig work on online platforms in the United States. In Doordash (2021), hourly pay is based only on time spent actively on gigs. In addition, zero work hour weeks are included in their calculations. For all studies besides ours and Pew (2021), *White* excludes Hispanic white. We define binary variables for 10+ and 20+ hours of work that equal 1 if an individual works at least that amount on average. The statistics presented here for other studies are partially derived from the author's calculations.

Table A4: Correlation of overestimation in forecasts and recalls across job outcomes

	Overestimation (Forecast - Actual)		
	Weekly Pay	Weekly Hours	Hourly Pay
Weekly Pay	1.00		
Weekly Hours	0.65***	1.00	
Hourly Pay	0.04	-0.31***	1.00

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

(A) Forecast

	Overestimation (Recall - Actual)				
	Weekly Pay	Net Weekly Pay	Weekly Hours	Hourly Pay	Net Hourly Pay
Weekly Pay	1.00				
Net Weekly Pay	0.77***	1.00			
Weekly Hours	0.54***	0.42***	1.00		
Hourly Pay	0.29***	0.16***	-0.14**	1.00	
Net Hourly Pay	0.16***	0.13**	-0.18***	0.70***	1.00

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

(B) Recall

Notes: We present the correlation matrix of overestimation of forecasts – Panel (A) – or recalls – Panel (B) – of job outcomes. Overestimation may be negative and is measured as the recall or forecast belief minus the actual job outcome for the same time period. Recall variables pool recalls of the week and the month before the baseline survey. Forecast variables refer to the first full week (starting on a Monday) following the baseline survey. We use an estimate of expected expenses to calculate net weekly pay and net hourly pay. Stars are used to denote statistical significance (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

Table A5: Heterogeneity analysis for overestimation of recalls of job outcomes

	Overestimation (Belief - Actual)				
	(1)	(2)	(3)	(4)	(5)
	Weekly Pay	Net Weekly Pay	Weekly Hours	Hourly Pay	Net Hourly Pay
Certain of Beliefs	22.5	18.5	2.68*	0.85	1.59*
	(19.9)	(19.9)	(1.48)	(0.95)	(0.93)
Excluded Group	70.2***	76.8***	3.55***	0.0021	1.26
	(18.2)	(18.0)	(1.34)	(0.86)	(0.84)
Observations	439	400	397	393	345

(A) Certainty of beliefs

	Overestimation (Belief - Actual)				
	(1)	(2)	(3)	(4)	(5)
	Weekly Pay	Net Weekly Pay	Weekly Hours	Hourly Pay	Net Hourly Pay
Full-Time Driver	-2.32	33.1*	-11.8***	2.56**	2.26**
	(17.7)	(17.7)	(1.49)	(0.99)	(0.98)
Excluded Group	89.4***	84.3***	7.48***	0.30	2.22***
	(8.44)	(8.61)	(0.57)	(0.39)	(0.39)
Observations	439	400	397	393	345

(B) Full-time workers

	Overestimation (Belief - Actual)				
	(1)	(2)	(3)	(4)	(5)
	Weekly Pay	Net Weekly Pay	Weekly Hours	Hourly Pay	Net Hourly Pay
No Household Budget	30.7*	19.4	4.18***	0.22	-0.14
	(17.9)	(18.0)	(1.35)	(0.88)	(0.86)
Struggling Financially	9.00	-1.81	1.93*	-0.28	-1.06
	(15.2)	(15.5)	(1.15)	(0.75)	(0.74)
Pay is Essential	61.9***	54.9**	2.71*	0.27	1.19
	(20.8)	(21.2)	(1.57)	(1.03)	(1.02)
Excluded Group	25.9	42.1**	1.69	0.55	2.06**
	(19.5)	(19.9)	(1.48)	(0.97)	(0.96)
Observations	439	400	397	393	345

(C) Financial need

Table A5: Heterogeneity analysis for overestimation of recalls of job outcomes (cont.)

	Overestimation (Belief - Actual)				
	(1)	(2)	(3)	(4)	(5)
	Weekly Pay	Net Weekly Pay	Weekly Hours	Hourly Pay	Net Hourly Pay
Age 18-34	4.91	-26.1	-0.043	2.26*	0.33
	(27.4)	(27.6)	(2.09)	(1.35)	(1.31)
Age 35-54	24.3	21.6	2.40	1.86	0.18
	(26.8)	(26.9)	(2.04)	(1.32)	(1.29)
White	3.22	7.75	-0.62	-0.80	-0.69
	(17.4)	(17.5)	(1.32)	(0.86)	(0.86)
Male	12.3	25.4	1.06	-0.68	0.74
	(16.1)	(16.3)	(1.20)	(0.77)	(0.77)
College Degree	-10.5	-10.3	-0.74	0.25	0.42
	(15.4)	(15.7)	(1.18)	(0.75)	(0.76)
HHold Income $0-$40k	10.4	13.4	2.08*	0.055	0.54
	(15.5)	(15.8)	(1.18)	(0.76)	(0.76)
Excluded Group	66.2**	73.5**	3.76	-0.40	2.08
	(32.6)	(32.8)	(2.49)	(1.61)	(1.58)
Observations	439	400	397	393	345

(D) Demographics

	Overestimation (Belief - Actual)				
	(1)	(2)	(3)	(4)	(5)
	Weekly Pay	Net Weekly Pay	Weekly Hours	Hourly Pay	Net Hourly Pay
Was Employed Full-Time	48.6***	61.8***	3.26**	0.54	0.53
	(17.5)	(17.9)	(1.34)	(0.86)	(0.86)
Was Employed Part-Time	-6.72	-1.17	1.31	-1.50	-0.86
	(20.6)	(20.7)	(1.58)	(1.02)	(1.00)
Was Self-Employed	54.6*	61.5*	2.03	-2.06	-1.42
	(32.9)	(32.5)	(2.62)	(1.64)	(1.64)
Has Other Gig Job	8.03	12.2	1.14	0.82	0.61
	(15.7)	(16.0)	(1.21)	(0.77)	(0.77)
Has Non-Gig Job	-43.4**	-50.7**	-2.32	-0.50	-0.65
	(20.3)	(20.4)	(1.53)	(0.99)	(0.98)
Excluded Group	74.1***	70.9***	4.20***	0.72	2.54***
	(13.1)	(13.2)	(1.00)	(0.65)	(0.64)
Observations	439	400	397	393	345

(E) Labor market

Notes: We regress overestimation of job outcome for recalls against worker characteristics. Robust standard errors in parentheses. Stars are used to denote the statistical significance of each coefficient (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). Overestimation of each outcome may be negative and is defined as the recall belief minus the actual job outcome for the same time period. We pool recalls of the week and the month before the baseline survey. *Certain of Beliefs* is a binary variable equal to 1 if a worker is totally or somewhat certain about their recalls. *Full-Time Worker* is equal to 1 if a gig worker, on average, works for at least 30 hours a week and is 0 otherwise. *Struggling Financially* is a binary variable equal to 1 if a worker reports either to struggle to pay the bills, get calls from collectors or is considering bankruptcy. *Gig Pay is Essential* is a binary variable equal to 1 if gig income is used primarily for purchasing essential goods such as food and housing. On Panel (E), we define binary variables relating to employment previous to working for the gig company (prefixed by "*Was*") and other employment currently (prefixed by "*Has*").

Table A6: Relationship of delivery and ride share gig experience with overestimation of job outcomes

	Overestimation (Belief - Actual)				
	(1)	(2)	(3)	(4)	(5)
	Weekly Pay	Net Weekly Pay	Weekly Hours	Hourly Pay	Net Hourly Pay
Experience Delivery (6-12 mo.)	17.1	-16.2	0.19	2.64**	0.10
	(23.6)	(24.6)	(1.86)	(1.17)	(1.16)
Experience Delivery (12+ mo.)	72.5***	53.5***	2.71**	3.19***	2.26***
	(16.9)	(17.2)	(1.31)	(0.83)	(0.83)
Experience Rideshare (6-12 mo.)	-15.3	34.0	1.77	-0.18	0.61
	(30.6)	(31.1)	(2.49)	(1.60)	(1.58)
Experience Rideshare (12+ mo.)	9.32	-0.96	0.31	-0.76	-0.0060
	(17.8)	(17.9)	(1.37)	(0.87)	(0.87)
Excluded Group	43.2***	61.3***	3.98***	-1.36*	1.21*
	(14.8)	(15.0)	(1.14)	(0.72)	(0.71)
Observations	439	400	397	393	345

Notes: We regress overestimation of job outcomes on binary variables for experience in ride share and delivery gig work. The excluded group has less than 6 months of experience in both ride share and delivery. Robust standard errors in parentheses. Stars are used to denote the statistical significance of each coefficient (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). Overestimation of each outcome is defined as the recall belief minus the actual job outcome for the same time period. We pool recalls of the week and the month before the baseline survey.

Table A7: Robustness checks for over-recalling of job outcomes

	Overestimation (Belief - Actual)				
	(1)	(2)	(3)	(4)	(5)
	Weekly Pay	Net Weekly Pay	Weekly Hours	Hourly Pay	Net Hourly Pay
One Company	-27.2*	-16.3	-0.96	-1.58**	-1.23*
	(14.8)	(15.1)	(1.13)	(0.72)	(0.71)
Excluded Group	102.1***	100.0***	6.23***	1.48***	3.17***
	(10.3)	(10.5)	(0.79)	(0.51)	(0.50)
Observations	439	400	397	393	345

(A) Number of companies

	Overestimation (Belief - Actual)				
	(1)	(2)	(3)	(4)	(5)
	Weekly Pay	Net Weekly Pay	Weekly Hours	Hourly Pay	Net Hourly Pay
Belief is a Round Number	68.2***	22.3	5.21***	0.91	0.65
	(16.1)	(15.5)	(1.25)	(0.72)	(0.70)
Excluded Group	43.0***	83.6***	2.06*	0.26	2.54***
	(13.5)	(9.59)	(1.06)	(0.52)	(0.48)
Observations	432	400	392	387	337

(B) Rounding up

	Overestimation (Belief - Actual)				
	(1)	(2)	(3)	(4)	(5)
	Weekly Pay	Net Weekly Pay	Weekly Hours	Hourly Pay	Net Hourly Pay
Pay for More Screenshots	36.3**	24.2	1.76	-0.48	-0.93
	(16.6)	(16.8)	(1.26)	(0.81)	(0.82)
Excluded Group	62.4***	74.7***	4.49***	1.05	3.36***
	(14.2)	(14.2)	(1.07)	(0.69)	(0.70)
Observations	439	400	397	393	347

(C) Payment for additional screenshots

Notes: We regress overestimation of job outcomes recalls against covariables. Robust standard errors in parentheses. Stars are used to denote the statistical significance of each coefficient (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). Overestimation of each outcome can be negative and is defined as the recall belief minus the actual job outcomes for the same time period. We pool recalls of the week and the month before the baseline survey. *One Company* is a binary variable equal to 1 if a worker has only worked for one gig company in the past 3 months. *Belief is a Round Number* is equal to 1 if a worker's recall belief is a multiple of 5 (for weekly hours, hourly wage and net hourly wage) or a multiple of 50 (for gross and net weekly pay) and 0 otherwise. *Pay for More Screenshots* is equal to 1 if a survey respondent accepted the offer of additional pay for submitting three additional weekly earnings screenshots from the gig platform app and 0 otherwise.

Table A8: Sensitivity of forecast error estimates to selective attrition

Share of misperception retained	Mean forecast error: Target in the full sample (1)	Mean forecast error: Required among nonrespondents (2)	Percentile rank: Forecast-error distribution (3)	Percentile rank: Recall-error distribution (4)
Panel A. Weekly pay ($ per week)				
One-half	55.32	0.64	27	29
One-quarter	27.66	-54.36	15	12
None	0.00	-109.37	8	5
Panel B. Weekly hours (hours per week)				
One-half	4.06	0.05	22	27
One-quarter	2.03	-3.99	12	10
None	0.00	-8.03	4	6

Notes: Forecast error is defined as the forecast minus the realized outcome, so negative values indicate underprediction. Among the 344 workers asked to provide forecasts, 171 completed the midline survey and 173 did not. The observed mean forecast errors among respondents are $110.64 for weekly pay and 8.12 hours for weekly hours. For each target full-sample mean forecast error, column (2) reports the nonrespondent mean that solves $\bar{e}^* = \frac{171}{344}\bar{e}_R + \frac{173}{344}\bar{e}_{NR}$, where $\bar{e}_R$ is the observed respondent mean and $\bar{e}_{NR}$ is the assumed nonrespondent mean. Columns (3) and (4) report the percentile rank of $\bar{e}_{NR}$ in the observed individual-level forecast-error and baseline recall-error distributions, respectively.

Table A9: Tests of positive selective recall and asymmetric updating (other outcomes)

	Recall Belief					
	Weekly Hours			**Hourly Pay**		
	(1)	(2)	(3)	(4)	(5)	(6)
Maximum	0.38***	0.34***	0.13	0.22***	0.20***	0.081
	(0.081)	(0.081)	(0.15)	(0.056)	(0.056)	(0.092)
Minimum	0.22**	0.22**		0.18***	0.18***	
	(0.10)	(0.10)		(0.067)	(0.068)	
Mean			0.47***			0.33***
			(0.18)			(0.11)
Observations	293	293	293	289	289	289
Demographic Controls		✓			✓	
p-value(Max=Min)	0.37	0.49		0.76	0.86	

(A) Selective Recall

	Belief$_t$			
	Weekly Hours		**Hourly Pay**	
	(1)	(2)	(3)	(4)
Belief$_{t-1}$	0.66***	0.62***	0.79***	0.79***
	(0.064)	(0.072)	(0.090)	(0.097)
$1\{\text{Actual}_t \geq \text{Belief}_{t-1}\}$	10.8***	10.2**	8.55***	9.46***
	(2.11)	(3.90)	(1.84)	(3.17)
$1\{\text{Actual}_t < \text{Belief}_{t-1}\}$	-2.25	-2.28	2.49	3.45
	(1.80)	(3.72)	(2.01)	(3.30)
Observations	149	149	140	140
Demographic Controls		✓		✓
p-value(Above=Below)	0.014	0.29	0.0033	0.045

(B) Belief Updating

Notes: Robust standard errors in parentheses. Stars are used to denote the statistical significance of each coefficient (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). Our set of demographic controls consists of binary variables for male, white, age between 18-34, age between 35-54, at least a college degree and household income below $40,000/year. Outcome variables in Panel (A) are recalls referring to the month before the baseline survey. Panel (A) shows regressions of recalls of job outcomes on functions of the 4 previous weeks of the actual job outcome. We define the minimum, the maximum and the mean for this set of four weeks. We present the p-value for a test of whether the coefficients for the maximum and the minimum variables are the same. In Panel (B), only participants that replied to our midline survey are included. $Belief_t$ is the recall of a job outcome of the first full week (starting on a Monday) after the baseline survey. $Belief_{t-1}$ is the pooled recall for each job outcome for the week or the month before the baseline survey. $Actual_t$ refers to the actual job outcome (obtained from a screenshot submitted from the gig platform app) of the first full week after the baseline survey. Each column shows the result of the regression of $Belief_t$ on $Belief_{t-1}$ and two binary variables: a variable equal to 1 if $Actual_t$ is above or equal to $Belief_{t-1}$, and a variable equal to 1 if $Actual_t$ is below $Belief_{t-1}$. There is no constant term in the regressions in Panel (B). We present the p-value for a test of whether the coefficients for the two indicator variables are the same.

Table A10: Relationship between high-earning weeks and recall of hours worked

	Recall Belief Weekly Hours	
	(1)	(2)
Last 4 Weeks (Actual)		
Maximum Hours	0.13	
	(0.15)	
Hours in Max Weekly Pay Week		0.28**
		(0.14)
Mean Hours	0.47***	0.34**
	(0.18)	(0.15)
Observations	293	288

Notes: Standard errors are reported in parentheses. Stars denote statistical significance (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). The dependent variable is recalled weekly hours referring to the month before the baseline survey. The sample is restricted to respondents with four weeks of observed hours and weekly pay. Maximum Hours and Mean Hours denote the maximum and mean of weekly hours across these four weeks. Hours in Max Weekly Pay Week denotes hours worked in the highest-pay week across these four weeks. Ties in maximum weekly pay are resolved using the last tied week's hours.

Table A11: Relationship between variance of pay and overestimation (other outcomes)

	Overestimation (Belief - Actual)			
	Weekly Hours		**Hourly Pay**	
	(1)	(2)	(3)	(4)
Last 4 Weeks (Actual)				
Coefficient of Variation (CV)	0.072**	0.077***	0.10***	0.10***
	(0.028)	(0.029)	(0.029)	(0.030)
Constant	4.03***	1.23	-1.47**	-1.91
	(1.36)	(3.67)	(0.72)	(2.32)
Observations	201	201	202	202
Demographic Controls		✓		✓
Average CV (SD/Mean)	0.41	0.41	0.19	0.19

Notes: Robust standard errors in parentheses. Stars are used to denote the statistical significance of each coefficient (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). Overestimation of a job outcome may be negative and is defined as the recall belief minus the actual job outcomes for the same time period. We show regressions of the overestimation of a job outcome against the coefficient of variation for the 4 previous weeks of that outcome. The coefficient of variation is the standard deviation over the mean. This variable is normalized so that a 1 unit increase is equal to an increase of 1 percentage point. We present the average coefficient of variation underneath each column. Our set of demographic controls consists of binary variables for male, white, age between 18-34, age between 35-54, at least a college degree and household income below $40,000/year.

Table A12: Tests of positive selective recall for forecasts

	Forecast Belief								
	Weekly Pay			**Weekly Hours**			**Hourly Pay**		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Last 4 Weeks									
Maximum	0.65***	0.63***	0.48***	0.42***	0.38***	0.27	0.17**	0.16**	0.0068
	(0.076)	(0.079)	(0.16)	(0.11)	(0.11)	(0.20)	(0.067)	(0.067)	(0.12)
Minimum	0.22**	0.23**		0.099	0.13		0.22***	0.23***	
	(0.10)	(0.11)		(0.14)	(0.14)		(0.079)	(0.080)	
Mean			0.37**			0.26			0.39***
			(0.19)			(0.23)			(0.13)
Observations	262	262	262	235	235	235	229	229	229
Control Variables		✓			✓			✓	
p-value(Max=Min)	0.015	0.023		0.17	0.29		0.73	0.62	

Notes: Robust standard errors in parentheses. Stars are used to denote the statistical significance of each coefficient (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). Forecast beliefs refer to the week after the baseline survey. Our set of demographic controls consists of binary variables for male, white, age between 18-34, age between 35-54, at least a college degree and household income below $40,000/year. We shows regressions of forecast of job outcomes on functions of the 4 previous weeks of the same job outcome. We define the minimum, the maximum and the mean for this set of four weeks. We present the p-value for a test of whether the coefficients for the maximum and the minimum variables are the same.

Table A13: Overestimation of job outcomes at the endline survey

	Overestimation (Recall - Actual)			
	(1)	(2)	(3)	(4)
	Weekly Pay	Weekly Hours	Hourly Pay	Net Hourly Pay
Constant	57.6***	4.51***	0.21	1.74***
	(11.4)	(0.79)	(0.53)	(0.45)
Observations	145	140	131	116

Notes: Robust standard errors in parentheses. Stars are used to denote the statistical significance of each coefficient (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). We regress the overestimation of recalls of job outcomes on a constant term, which then measures the average of the dependent variable. *Overestimation* is the recall belief minus the actual job outcome. Thus, a positive (negative) value implies overestimation (underestimation) of a job outcome. The recalls are for the weekly average of the last month. All data is from the endline survey.

Table A14: Relationship between net hourly pay and reservation wage (means)

	Net Hourly Pay	
Mean	Belief	Actual
Panel (A): Full Sample		
Reservation Wage (w_R)	14.91	14.91
Net Hourly Pay (w)	15.46	12.38
$w - w_R$	.55	-2.53
Panel (B): Outside Option Less Likely to be Gig		
Reservation Wage (w_R)	14.15	14.15
Net Hourly Pay (w)	14.84	12.3
$w - w_R$	.69	-1.85
Panel (C): Reservation Wage is Overestimated		
Reservation Wage (w_R)	14.91	13.24
Net Hourly Pay (w)	15.46	12.38
$w - w_R$	.55	-.85
Panel (D): Wage at Other or Previous Jobs		
Wage in Other Jobs (w_R)	17.76	17.76
Net Hourly Pay (w)	15.46	12.38
$w - w_R$	-2.30	-5.38
Panel (E): Maximum Weekly Net Hourly Pay		
Reservation Wage (w_R)	14.91	14.91
Net Hourly Pay (w)	15.46	14.3
$w - w_R$	.55	-.61

Notes: Belief w is the pooled net hourly pay recall of the week and the month before the baseline survey. Actual w is the actual expected net hourly pay referring to the same period. Reservation wage proxy w_R is the answer to the following question: "What is the lowest acceptable hourly pay after taxes and expenses that would accept to keep working for [gig company]?". In the first two rows of each panel, we calculate the mean of w_R and w. In the final row of each panel, we calculate the difference between w and w_R. Panel (A) is our full sample and base measure. In Panel (B), we include only workers that did not do gigs before their current gig job and that do not work for other gig companies. In Panel (C), we assume w_R is measured with half as much error as w by gig workers. In Panel (D), we use the worker's self-reported gross hourly wages in either a previous job or in another current job. This measure is elicited in 5-dollar bins, for which we take the midpoint. In Panel (E), we assume the actual w is the maximum weekly average of the net hourly pay in the past 4 weeks.

Figure A1: Elicitation of recall of job outcomes



We will now ask you questions about your pay and amount of work you do for **DoorDash**.

Please answer to the best of your ability. Your answers will be aggregated with those of other participants to create insights about working for online gig platforms.

Think about the **last month** you worked for DoorDash.

Please answer the following questions considering **only DoorDash**, and do not use commas in your numerical answers.

Consider tips, bonuses, promotions and platform fees when thinking about your pay.

How many hours do you work per week?

_____ hours per week

How much do you get paid **per hour** on average, **before expenses and taxes**?

$ _____ /hour

How much do you get paid **per week**, **before expenses and taxes**?

$ _____ /week

Out of your total pay **before expenses and taxes**, what percentage comes from bonuses and tips?

_____ %

For this question, consider only the time in which you are either picking up deliveries or passengers or bringing them to their destination.

How much do you get paid **per hour actively working on gigs** on average, **before expenses and taxes**?

$ _____ /hour

How much do you spend on **gas** per week?

$ _____ /week

How much do you spend on **expenses and taxes** per week?

$ _____ /week

How much do you get paid **per hour** on average, **after expenses and taxes**?

$ _____ /hour

Figure A2: Elicitation of forecast of job outcomes in baseline survey

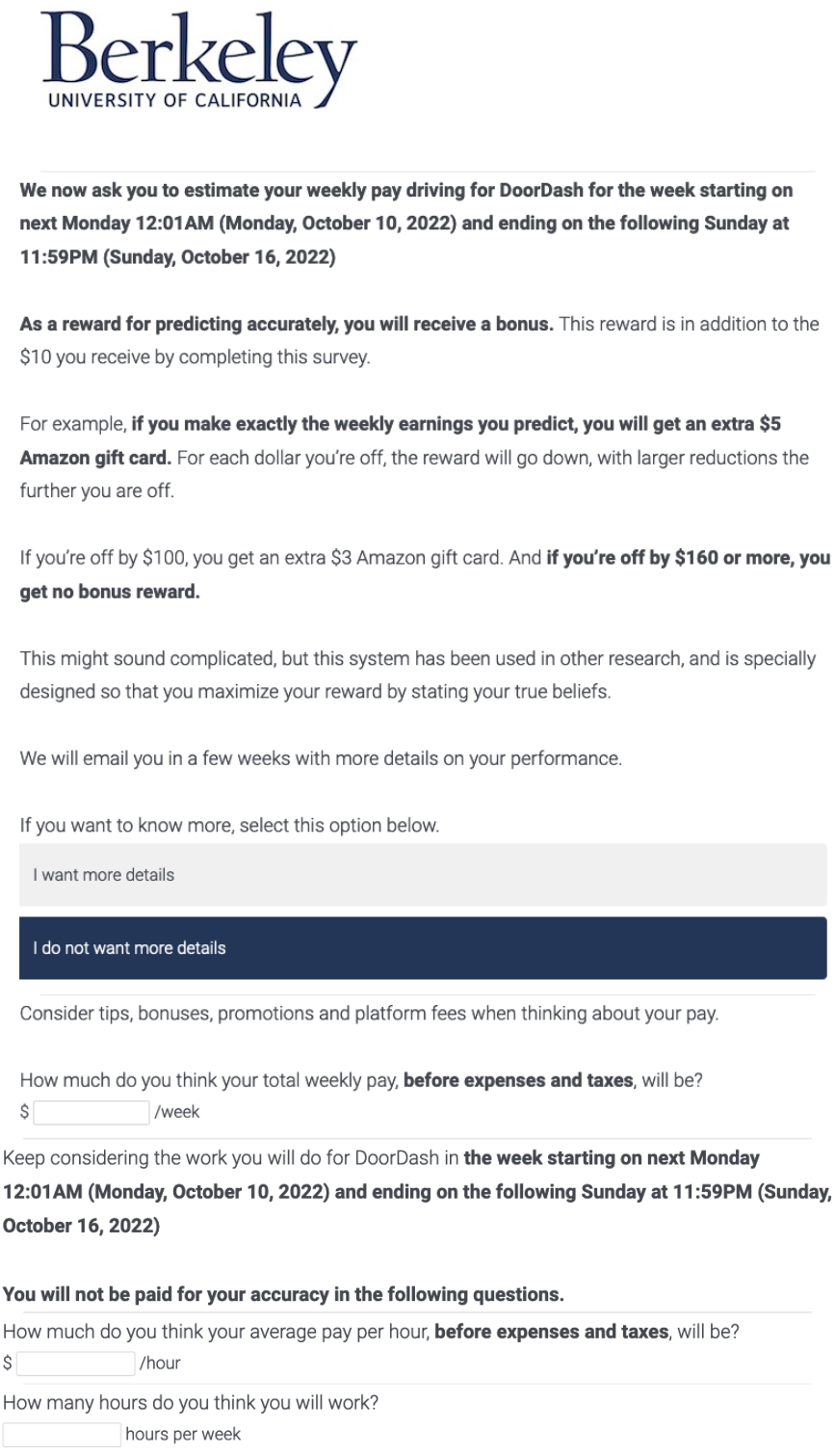
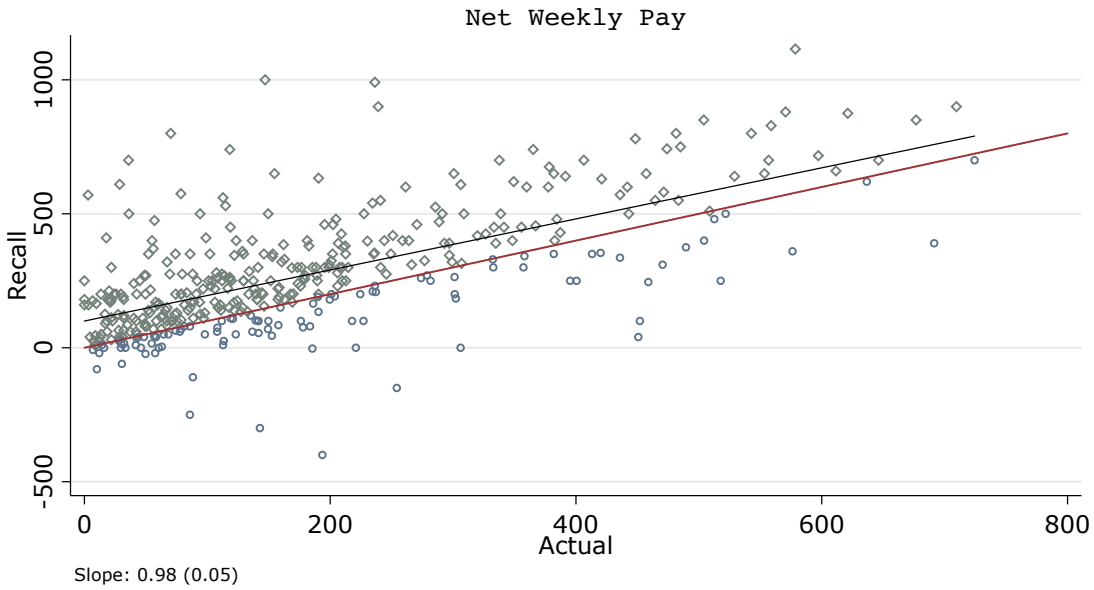
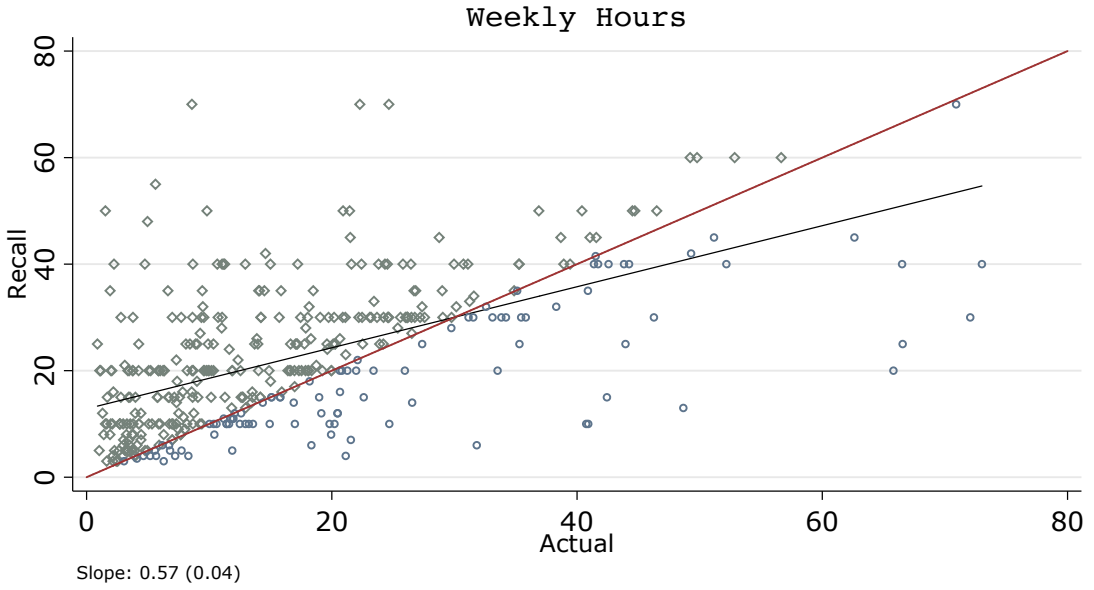


Figure A3: Scatter plots relating recalls and actual job outcomes

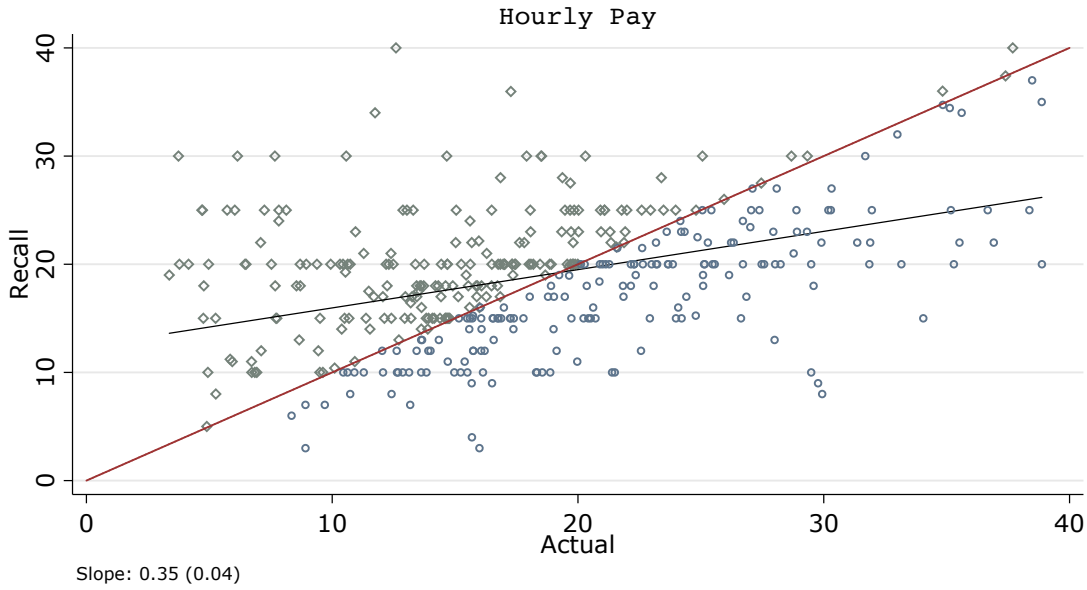


(A) Net Weekly Pay

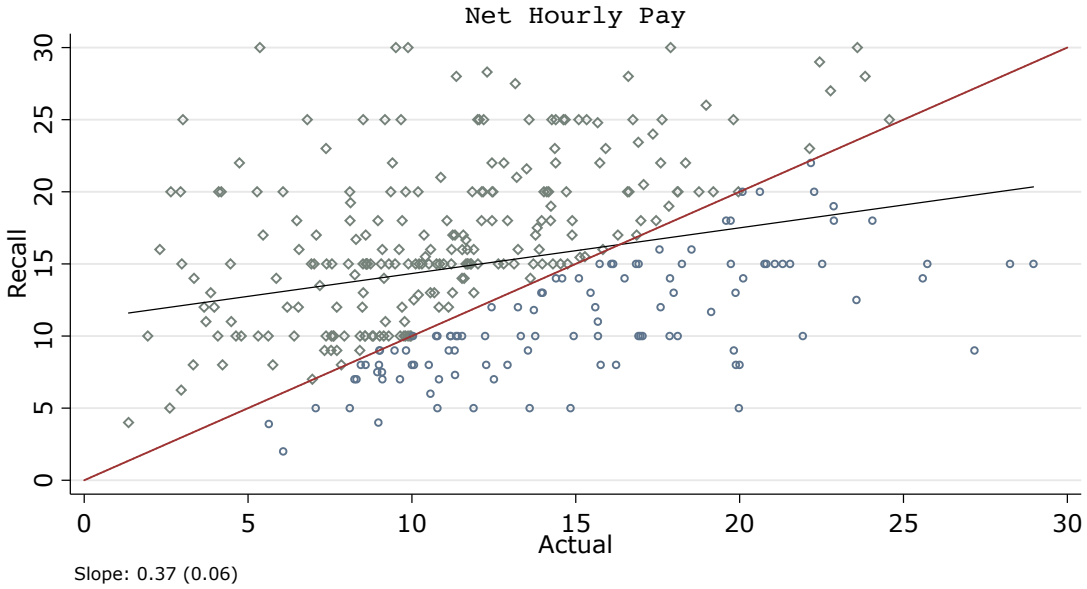


(B) Weekly Hours

Figure A3: Scatter plots relating recalls and actual job outcomes (cont.)



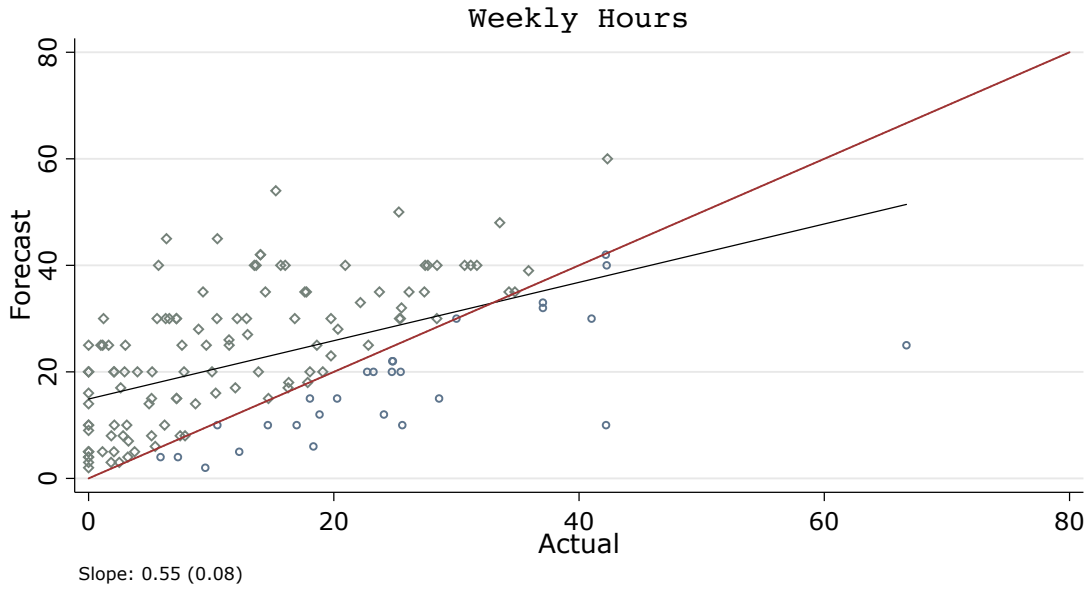
(C) Hourly Pay



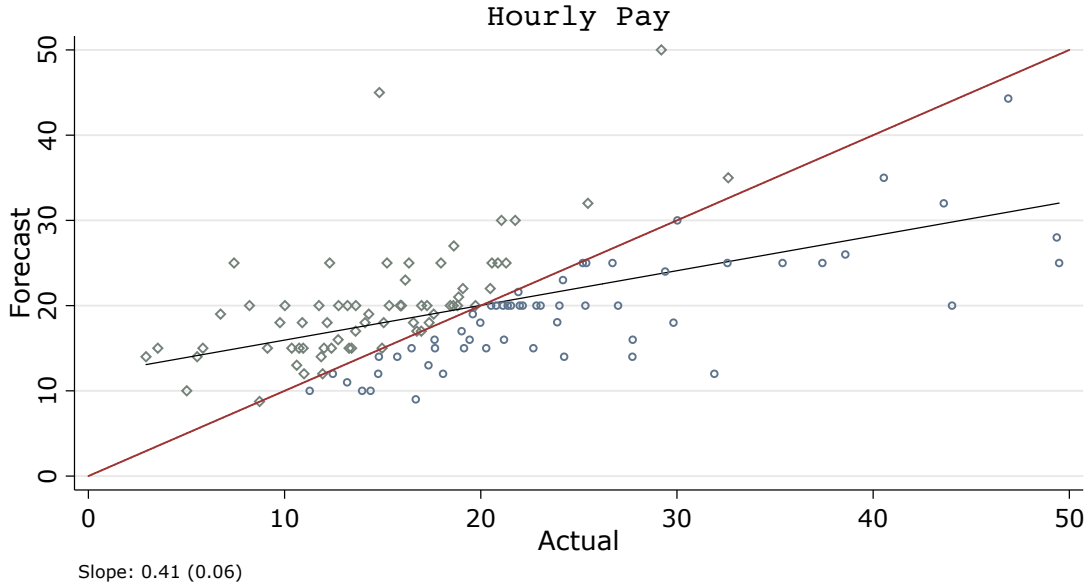
(D) Net Hourly Pay

Notes: In each graph, we draw a 45 degree line and the estimated slope of the regression of the belief on the actual job outcome. Underneath each graph, we present the slope from the regression of the recall belief on the actual outcome, with the associated standard error in parentheses. We exclude outliers from each graph. Points above the 45 degree line indicate overestimation, whereas points below the 45 degree line indicate underestimation of the job outcome. We pool recalls of the week and the month before the baseline survey. The actual job outcomes pay refers to the same time period as the recalls and are obtained using information from screenshots from gig platform apps.

Figure A4: Scatter plots relating forecasts and actual job outcomes



(A) Weekly Hours



(B) Hourly Pay

Notes: In each graph, we draw a 45 degree line and the estimated slope of the regression of the belief on the actual job outcome. Underneath each graph, we present the slope from the regression of the forecast belief on the actual outcome, with the associated standard error in parentheses. We exclude outliers from each graph. Points above the 45 degree line indicate overestimation, whereas points below the 45 degree line indicate underestimation of the job outcome. Forecast refer to the week following the baseline survey. The actual job outcomes pay refers to the same time period as the forecasts and are obtained using information from screenshots from gig platform apps.

Figure A5: Relationship between forecast and recall overestimation of job outcomes



Notes: We draw the fitted line of the regression relating the forecast overestimation and the recall overestimation of job outcomes. The slope and its standard error are shown underneath the plot. We exclude outliers from these graphs. We pool recalls of the week and the month before the baseline survey. The forecast refers to the first full week (starting on a Monday) following the baseline survey. Overestimation is measured as the recall or forecast minus the actual job outcome.

Figure A6: Relationship between forecast and recall beliefs



Notes: In each graph, we draw the fitted line of the regression relating the forecast belief and the recall belief of a job outcome. The slope and its standard error are shown underneath each plot. We exclude outliers from each graph. We pool recalls of the week and the month before the baseline survey. The forecast refers to the first full week (starting on a Monday) following the baseline survey.

Figure A7: Categories of costs considered in expenses calculation by gig workers

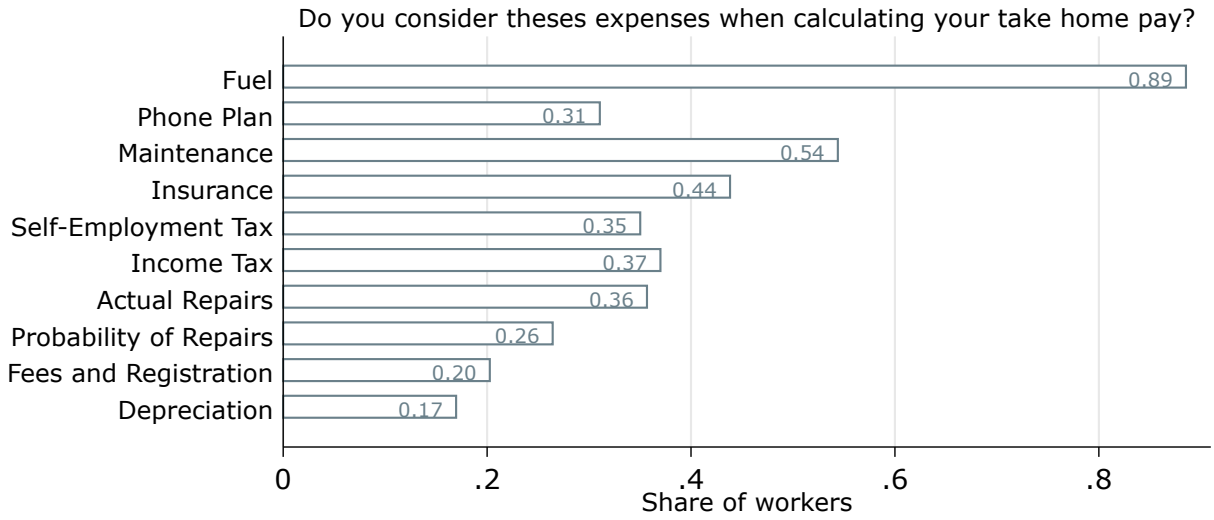


Figure A8: Main day of the month when workers pay major bills

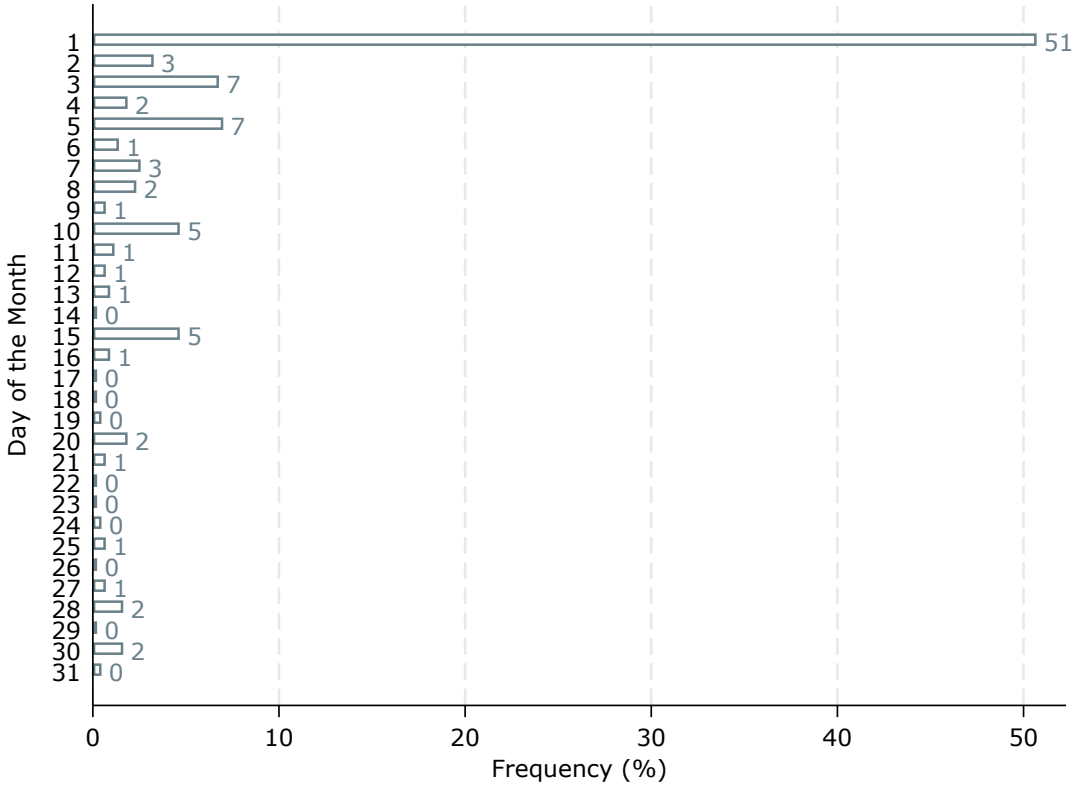
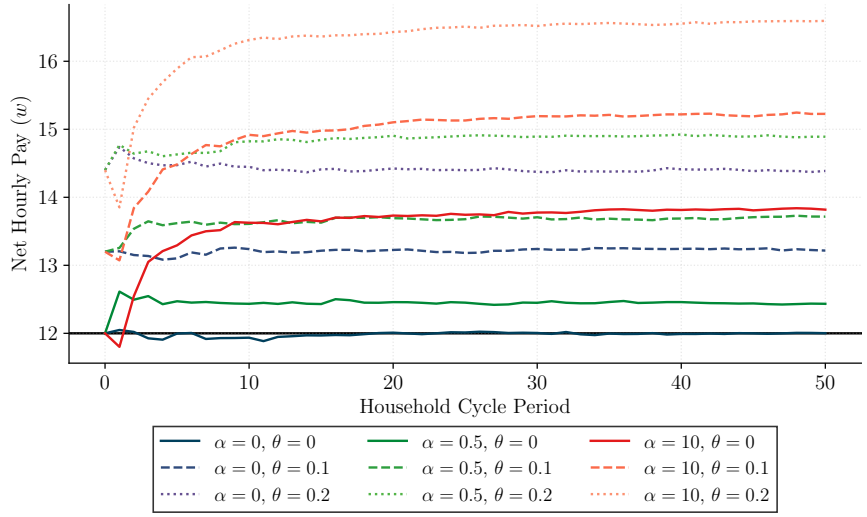
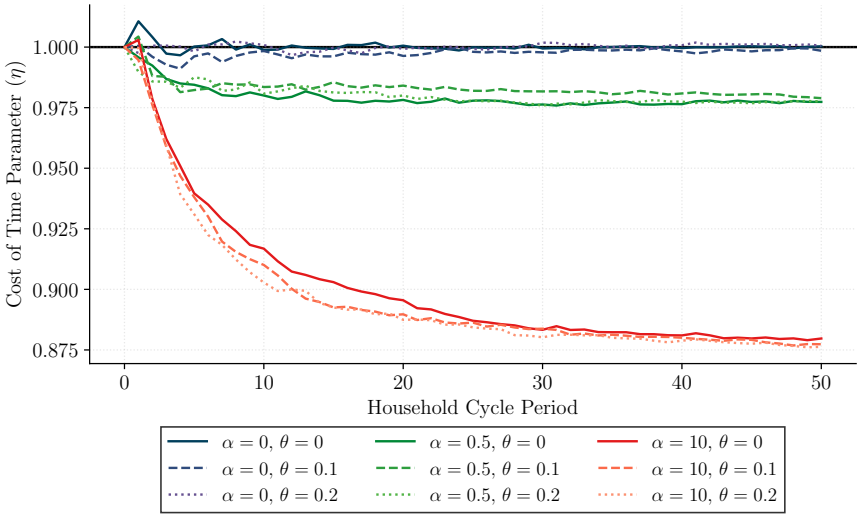


Figure A9: Simulation of learning models for different combinations of biases



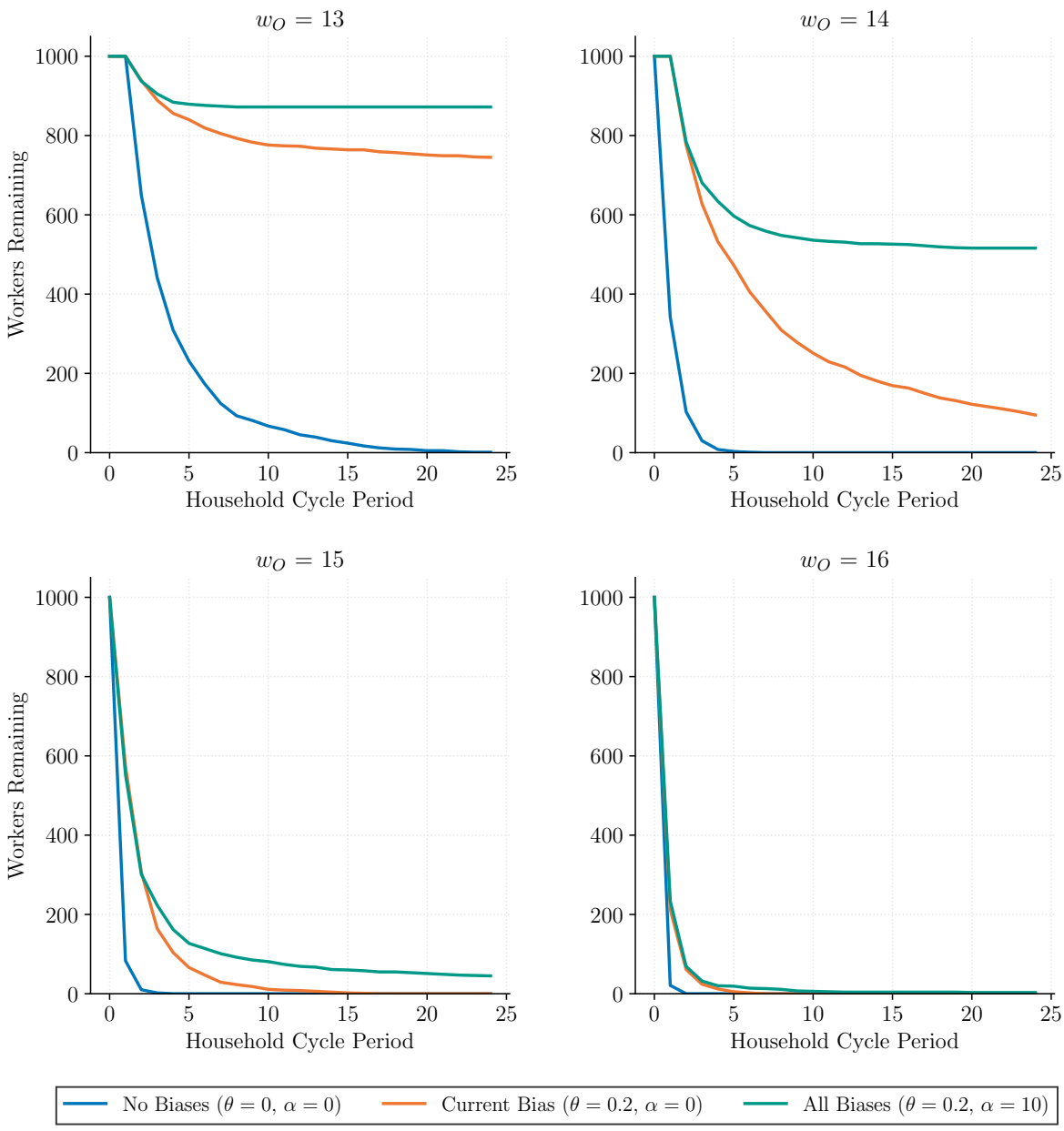
(A) Net Hourly Gig Pay



(B) Work Disutility Parameter

Notes: We numerically plot one realization of the learning model described in Section 5.1 and Appendix F for different values of the memory bias parameter α and hourly net pay overestimation parameter θ. In Panel (A), we plot the posterior mean for the net hourly pay at a gig job $\tilde{E}[w_{G,t+1}]$. In Panel (B), we plot the posterior mean for the work disutility shifter η, $\tilde{E}[\eta_{t+1}]$. We assume the true expected gig net hourly pay is $\mu_G = 12$, the true expected η is 1, and the standard deviations for the signals of net hourly pay $w_{G,t}$ and the cost of effort shifter η_t are, respectively, $\sigma_\varepsilon = 4$ and $\sigma_\nu = 0.2$. In addition, the prior means are $\mu_0 = 12$ and $\eta_0 = 1$, with the prior standard deviation being equal to 10 for w_G and 0.5 for η.

Figure A10: Job choice over time in behavioral labor supply model for different combinations of biases



Notes: We numerically solve the model described in Section 5.1 for different values of the memory bias parameter α and hourly net pay overestimation parameter θ. We simulate the job choices of 1,000 gig workers across household budget cycles. We plot how many workers, at each cycle, choose to remain at their gig job for different combinations of α, θ and the non-flexible job hourly pay w_O. Hours worked at the non-flexible job $\bar{h}$ equal 15. The job choice decision involves comparing the perceived gig job utility $\tilde{E}[V_G]$ and the non-flexible job utility $E[V_O]$ over the next household cycle. This decision is detailed in Appendix F. We assume that the consumption utility function is $u(c) = 0.85 \cdot c^{0.7}$, the cost of effort function is $c(h) = h^{1.2}$, the savings utility function is $g(s) = 1.5 \cdot \text{sgn}(s) \cdot |s|^{0.6}$, $r = 0$, the true expected gig net hourly pay is $\mu_G = 12$, and the standard deviations for the signals of net hourly pay $w_{G,t}$ and the work disutility shifter η_t are, respectively, $\sigma_\varepsilon = 2$ and $\sigma_\nu = 0.1$. The prior means are $\mu_0 = 12$ and $\eta_0 = 1$, with the prior standard deviation being equal to 10 for w_G and 0.5 for η.

Appendix B: Details of Expenses Calculation

We now provide details of our calculations of expected expenses associated with gig work, which we use to estimate the actual expected net hourly and weekly pay. Only variable costs are considered: fuel, maintenance and repair, variable depreciation, self-employment taxes, and federal income taxes. We obtain estimates of operating costs from the AAA Your Driving Costs 2022 booklet. This guide has been published since 1950 and uses a proprietary methodology to calculate the costs of owning and operating a *new* car in the United States over five years.

This source has estimates of per mile operating costs for 9 different car categories: small sedan, medium sedan, subcompact SUV, compact SUV, medium SUV, midsize pickup, half-ton pickup truck, hybrid, and electric. We use expected maintenance and repair costs from this guide. We only use its estimates of fuel costs for hybrid and electric cars. For other categories, we average the AAA's gas price for the three months before the baseline survey. This is combined with information on fuel efficiency (miles per gallon) taken from the guide. Only the variable part of depreciation, resulting from driving additional miles, is considered. We calculate this cost by taking the increase, estimated in the guide, in total depreciation costs from driving 15,000 to 20,000 miles and dividing that by 5 thousand. In this way, we obtain an estimate of variable depreciation per mile for each car category.

To account for variation in costs with car age, we adjust the reported costs in the Your Driving Costs guide based on information from CarEdge. We use three car age groups: between 0 and 5 years, 6 and 10 years, and 10 years and above. We then calculate how (i) maintenance and repair costs and (ii) variable depreciation costs increase with car age. For each car category, we take an average of this variation for the top five car models.[32] On

[32]For each respective car category, these are: small sedan (Honda Civic, Hyundai Elantra, Nissan Sentra, Toyota Corolla, Volkswagen Jetta), medium sedan (Chevrolet Malibu, Honda Accord, Hyundai Sonata, Nissan Altima, Toyota Camry), subcompact SUV (Chevrolet Trax, Honda HR-V, Hyundai Kona, Jeep Compass, Subaru Crosstrek), compact SUV (Chevrolet Equinox, Ford Escape, Honda CR-V, Nissan Rogue, Toyota RAV4), medium SUV (Chevrolet Traverse, Ford Explorer, Subaru Outback, Jeep Grand Cherokee, Toyota Highlander), midsize pickup (Chevrolet Colorado, Ford Ranger, Honda Ridgeline, Jeep Gladiator, Toyota Tacoma), half-ton pickup truck (Chevrolet Silverado, Ford F-150, Nissan Titan, Ram 1500, Toyota

average across car categories, and relative to the costs for a car that is between 0 and 5 years old, depreciation costs are estimated to be 84% (43%) of the initial ones from years 6 to 10 (after year 10). According to our estimates, repair and maintenance costs for cars between 6 and 10 years old are, on average, 2.75 times higher than those for cars from 0 to 5 years old. This number is 3.76 for cars older than 10 years. We then multiply these factors by the estimates of depreciation and maintenance and repair from the AAA guide. Fuel expenses are assumed to be the same regardless of the age of the car.

We assume workers drive, on average, for 10 miles a hour. This allows us to turn per mile costs from the Your Driving Costs guide into per hour costs. We calculate self-employment taxes by first subtracting gross earnings per hour by the IRS mileage rate deduction ($.585/mile x 10 miles) and, following IRS procedure, multiplying the result by 0.925 and then by 15.3%.

Income taxes per hour are estimated by using the reported yearly household income, calculating how much of that income comes from gig work and then applying the standard deduction for 2022 for single filers. We assume gig workers work in half of the weeks of the year and make their average weekly earnings in each of these weeks. We divide the result by an estimate of total miles driven per year, still assuming that 10 miles are driven per hour. In our calculations, income taxes are, on average, 1% of total expected costs.

To determine how much of the driver's total pay is going towards expected expenses, we subtract fuel, maintenance and repairs, depreciation, self-employment tax, and income taxes from their gross earnings per hour. Using this method, we calculate our post-expenses and taxes earnings per hour and convert it into a share of total gross pay. Table B1 provides an example of how this calculation is done, considering a driver of a 2022 Honda Accord who earns $20/hour and works 20 hours per week. Using this estimated expenses share, we calculate the actual expected net hourly and weekly pay. We do not calculate expected expenses for workers who rent a car to do gig work or who use a bike or scooter.

Tundra), hybrid vehicle (Ford Explorer, Honda CR-V, Hyundai Ioniq, Toyota Prius Liftback, Toyota RAV4) and electric car (BMW i3, Chevrolet Bolt, Hyundai Kona Electric, Nissan Leaf, Tesla Model 3)

Table B1: Example of expected expenses calculation

Category	Calculation	Value
Gross Earnings (1)		\$20/hour
Fuel (2)		-\$1.57/hour
Maintenance and Repair (3)		-\$1.04/hour
Variable Depreciation (4)		-\$0.52/hour
Pre-Tax Net Earnings (5)	(1) - (2) - (3) - (4)	\$16.86/hour
Expenses Deductions (6)		- \$5.85/hour
Self-Employment Tax (7)	[(1) - (6)]*0.925*15.3%	-\$2.00/hour
Federal Income Tax (8)		-\$0.19/hour
Post-Tax Net Earnings (9)	(5) - (7) - (8)	\$14.66/hour
Share of Expenses (10)	[(1) - (9)] / (1)	26.6%

Notes: Example of calculation used to estimate the expected expenses share out of total pay. We consider a driver of a 2022 Honda Accord who makes \$20/hour, works 20 hours a week and drives, on average, for 10 miles per hour. Our cost measures for maintenance, repair and variable depreciation are based on the AAA Your Driving Costs 2022 guide. Fuel costs are the average of gas prices from AAA in the three months before the baseline survey. We adjust for variation of maintenance, repair and depreciation costs over a car's lifespan by using information from CarEdge. We apply the IRS mileage rate deduction in 2022 (\$.585/mile) to calculate self-employment taxes and we estimate federal income taxes by combining reported yearly household income, an estimate of gig income over the year and by applying the standard deduction for 2022.

Appendix C: Beliefs about Other Workers

One important aspect of optimistic beliefs is overplacement (Healy and Moore, 2007): the belief that one is more skilled relative to others than one actually is. To analyze this phenomenon in our setting, we ask, for a sub-sample, beliefs about job outcomes for other workers. In particular, for half of the survey respondents in the control group of the information treatment described in Appendix E (or one quarter of the full sample), we elicit beliefs about job outcomes for the average worker for the same gig company in the baseline survey.[33] See Figure C1 for an example of the questions we ask.

Our first set of results are shown in Figure C2. We compare, divided by job outcome, the average (i) actual outcome, (ii) recall belief for oneself and (iii) recall belief for the average worker of the same gig company. We find that workers believe the average worker works longer hours and has a higher weekly pay. This effect is very large, and implies a belief of the average worker working full-time for the gig company, which is not correct for our sample. In addition, workers believe the average worker has a higher gross hourly pay and a similar net hourly pay to themselves.

We then subtract each worker's belief for other workers from the recall belief about their own job performance. We next calculate leave-out means of the actual job outcomes. Then, we subtract each worker's actual job outcome by this leave-out mean. Finally, we compare these two differences to test for overplacement, which exists if workers believe their job outcome is higher relative to others than it actually is.

Results are shown in Table C1. We find no evidence of overplacement on hourly pay and find evidence of underplacement for weekly hours. Finally, we ask workers whether they believe other workers in the same gig company misunderstand their pay: 72% of workers believe it's likely others overestimate pay, versus 43% for others underestimating pay.

[33] These beliefs are elicited for everyone in the midline and endline surveys.

Table C1: Beliefs of job outcomes for self versus other workers in same gig company

Pay Relative to Others (Belief)	**Summary Statistics**					
	Mean	P25	Median	P75	Std. Dev.	N
Hourly Wage	-1.34	-4.75	0.00	2.00	7.07	108
Net Hourly Wage	0.21	-2.00	0.00	3.00	4.91	99
Weekly Pay	-184.91	-300.00	-150.00	0.00	289.76	110
Weekly Hours	-11.92	-20.00	-10.00	-3.00	14.89	113

(A) Belief

Pay Relative to Others (Belief - Actual)	**Summary Statistics**					
	Mean	P25	Median	P75	Std. Dev.	N
Hourly Wage	-1.52	-7.23	-0.73	4.12	9.85	98
Net Hourly Wage	0.27	-2.72	0.95	4.47	6.47	77
Weekly Pay	-147.35	-292.57	-108.03	65.94	341.45	108
Weekly Hours	-9.47	-17.69	-9.43	-0.04	14.13	99

(B) Belief - Actual

Notes: We present summary statistics of variables comparing outcomes for oneself and for other workers. Recalls are pooled for the week and the month before the baseline survey. In Panel (A), we compare recall beliefs for oneself versus for the average worker. A negative value means that a worker believes the job outcomes for the average worker is higher than for themselves. On Panel (B), we do a double difference: we subtract the recall comparison of self and others from Panel (A) with the difference of actual job outcomes for oneself minus the leave-out mean of the same outcome for other workers.

Figure C1: Elicitation of beliefs about the average gig worker



For the next four questions, consider **the group of people who worked for DoorDash in the last month** and who will reply to this survey.

Please answer the following questions considering only the work they do for **DoorDash**, and do not use commas in your numerical answers.

Consider tips, bonuses, promotions and platform fees when thinking about pay.

How much do you think **this group** gets paid **per week**, **before expenses and taxes**?

$ _____ /week

How much do you think **this group** gets paid **per hour** on average, **after expenses and taxes**?

$ _____ /hour

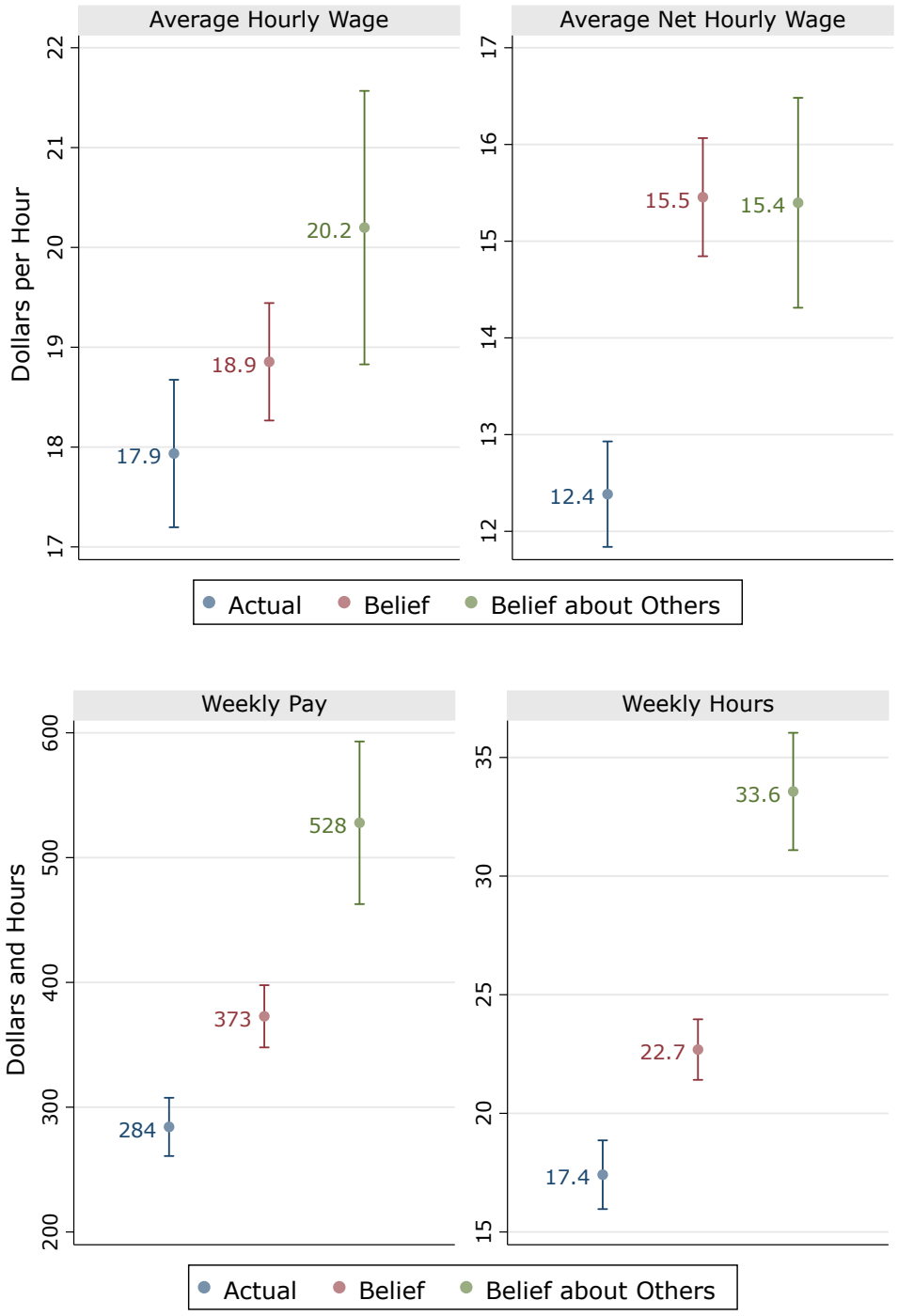
How many hours do you think **this group** works per week on average?

_____ hours per week

How much do you think **this group** gets paid **per hour** on average, **before expenses and taxes**?

$ _____ /hour

Figure C2: Comparison of actual job outcomes with self beliefs and beliefs about others



Notes: We plot the mean and the 95% confidence intervals of three measures pooled for the week and the month before the baseline survey: the actual job outcome, the recall belief of the job outcome for oneself and the belief of the job outcome for the average gig worker at the same company.

Appendix D: Additional Analyses and Robustness

In our baseline survey, we asked people to recall either the last week or the last month in which they worked for the gig company. Table D1 shows that overestimation is generally larger when workers are asked to recall an average of the last 4 weeks, compared with just last week. Figure D1 shows the results of regressions of recall beliefs for the past month on the last four weeks of actual realizations of the same outcome, ordered by recency, and a constant. For gross weekly pay, we find that recent periods influence beliefs more than older ones. These findings are consistent with more recent events being better recalled than older ones.

We now provide some robustness checks to how we measure the job performance of gig workers in our sample. Because gross weekly pay is salient and is reported to workers with bonuses, tips and fees included, we do not believe it is subject to credible concerns on this front. In contrast, measuring work hours is not as straightforward. We measure work hours as *online* hours, which is the total time a gig worker has the platform app turned on and is available for gigs. We believe his measure captures the nature of a standard job in which only part of the time is spent actively working.

Yet, gig workers may believe active hours, which includes only the time spent actively working on gigs, is the correct measure of labor supply. In this case, there will be a discrepancy between our definition and theirs. We will *underestimate* the overestimation of weekly hours but overstate the overestimation of hourly pay, as labor supply appears in the denominator of this variable.

To empirically access how much the definition of work hours matters, we provide a robustness check in Table D2. Our regressions include binary variables for behaviors that might lead to inaccurate measures of actual work hours, such (i) multi-tapping, or having more than one gig platform app active at the same time; (ii) being online on the app but with no intent to actually do gigs; (iii) not considering the time spent waiting for gigs as work or (iv) thinking commuting is work. We find that overestimation of recalls of job outcomes is

robust to excluding gig workers who report doing any of these behaviors. In other words, our overestimation results are not the result of a mismatch between ours' and the gig workers' definition of work hours.

Our measure of actual expenses is based on expected costs calculated at the car category and age level. This measure is therefore less reliable than job outcomes derived directly from screenshots. There are however a few ways in which we may be underestimating the expenses involved in gig work. First, our calculations ignore fixed costs such as insurance, registration fees, and non-variable depreciation (that is, depreciation not due to driving more miles). This means we likely underestimate costs for drivers who buy cars primarily for doing gig work. In addition, we ignore state income taxes and assume drivers can deduct miles driven when calculating self-employment taxes. This deduction is known to reduce self-employment taxes. Yet, this deduction is not available to 47% of gig drivers in our sample, since they report not recording their miles.

Table D1: Overestimation of job outcomes by recall period

	Overestimation (Recall - Actual)				
	Weekly Pay	Net Weekly Pay	Weekly Hours	Hourly Pay	Net Hourly Pay
Last Week	66.8***	71.5***	3.33***	0.81	2.97***
	(13.0)	(13.2)	(0.95)	(0.62)	(0.60)
Last Month	99.4***	102.1***	7.05***	0.64	2.35***
	(8.96)	(9.15)	(0.69)	(0.45)	(0.45)
Observations	439	400	397	393	345

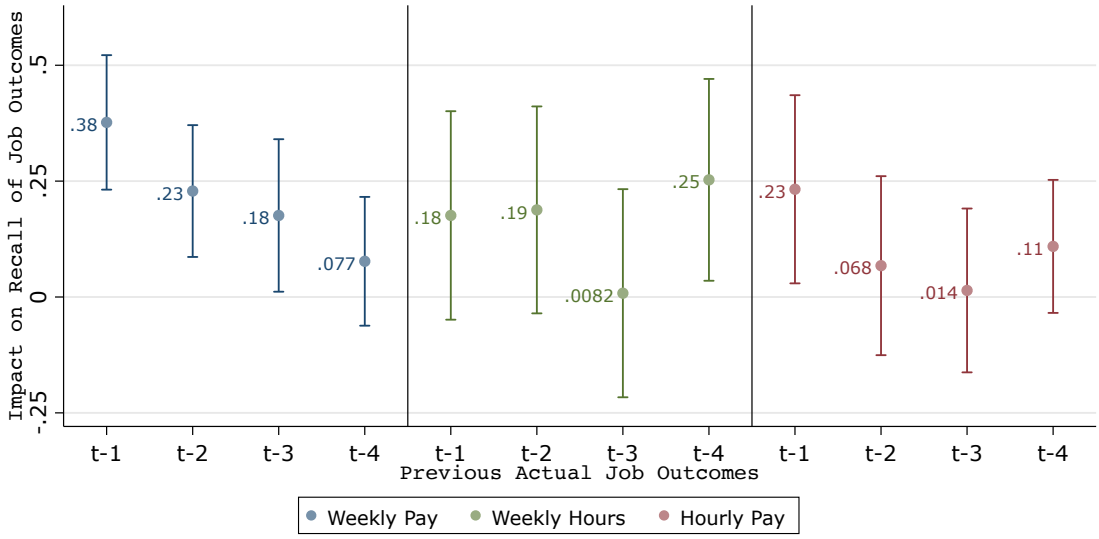
Notes: Robust standard errors in parentheses. Stars are used to denote the statistical significance of each coefficient (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). The regressions include no constant term. Overestimation of each outcome is defined as the recall belief minus the actual job outcome for the same time period. *Last Week* (*Last Month*) is a binary variable equal to 1 if the worker was asked to recall job outcomes for the last week (last month) in the baseline survey. We use an estimate of expected expenses to calculate net weekly pay and net hourly pay.

Table D2: Overestimation of job outcomes depending on definition of work hours

	Overestimation (Belief - Actual)				
	(1)	(2)	(3)	(4)	(5)
	Weekly Pay	Net Weekly Pay	Weekly Hours	Hourly Pay	Net Hourly Pay
Online but Not Working	28.8	23.6	3.97***	-0.24	-0.65
	(19.1)	(19.8)	(1.46)	(0.94)	(0.92)
Multiple Apps	56.5***	32.3*	1.31	3.20***	2.41***
	(16.1)	(16.6)	(1.26)	(0.79)	(0.78)
Considers Wait Not Work	-10.9	9.50	-0.15	0.076	0.65
	(16.2)	(16.6)	(1.24)	(0.79)	(0.78)
Considers Commute Work	-0.014	-5.23	-0.19	-0.12	0.64
	(15.8)	(16.3)	(1.21)	(0.77)	(0.76)
Excluded Group	70.6***	77.0***	4.79***	-0.17	1.51**
	(12.9)	(13.1)	(0.99)	(0.63)	(0.62)
Observations	439	400	397	393	345

Notes: We regress overestimation of job outcomes recalls against covariables. Robust standard errors in parentheses. Stars are used to denote the statistical significance of each coefficient (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). Overestimation of each outcome is defined as the recall belief minus the actual job outcomes for the same time period. We pool recalls of the week and the month before the baseline survey. *Online but Not Working* is a binary variable equal to 1 if a worker, either all the time or frequently, reports being online in the gig platform app with no intention of accepting gigs. *Multiple Apps* is a binary variable equal to 1 if a worker, either all the time or frequently, reports being online in more than one gig platform app at the same time. *Considers Wait Not Work* is equal to 1 if a worker considers little or none of the time spent waiting for rides as work hours and 0 otherwise. *Considers Commute Work* is equal to 1 if a worker considers all or most of the time spent commuting before and after a shift as work hours and 0 otherwise.

Figure D1: Estimated recall belief weights by recency of actual job outcomes



Notes: We plot the coefficients from regressions of recall beliefs for the past month on the last four weeks of actual realizations of the same outcome, ordered by recency, and a constant (omitted from the graph). Only recalls of the last month and workers who submitted four weeks of screenshots containing data on actual job performance are included. The notation for weeks on the x-axis ($t-1$, $t-2$, $t-3$, $t-4$) do not necessarily refer to four previous consecutive weeks, but only indicate the relative ordering of the weeks.

Appendix E: Details on the Information Treatment

We now detail our randomized information treatment. Our treatment is inspired by previous work aiming to change beliefs and behaviors through de-biasing interventions (Cullen and Perez-Truglia, 2022; Card et al., 2012; Bottan and Perez-Truglia, 2020). In the baseline survey, workers are randomly assigned to the treatment group with a 50% probability.[34] In that case, workers are required to manually input the weekly pay and weekly hours information from all screenshots they submit. This was done so we could provide feedback on their beliefs during the baseline survey. This also has the added effect of forcing them to face the information contained in the screenshots.

Then, on a single page, we explain the following: (i) how we calculate their gross hourly pay and its value; (ii) show how we calculate their expected expenses share, given their car; (iii) calculate their actual net hourly pay based on this information; (iv) compare the actual net hourly pay with their recall, informing them if they are under or overestimating it. We provide gig workers with an informative signal. Despite this, the signal is not fully accurate at the individual level due to noise in outcomes and in measuring expenses.

The next page informs them that gig workers in our sample often overestimate their job outcomes. We then provide them with a brief explanation of the concept of overconfidence. We find that 72% of gig workers say they plan to use in practice the information we provide, and that 81% say they agree partially or entirely with the information presented to them. Figure E1 shows an example of our information treatment. During the midline survey, we give the treatment group the same information pages presented at baseline. The purpose of this is to reinforce the information initially presented to them. Control group gig workers receive the same information, but only at the end of the endline survey.

If a worker does not use a car in his gig work, or if they rent a car to do gigs, we provide an alternative version of the information treatment, as we believe our measure of expected expenses would then be inadequate. In particular, we show these workers similar

[34] Table E1 shows a randomization balancing test.

information, but concerning their gross weekly pay. This happens for less than 5% of our sample. This group is excluded from all analysis concerning the information treatment.

E.1 Estimation Strategy

Our treatment should have heterogeneous effects depending on which direction we correct gig workers' beliefs. In other words, treatment effects should differ based on whether we tell gig workers they overestimate (*Bad Signal*) or underestimate (*Good Signal*) their net hourly pay. As a result, we estimate treatment effects separately based on whether the initial overestimation of net hourly pay is positive or negative. Our main specification is:

$$y_{it} = \beta_0 + \beta_1 Over_i + \beta_2 Bad\ Signal_i + \beta_3 Good\ Signal_i + X_{i0}\Gamma + \varepsilon_{it} \tag{6}$$

where y_{it} are belief or labor market outcomes for individual i at period t. Period t is a post-information treatment period. $Over_i = 1$ if initial overestimation of net hourly pay is positive and 0 if it is negative. Our two variables of interest here are $Bad\ Signal_i$ and $Good\ Signal_i$: $Bad\ Signal_i$ is equal to 1 if an individual is in the treatment group and $Over_i = 1$; $Good\ Signal_i$ if an individual is in the treated group and $Over_i = 0$. Finally, X_{i0} is the covariates matrix, composed of binary variables for male, white, age between 18-34, age between 35-54, at least a college degree and household income at most \$40k/year. In addition, it includes the pre-treatment outcome y. The variable ε_{it} is the regression error. We use robust standard errors in all regressions.

In the above specification, the treatment effect is identified by comparing all workers who overestimate (underestimate) their net pay in the treatment group with all workers who overestimate (underestimate) it in the control group. By doing so, we ignore the magnitude of the mistakes gig workers make. In order to take that into account, we use the following alternative specification:

$$y_{it} = \beta_0 + \beta_1 Mis_i + \beta_2 Treat_i + \beta_3 Treat_i \cdot Mis_i + X_{i0}\Gamma + \varepsilon_{it} \tag{7}$$

where $Treat_i$ is the treatment binary variable, equal to 1 if an individual is in the treatment group, Mis_i is the initial overestimation (recall belief minus actual) of net hourly pay. In equation (7), β_2 identifies the intercept of the treatment effect, while β_3 identifies its slope: it measures how the treatment effect varies depending on the value of the initial overestimation of net hourly pay.

E.2 Effect on Beliefs and Labor Market Outcomes

We now analyze the effects of our randomized information treatment on beliefs about gig job outcomes. After the information treatment, all of the beliefs we elicit about job outcomes concern periods following the treatment. Thus, effects on beliefs are also dependent on how job market outcomes change as a result of our treatment. For instance, our treatment may affect a worker's choice of hours. As a result, actual hourly pay can change in ways that should alter beliefs about net hourly pay. Many settings for information treatments, in contrast, do not allow the subject to influence the actual outcome after learning its value. The following results should be analyzed with this in mind.

After the information treatment, workers in the treatment group can review their incentivized forecast of weekly pay for the next week. We expect workers who are told they overestimate their net hourly pay to lower their forecast. This is because they might decide to work fewer hours in the future and also because they can lower their belief of their expected hourly pay. The opposite should happen for workers initially underestimating their pay.

Table E2 partially confirms this. In column (1), we regress the change in weekly pay forecast against the initial net hourly pay overestimation. On average, workers who review their pay forecast decrease it by \$10. In addition, a raise in initial overestimation by \$1/hour is associated with a decrease in the forecast of around \$1.6, which is significant at 10%. In column (2), we test whether workers are more likely to review their forecasts when told they are making larger mistakes (in absolute value). This appears to be the case.

In Table E3, we present the average overestimation of job outcomes at our midline (one week recall) and endline surveys (one month recall), separately by the treatment and the control groups. Our midline survey has an overall lower level of misestimation. We believe this is due to us (unlike in the baseline survey) requesting a recall of a specific week, explicitly stating the beginning and end dates. Thus, it is likely that a sizable share of workers directly checked their actual job outcomes in that week and input that information. We do not observe this pattern at the endline survey, where we elicit one month recalls.

We then examine how the information treatment affects beliefs when the initial level of overestimation is considered. Results are shown in Table E4. Our treatment does not seem to affect recall beliefs of net hourly pay in the expected way. This might partially be due to changes in behaviors (in response to our intervention) happening at the same time as our information provision. Alternatively, our information treatment might change beliefs immediately, but this effect may fade away over time. Table E5 presents an alternative specification for measuring effects on beliefs, using a continuous measure of net hourly pay overestimation. Our findings are similar.

In Table E6, we estimate the effect of the information treatment on labor supply and on the probability of holding a job outside the gig company. Effects are measured at the endline survey, distributed between two to five months after the treatment. Labor supply is an average of the weekly hours worked for the gig company for all weeks collected during the endline survey. It includes zeroes for those no longer working for the gig company.

The top panel uses separate coefficients for good and bad signals, while the bottom one uses a continuous measure of misestimation. Workers who initially underestimated their earnings were less likely to hold other jobs by the endline survey, with this effect being statistically significant. However, there was no significant effect for those receiving a so-called bad signal. The effect on weekly hours worked was not statistically significant in either case.

Figure E1: Example of information treatment

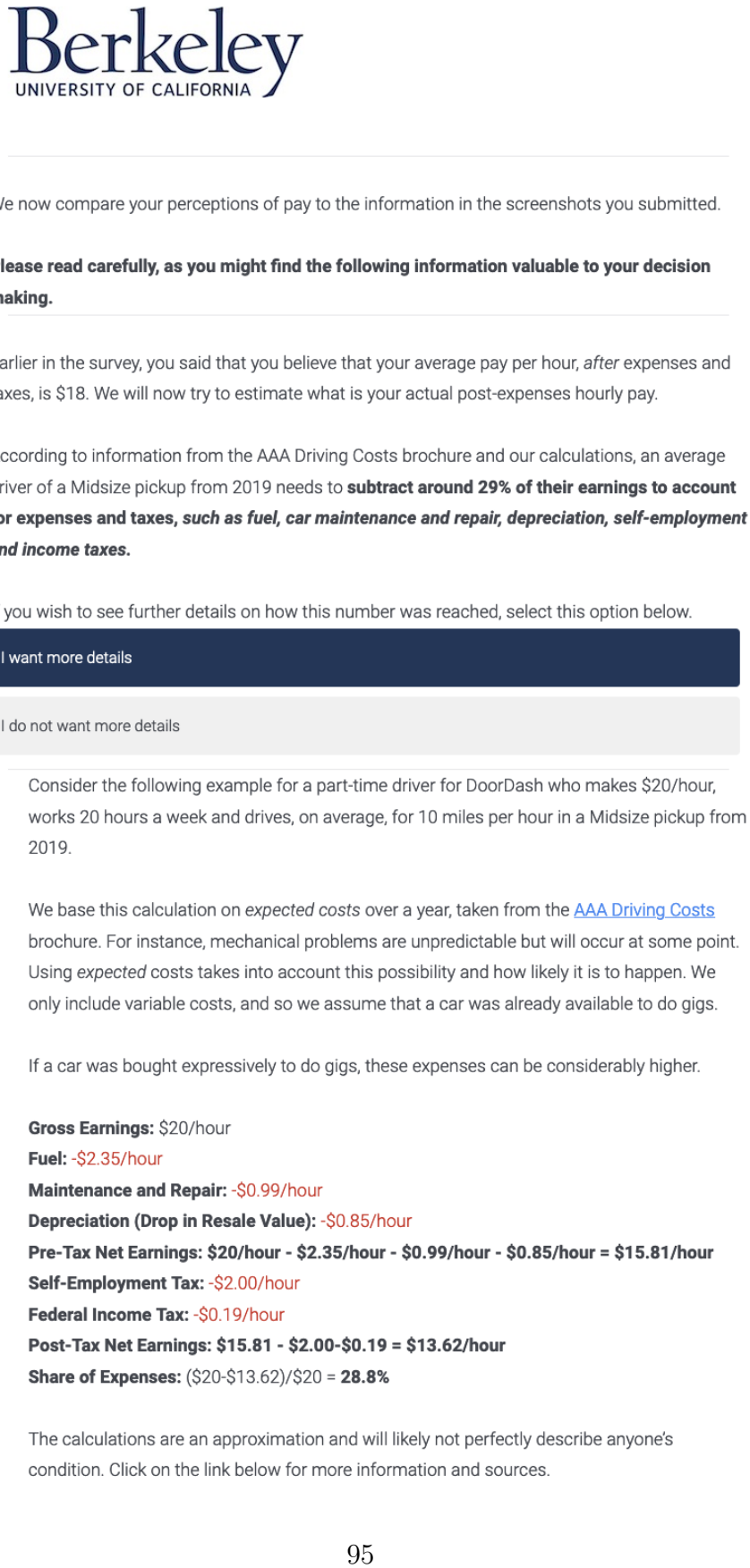


Figure E1: Example of information treatment (Cont.)

In the **screenshots you submitted for last week**, you worked approximately **9.2 hours** for DoorDash. Your total pay in this time were **$155.12**.

Assuming you were working for DoorDash the whole time you were online on the app, your **actual** average pay per hour, *before* expenses and taxes, is found by dividing your total pay by the time you worked during these weeks. This is equal to **$16.86 per hour**.

Using the expenses share of 28.8%, we can estimate that your **actual** average take home pay (or hourly pay *after* expenses and taxes) to be around **$12**.

This is 33% lower than your assessment ($18).

This might indicate that you are *overestimating your take home pay.*

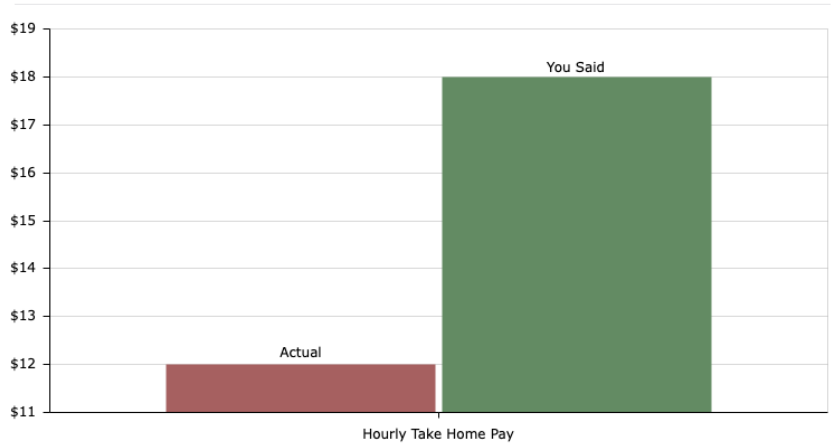


Figure E1: Example of information treatment (Cont.)

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We will now to share some of the preliminary results we have obtained from our study.

Following similar calculations as the ones we just presented, we found that previous respondents to our survey have, on average, **overestimated** their pay and **underestimated** their expenses.

According to our data, drivers, on average, overestimate their average hourly pay by **8.3%**, their average hourly pay after expenses by **23.8%** and their weekly pay by **23.9%.**

While this might seem weird, as it can imply that drivers do not always have a good understanding of how much money they make on gig economy jobs, **we believe they can be explained by the psychological concept of *overconfidence*.**

Overconfidence means that, in many situations, people have a tendency to be too optimistic and too certain about important aspects of their lives. For instance, three-quarters of US drivers consider themselves better-than-average drivers.

These concepts matter more in situations where understanding something is a hard task. In our view, that is exactly the case for calculating pay in gig economy jobs, as pay tends to vary a lot depending on time of day, expenses and market conditions each day.

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Recall that we asked you to predict your weekly pay driving for DoorDash for the week starting on next Monday 12:01AM (Monday, October 10, 2022) and ending on the following Sunday at 11:59PM (Sunday, October 16, 2022). As a reward for predicting accurately, you will receive a bonus.

Before, you predicted a total weekly pay equal to $ **before expenses and taxes**.

If you want to change your prediction, please write it down in the box below.
$ _____ /week

Table E1: Balancing test for randomized information treatment

Baseline Survey	**Full Sample**			**Control**		**Treatment**		
	N	Mean	SD	Mean	SD	Mean	SD	**Diff.**
Age 18-34	454	42.07	49.42	41.95	49.45	42.20	49.50	0.3
Age 35-54	454	47.14	49.97	48.73	50.09	45.41	49.90	-3.3
White	454	72.69	44.61	75.42	43.15	69.72	46.05	-5.7
Male	454	41.19	49.27	40.25	49.15	42.20	49.50	1.9
College Degree	454	38.55	48.72	41.10	49.31	35.78	48.05	-5.3
HHold Income 0−40k	454	54.19	49.88	54.66	49.89	53.67	49.98	-1.0
No Household Budget	454	21.59	41.19	18.22	38.68	25.23	43.53	7.0*
Struggling Financially	454	42.73	49.52	41.53	49.38	44.04	49.76	2.5
Gig Pay is Essential	454	83.48	37.18	83.47	37.22	83.49	37.22	0.0
Has Other Gig Job	454	35.68	47.96	36.44	48.23	34.86	47.76	-1.6
Has Non-Gig Job	454	17.18	37.76	17.80	38.33	16.51	37.22	-1.3
Employed Full-Time Prior to Gig	454	36.34	48.15	38.98	48.87	33.49	47.30	-5.5
Employed Part-Time Prior to Gig	454	20.26	40.24	18.64	39.03	22.02	41.53	3.4
Unemployed Prior to Gig	454	10.13	30.21	8.47	27.91	11.93	32.48	3.5
Experience Delivery (12+ mo.)	454	57.49	49.49	56.78	49.64	58.26	49.43	1.5
Experience Rideshare (12+ mo.)	454	23.57	42.49	23.31	42.37	23.85	42.72	0.5

(A) Covariates

Baseline Survey	**Full Sample**			**Control**		**Treatment**		
	N	Mean	SD	Mean	SD	Mean	SD	**Diff.**
Hourly Pay	400	17.94	7.52	18.36	7.24	17.47	7.80	-0.9
Hourly Net Pay	360	12.38	5.26	12.50	4.85	12.27	5.65	-0.2
Weekly Pay	445	284.20	250.44	261.02	217.48	308.76	279.61	47.7**
Weekly Hours	404	17.41	14.81	16.70	13.95	18.17	15.66	1.5
Expenses Share	408	32.19	5.19	32.42	5.17	31.95	5.21	-0.5
Hourly Pay (Recall)	436	18.85	6.25	18.98	6.31	18.72	6.19	-0.3
Net Hourly Pay (Recall)	412	15.46	6.31	15.73	6.21	15.15	6.42	-0.6
Weekly Pay (Recall)	439	372.82	265.80	371.20	259.91	374.57	272.62	3.4
Weekly Hours (Recall)	440	22.69	13.63	22.68	13.57	22.69	13.73	0.0
Hourly Pay (Forecast)	324	19.36	6.80	18.87	6.32	19.92	7.31	1.1
Weekly Pay (Forecast)	344	376.79	324.95	354.20	313.60	402.77	336.65	48.6
Weekly Hours (Forecast)	333	23.24	14.05	22.19	13.55	24.47	14.56	2.3

(B) Outcomes

Notes: Data in this table is collected in our baseline survey. We first show the full sample. We then divide it into the control and the treatment groups for our information treatment. The treatment group receives detailed information on whether they overestimate or underestimate their net hourly pay. The last column shows the difference in means between the control and the treatment groups for each variable. Stars are used to denote the statistical significance of this difference (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

Table E2: Effect of information treatment on reviewing incentivized forecast of weekly pay

	Reviews Weekly Pay Forecast	
	(1)	(2)
	Change in Forecast	Has Reviewed Forecast
Overestimation Net Hourly Pay	-1.60*	
	(0.83)	
Abs(Overestimation Net Hourly Pay)		0.014*
		(0.0075)
Constant	-10.9*	0.17***
	(6.06)	(0.057)
Observations	147	149

Notes: Robust standard errors in parentheses. Stars are used to denote the statistical significance of each coefficient (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). Only those in the information treatment group are included in the regressions, as they were the ones allowed to review their forecast of weekly pay. *Overestimation Net Hourly Pay* is the overestimation in net hourly pay, which is defined as the recall belief minus the actual net hourly pay. Recall of net hourly pay is pooled for the week and the month before the baseline survey, with the actual job outcomes referring to the same time period for each worker. *Abs(Overestimation Net Hourly Pay)* is the absolute value of the overestimation in net hourly pay. *Change in Pay Forecast* measures the within-individual difference in the weekly pay forecast before and after receiving the information treatment. *Has Reviewed Pay Forecast* is a binary variable equal to 1 if the individual has chosen to review their weekly pay forecast immediately after the information treatment.

Table E3: Effect of information treatment on overestimation of job outcomes

	Overestimation (Recall - Actual)			
	(1)	(2)	(3)	(4)
	Weekly Pay	Weekly Hours	Hourly Pay	Net Hourly Pay
Treatment	-4.66	0.21	0.35	2.21***
	(13.4)	(0.91)	(0.84)	(0.67)
Control	17.0	1.62*	2.12***	2.39***
	(12.2)	(0.83)	(0.74)	(0.64)
Observations	172	162	161	131

(A) One to Two Weeks After Treatment

	Overestimation (Recall - Actual)			
	(1)	(2)	(3)	(4)
	Weekly Pay	Weekly Hours	Hourly Pay	Net Hourly Pay
Treatment	57.2***	4.39***	-0.46	0.98
	(16.8)	(1.17)	(0.82)	(0.66)
Control	57.9***	4.61***	0.70	2.37***
	(15.6)	(1.07)	(0.70)	(0.61)
Observations	145	140	131	116

(B) Two to Five Months After Treatment

Notes: Robust standard errors in parentheses. Stars are used to denote the statistical significance of each coefficient (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). We regress the overestimation of recalls of job outcomes on two binary variables: one for the treatment group and one for the control group in the information treatment. We omit the constant term in these regressions. *Overestimation* is the recall belief minus the actual job outcome. A positive (negative) value implies overestimation (underestimation) of a job outcome. The recalls are for the last week in Panel (A) – measured in our midline survey – and for the last month in Panel (B), measured in our endline survey.

Table E4: Effect of information treatment on beliefs of job outcomes

	Belief Net Hourly Pay		Belief Other Net Hourly Pay		Belief Expenses Share	
	(1)	(2)	(3)	(4)	(5)	(6)
Good Signal	2.41	1.07	2.91	2.37	-9.40**	-9.76*
	(1.90)	(1.91)	(1.90)	(1.79)	(4.72)	(4.94)
Bad Signal	0.072	0.81	-1.31	-1.31	4.45	3.84
	(1.30)	(1.26)	(1.16)	(1.24)	(3.14)	(3.53)
Observations	136	136	141	141	145	145
Baseline Outcome		✓		✓		✓
Demographic Controls		✓		✓		✓
p-value(Treatment No Effect)	0.45	0.69	0.17	0.23	0.054	0.080

(A) One to Two Weeks After Treatment

	Belief Net Hourly Pay		Belief Other Net Hourly Pay		Belief Expenses Share	
	(1)	(2)	(3)	(4)	(5)	(6)
Good Signal	2.22	1.23	-0.68	-0.95	-2.05	0.74
	(1.72)	(1.58)	(1.37)	(1.33)	(4.11)	(4.53)
Bad Signal	1.23	1.85*	-0.77	-0.85	-4.57	-5.54**
	(1.20)	(1.06)	(0.82)	(0.88)	(2.83)	(2.73)
Observations	129	128	132	132	136	136
Baseline Outcome		✓		✓		✓
Demographic Controls		✓		✓		✓
p-value(Treatment No Effect)	0.26	0.17	0.57	0.49	0.24	0.13

(B) Two to Five Months After Treatment

Notes: Robust standard errors in parentheses. Stars are used to denote the statistical significance of each coefficient (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). We estimate versions of $y_{it} = \beta_0 + \beta_1 Over_i + \beta_2 BadSignal_i + \beta_3 Good\ Signal_i + X_{i0}\Gamma + \varepsilon_{it}$, where y_{it} is the outcome variable for individual i at period t. $Over_i = 1$ if initial overestimation of net hourly pay is positive and 0 otherwise. Our two variables of interest here are $Bad\ Signal_i$ and $Good\ Signal_i$: $Bad\ Signal_i$ is equal to 1 if an individual is in the treatment group and $Over_i = 1$; $Good Signal_i$ if an individual is in the treated group and $Over_i = 0$. Individuals in the treatment group were told whether they misestimated their actual net hourly pay. X_{i0} is the covariates matrix. Our set of demographic controls consists of binary variables for male, white, age between 18-34, age between 35-54, at least a college degree and household income below $40,000/year, in addition to the outcome variable in the baseline survey. We test whether all treatment variables (*Good Signal* and *Bad Signal*) are jointly significant and provide a p-value for this test for each model. *Beliefs* dependent variables refer to the recall for net hourly pay (models (1) and (2)), the recall for expenses share (models (5) and (6)) and the belief of hourly net pay for the average worker in the same gig company (models (3) and (4)). The recalls are for the last week in Panel (A) – measured in our midline survey – and for the last month in Panel (B), measured in our endline survey.

Table E5: Effect of information treatment on beliefs of job outcomes (alternative specification)

	Belief Net Hourly Pay		Belief Other Net Hourly Pay		Belief Expenses Share	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	1.53	0.59	0.70	0.53	-2.19	-2.51
	(1.13)	(1.08)	(1.13)	(1.13)	(3.03)	(3.28)
Treatment · Overestimation	-0.26*	0.0025	-0.24	-0.23	0.65	0.61
	(0.16)	(0.15)	(0.17)	(0.16)	(0.40)	(0.40)
Observations	136	136	141	141	145	145
Baseline Outcome		✓		✓		✓
Demographic Controls		✓		✓		✓
p-value(Treatment No Effect)	0.14	0.85	0.37	0.36	0.28	0.32

(A) One to Two Weeks After Treatment

	Belief Net Hourly Pay		Belief Other Net Hourly Pay		Belief Expenses Share	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	1.97*	1.16	-0.75	-1.01	-4.78*	-3.96
	(1.08)	(0.99)	(0.82)	(0.82)	(2.57)	(2.57)
Treatment · Overestimation	-0.17	0.069	-0.019	0.019	0.22	-0.018
	(0.14)	(0.13)	(0.12)	(0.12)	(0.39)	(0.41)
Observations	129	128	132	132	136	136
Baseline Outcome		✓		✓		✓
Demographic Controls		✓		✓		✓
p-value(Treatment No Effect)	0.17	0.27	0.49	0.42	0.17	0.22

(B) Two to Five Months After Treatment

Notes: Robust standard errors in parentheses. Stars are used to denote the statistical significance of each coefficient (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). We estimate equation (7) defined in Appendix E. We include in our regressions but do not report the constant term and *Overestimation*. *Overestimation* is defined as the recall belief of net hourly pay minus the actual net hourly pay, pooled for one week and one month before the baseline survey. *Treatment* is a binary variable equal to 1 if the worker was assigned to our information treatment at the baseline survey. Our set of demographic controls consists of binary variables for male, white, age between 18-34, age between 35-54, at least a college degree and household income below $40,000/year, in addition to the outcome variable in the baseline survey. We test whether all treatment variables (*Treat and Treat · Overestimation*) are jointly significant and provide a p-value for this test for each model. *Beliefs* dependent variables refer to the recall for net hourly pay (models (1) and (2)), the recall for expenses share (models (5) and (6)) and the belief of hourly net pay for the average worker in the same gig company (models (3) and (4)). The recalls are for the last week in Panel (A) – measured in our midline survey – and for the last month in Panel (B), measured in our endline survey.

Table E6: Effect of information treatment on labor market decisions

	Other Jobs		**Weekly Hours**	
	(1)	(2)	(3)	(4)
Good Signal	-0.28**	-0.28**	-1.62	-1.12
	(0.11)	(0.12)	(3.16)	(1.97)
Bad Signal	0.037	0.055	1.84	0.91
	(0.083)	(0.082)	(2.24)	(2.16)
Observations	168	168	151	142
Baseline Outcome		✓		✓
Demographic Controls		✓		✓
p-value(Treatment No Effect)	0.041	0.045	0.63	0.79

(A) Main specification

	Other Jobs		**Weekly Hours**	
	(1)	(2)	(3)	(4)
Treatment	-0.12*	-0.11	0.18	-0.55
	(0.071)	(0.070)	(1.94)	(1.52)
Treatment · Overestimation	0.022**	0.024**	0.083	0.17
	(0.011)	(0.011)	(0.25)	(0.27)
Observations	168	168	151	142
Baseline Outcome		✓		✓
Demographic Controls		✓		✓
p-value(Treatment No Effect)	0.080	0.056	0.93	0.80

(B) Alternative specification

Notes: Robust standard errors in parentheses. Stars are used to denote the statistical significance of each coefficient (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). We estimate our main and alternative specifications, defined in Appendix E. *Overestimation* is defined as the recall belief of net hourly pay minus the actual net hourly pay, pooled for one week and one month before the baseline survey. *Treatment* is a binary variable equal to 1 if the worker was assigned to our information treatment at the baseline survey. *Bad Signal* is equal to 1 if an individual is in the treatment group and *Overestimation* is strictly positive, and equal to 0 otherwise. *Good Signal* is equal to 1 if an individual is in the treatment group and *Overestimation* is non-positive, and equal to 0 otherwise. Our set of demographic controls consists of binary variables for male, white, age between 18-34, age between 35-54, at least a college degree and household income below $40,000/year, in addition to the outcome variable in the baseline survey. We test whether all treatment variables are jointly significant and provide a p-value for this test for each model. The dependent variable for models (3) and (4) is the weekly hours worked for the main gig company during the weeks collected in the endline survey (including zeroes). *Other Jobs* is a binary variable equal to 1 if the worker has another gig or non gig job.

Appendix F: Additional Model Discussion

This section expands on additional aspects of our behavioral labor supply model. We first detail how workers make labor supply and consumption decisions in the non-flexible job. We then describe how gig workers make job choice decisions. Next, we describe the learning process for the work disutility shifter η, which influences workers' labor supply choices. Finally, we present additional model simulation results.

F.1 Non-Flexible Job Optimization Problem

We assume the non-flexible job has a constant net hourly pay of w_O, which the worker correctly observes. At this job, the weekly labor supply $h_{O,t}$ is fixed and equals $h_{O,t} = \bar{h}$ for all t. At the beginning of a household budget cycle, in period t, the non-flexible job worker maximizes his value function by choosing c_t, c_{t+1} and s_{t+1}:

$$\max_{c_t, c_{t+1}, s_{t+1}} u(c_t) + u(c_{t+1}) + g(s_{t+1}) - \eta_t c(\bar{h}) - c(\bar{h}) \tag{8}$$

such that

$$c_t + c_{t+1} + s_{t+1} = w_O \cdot \bar{h} + w_O \cdot \bar{h} + (1 + r) \cdot s_{t-1} \tag{9}$$

where we made the simplifying assumption that this worker can correctly forecast η_{t+1}, and used the fact that $E[\eta_{t+1}] = 1$. The first-order conditions imply that

$$u'(c_t^*) = u'(c_{t+1}^*) = g'(s_{t+1}^*) \tag{10}$$

In period $t + 1$, the worker will not revise their plans for c_{t+1} and s_{t+1}, as their labor income is outside of their control and the realization of η_{t+1} affects only the fixed component of their utility.

F.2 Job Choice Decision

At the start of each household budget cycle, before learning η_t and $w_{G,t}$, gig workers decide whether they want to move to the non-flexible job O. They make this decision by comparing the perceived expected payoff from one two-period cycle in the gig job ($\tilde{E}[V_G]$) and in the non-flexible job ($E[V_O]$), choosing whichever job provides them with the highest amount of perceived utility.

Define the perceived expected payoff from the gig job $\tilde{E}[V_G]$ as

$$\tilde{E}[V_G] = \tilde{E}\left[u(\tilde{c}_t^*) + u(\tilde{c}_{t+1}^*) + g(\tilde{s}_{t+1}) - \eta_t c(\tilde{h}_t^*) - \eta_{t+1} c(\tilde{h}_{t+1}^*)\right] \tag{11}$$

where $\tilde{E}[\cdot]$ is the perceived expected value function implied by the learning processes described in equations (5) and (13). It denotes the need to estimate η and w_G, which affect the decision of how much to consume and work in each period. In addition, this expectation takes into account the effects of overestimating current net pay ($\theta > 0$). Define the *actual* payoff from the gig job $E[V_G]$ as the expected value of the gig job under correct beliefs ($\tilde{E}[\eta_t] = E[\eta_t]$, $\tilde{E}[w_{G,t}] = E[w_{G,t}]$ and $\theta = 0$) but biased decisions $\tilde{c}_t^*$, $\tilde{c}_{t+1}^*$, $\tilde{s}_{t+1}$, $\tilde{h}_t^*$, $\tilde{h}_{t+1}^*$.

Using the fact that $E[\eta_t] = 1$ for any t, define the expected payoff from the non-flexible job $E[V_O]$ as

$$E[V_O] = u(c_t^*) + u(c_{t+1}^*) + g(s_{t+1}^*) - c(\bar{h}) - c(\bar{h}) \tag{12}$$

F.3 Learning the Work Disutility Shifter η

Instead of directly posing the problem of forecasting work hours, we assume the gig worker tries to learn the true mean of η, which is a shifter for work disutility and directly influences the labor supply schedule at each wage.

We assume $\eta_t = 1 + \nu_t$, where $\nu_t \sim N(0, \sigma_\nu^2)$. Assume σ_ν^2 is known. Assume the worker has a prior $\tilde{\eta}_0$ such that $\tilde{\eta}_0 \sim N(\eta_0, \sigma_{\eta 0}^2)$. The gig worker receives an unbiased signal for

η in each period.[35] There is a bias involved in processing signals, which depend on their rank within the distribution of observed signals. In particular, let $\psi_{t,j} = \frac{1}{1+\alpha F_t(\eta_j)}$, where $\alpha \geq 0$ and $F_t(\eta_j)$ is the empirical cumulative distribution function for signals up to period t, referring to η_j. Thus, more weight is given to lower values of η, which imply a higher desire to work given other factors.

We can then modify the standard formula for normal Bayesian learning to find the perceived posterior mean of η_{t+1}, which we call $\tilde{E}[\eta_{,t+1}]$:

$$\tilde{E}[\eta_{t+1}] = \frac{\sigma_\nu^2 \eta_0 + \sigma_{\eta 0}^2 \sum_{j=1}^{t} \psi_{t,j} \eta_j}{\sigma_\nu^2 + \sigma_{\eta 0}^2 \sum_{j=1}^{t} \psi_{t,j}} \tag{13}$$

If $\alpha = 0$, we return to the standard Bayesian learning model, such that the posterior mean eventually converges to the true mean of 1. It is easy to see that $\alpha > 0$ leads to a persistent bias in the posterior mean. In particular, lower η signals get overweighted, such that the posterior mean $\tilde{E}[\eta_{t+1}]$ will tend to be underestimated.

F.4 Additional Model Simulation Results

Figure F1 plots the evolution of workers' beliefs about net hourly pay ($\tilde{E}[w_{G,t+1}]$) for different combinations of gig pay misperceptions (θ), memory bias (α), and wage signal standard deviation (σ_ε). Panel (A) shows patterns without memory bias ($\alpha = 0$). In this case, increasing wage signal variance only affects belief volatility. Panel (B) adds a memory bias. With low signal variance, beliefs show a moderate upward bias. However, as signal variance increases, and fixing the value of θ, the size of the net gig pay overestimation greatly increases.

These patterns emerge as higher wage variance produce more extreme signals. Workers then overweight past high wage realizations relative to low ones, and selective memory bias becomes more consequential. The results exemplify the idea that increased flexibility in

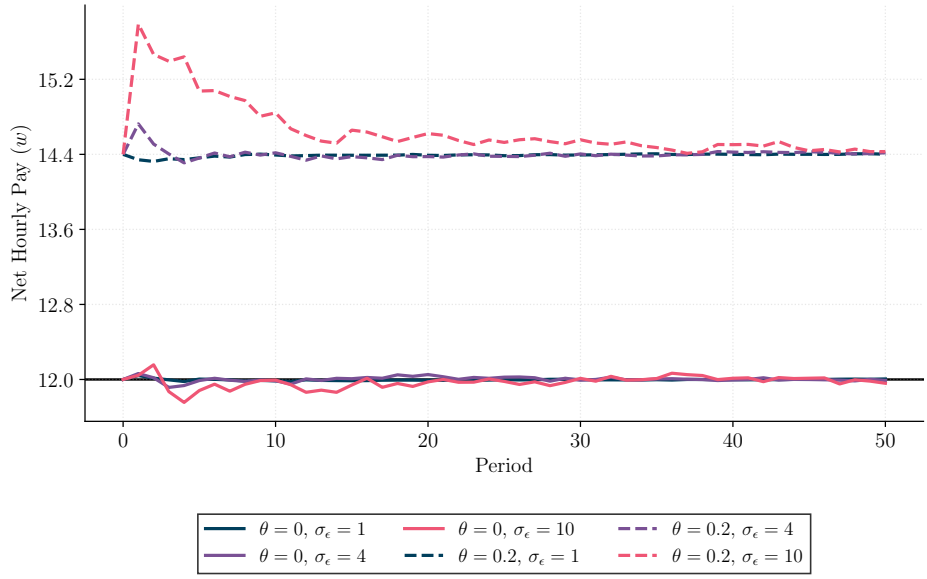
[35] Intuitively, we can assume he observes, in period t, $\tilde{h}_t^*$, $w_{G,t}(1+\theta)$ and $\tilde{c}_t^*$. He can then back out η_t from the first-order conditions, such that $\eta_t = \frac{w_{G,t}(1+\theta) \cdot u'(\tilde{c}_t^*)}{c'(\tilde{h}_t^*)}$.

pay structures, which typically increases wage variance, may exacerbate the consequences of behavioral biases in workers' learning processes.

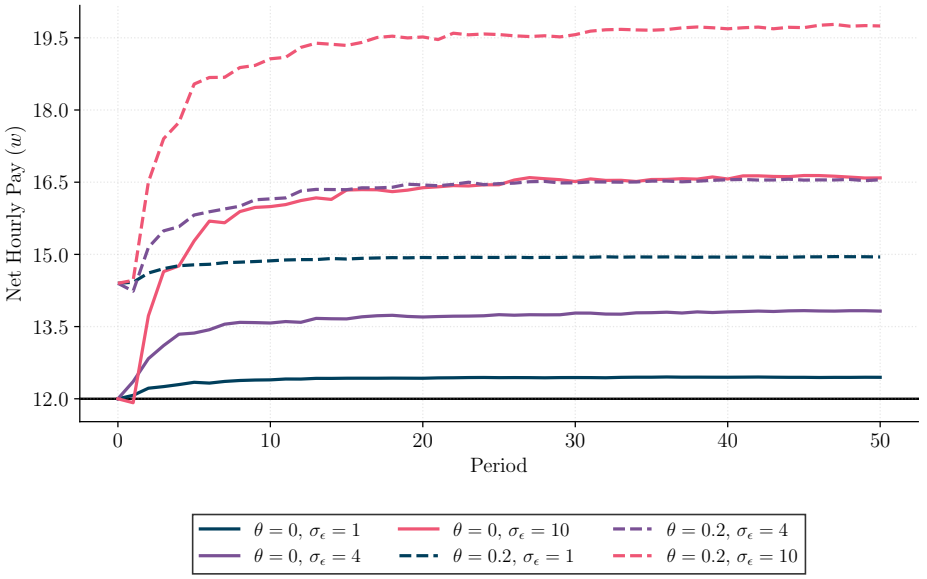
In Figure F2, we solve our model under different combinations of behavioral biases and then present workers' resulting labor supply decisions. Details of the simulation are given in the figure's footnotes. We plot hours worked in periods t and $t+1$ for three cases: no biases, net gig pay misperception only ($\theta = 0.2$, $\alpha = 0$), and net gig pay misperception with a positive memory bias ($\theta = 0.2$, $\alpha = 10$). Unbiased workers choose similar hours across both periods and across cycles. Adding net gig pay misperceptions increases average hours work substantially, as workers overestimate the returns to working. Adding a memory bias leads to a sizable divergence between h_t and h_{t+1}, which widens over time with biased learning of future expected pay and propensity to work. This represents an inefficient allocation of labor supply across the two periods, with work being backloaded to the end of the cycle.

Figure F3 illustrates whether gig workers, in our model, overestimate various outcomes under different combinations of behavioral biases. Panel (A) shows the overestimation of hours worked in period $t + 1$. With no biases or with $\theta > 0$ only, workers make minimal errors forecasting future hours worked. However, adding biased learning ($\alpha = 10$) leads to substantial hours over-prediction, with it growing over time. Panel (B) present a similar story qualitatively, but for the overestimation of net hourly gig pay. Panel (C) demonstrates how biases compound in workers' predictions of their net labor income. Panel (D) then reveals that biased workers significantly over-forecast their savings, which, if translated to real-world terms, can have important implications for household financial planning.

Figure F1: Simulation of learning models for different combinations of biases and wage signal variance



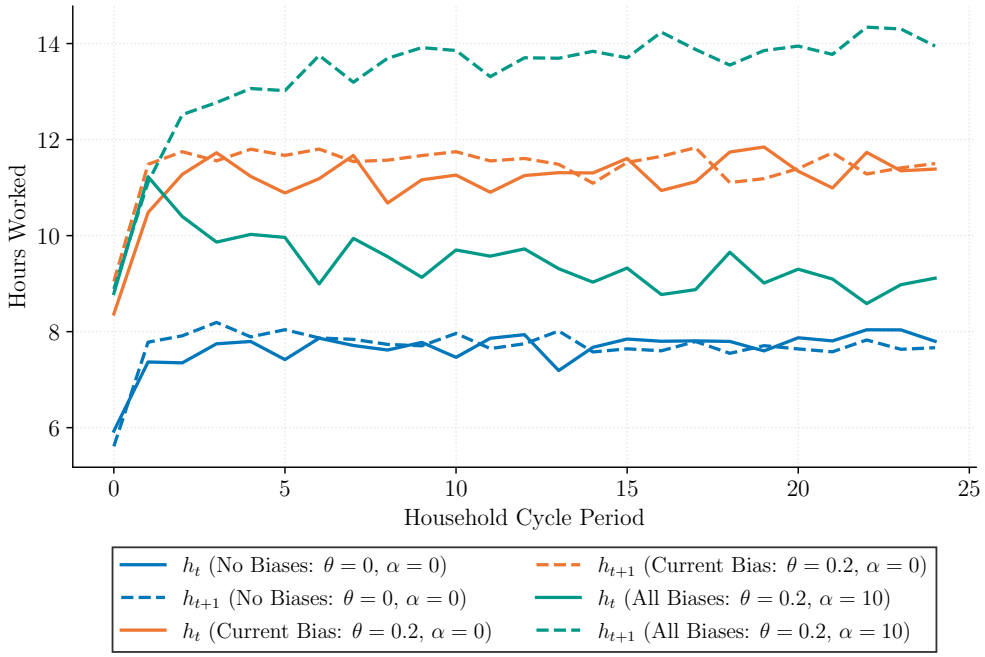
(A) Learning without Biased Memory ($\alpha = 0$)



(B) Learning with Biased Memory ($\alpha = 10$)

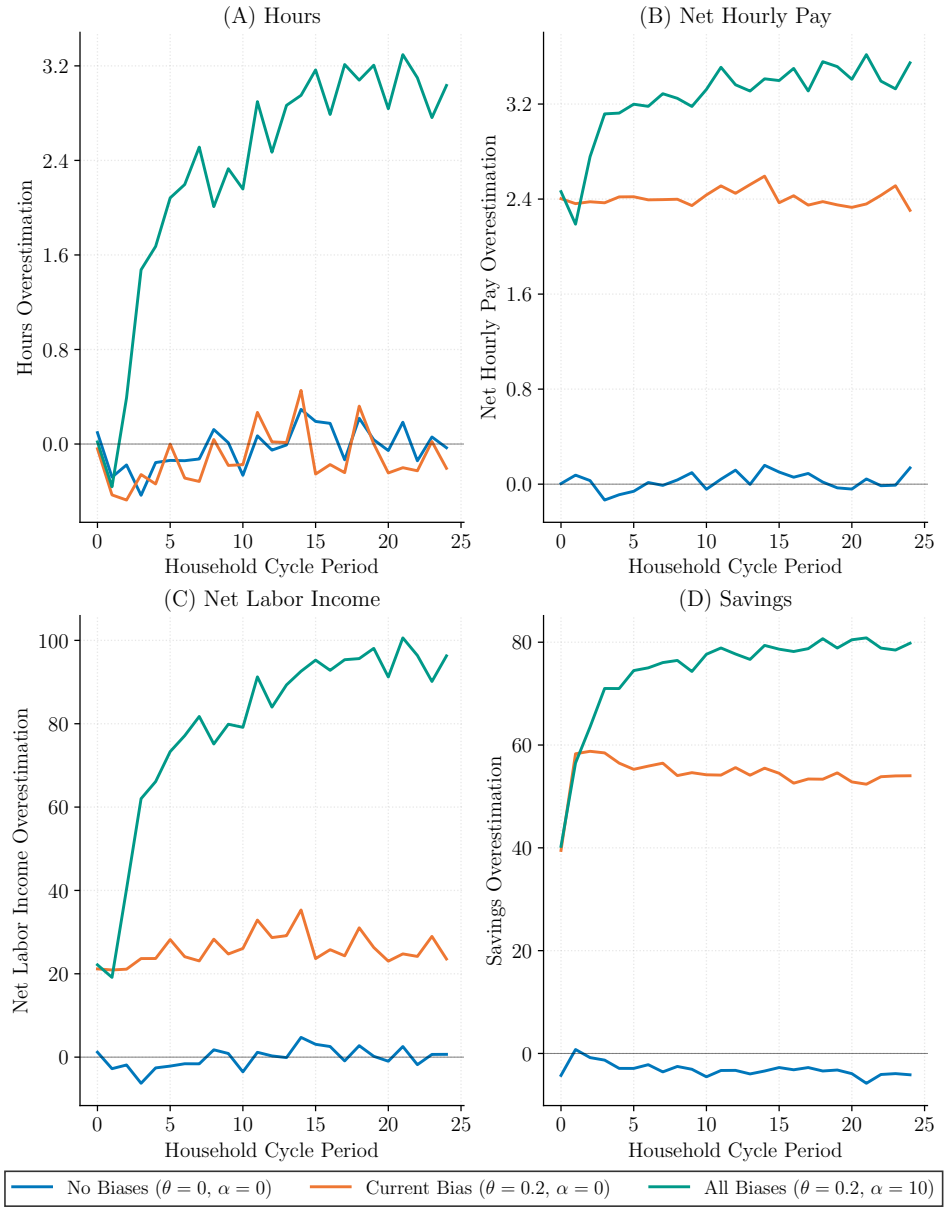
Notes: We plot one realization of the learning model described in Section 5.1 and Appendix F for different values of the of the memory bias parameter α, hourly net pay overestimation parameter θ and the standard deviation σ_ε of the net pay signal $w_{G,t}$. In Panel (A), we plot the posterior mean for the net hourly pay at a gig job $\tilde{E}[w_{G,t+1}]$ after each signal for different combinations of θ and σ_ε when $\alpha = 0$. In Panel (B), we plot the same posterior mean, but for different combinations of θ and σ_ε when $\alpha = 10$. We assume the true expected gig net hourly pay is $\mu_G = 12$. The prior means are $\mu_0 = 12$ and $\eta_0 = 1$, with the prior standard deviations being equal to 10 for w_G and 0.5 for η.

Figure F2: Simulation of behavioral labor supply model for different combinations of biases



Notes: We numerically solve the model described in Section 5.1 for different values of the memory bias parameter α and hourly net pay overestimation parameter θ. We plot the labor supply in both periods of each household cycle across cycles. We solve 10,000 realizations and plot the median one for each household cycle period. We assume that the consumption utility function is $u(c) = 0.85 \cdot c^{0.7}$, the cost of effort function is $c(h) = h^{1.2}$, the savings utility function is $g(s) = 1.5 \cdot \text{sgn}(s) \cdot |s|^{0.6}$, $r = 0$, the true expected gig net hourly pay is $\mu_G = 12$, and the standard deviations for the signals of net hourly pay $w_{G,t}$ and the work disutility shifter η_t are, respectively, $\sigma_\varepsilon = 2$ and $\sigma_\nu = 0.1$. The prior means are $\mu_0 = 12$ and $\eta_0 = 1$, with the prior standard deviations being equal to 10 for w_G and 0.5 for η.

Figure F3: Overestimation of job outcomes in simulation of behavioral labor supply model for different combinations of biases



Notes: We numerically solve the model described in Section 5.1. In all four panels, we plot the overestimation of a variable across cycles for different combinations of the memory bias parameter α and the hourly net pay overestimation parameter θ. In Panel (A), we plot the overestimation of hours worked at $t+1$ ($\tilde{h}^p_{t+1} - \tilde{h}^*_{t+1}$). In Panel (B), we plot the overestimation of expected net hourly pay at $t+1$ ($\tilde{E}[w_{G,t+1}] - E[w_{G,t+1}]$). In Panel (C), we plot the overestimation of net labor income at $t+1$ ($\tilde{E}[w_{G,t+1}] \cdot \tilde{h}^p_{t+1} - E[w_{G,t+1}] \cdot \tilde{h}^*_{t+1}$). In Panel (D), we plot the overestimation of savings at $t+1$ ($s^p_{t+1} - s^*_{t+1}$). We solve 10,000 realizations and plot the median one for each household cycle period. We assume that the consumption utility function is $u(c) = 0.85 \cdot c^{0.7}$, the cost of effort function is $c(h) = h^{1.2}$, the savings utility function is $g(s) = 1.5 \cdot \text{sgn}(s) \cdot |s|^{0.6}$, $r = 0$, the true expected gig net hourly pay is $\mu_G = 12$, and the standard deviation for the signals of net hourly pay $w_{G,t}$ and the cost of effort shifter η_t are, respectively, $\sigma_\varepsilon = 2$ and $\sigma_\nu = 0.1$. The prior means are $\mu_0 = 12$ and $\eta_0 = 1$, with the prior standard deviations being equal to 10 for w_G and 0.5 for η.